

# States, Evolution Laws, Flows, First-Order Systems, Vector Fields, and Phase Portraits

Thursday, August 27, 2026

## Idea: A dynamical model

A deterministic dynamical model consists of

$$\boxed{\text{state}} \quad + \quad \boxed{\text{evolution law}} \quad + \quad \boxed{\text{initial condition}}.$$

The state records the information needed to apply the evolution law.

## Definition: Continuous-time dynamical system

Let  $n \geq 1$ , let  $X \subseteq \mathbb{R}^n$ , let  $J \subseteq \mathbb{R}$  be an interval, and let  $f : J \times X \rightarrow \mathbb{R}^n$ . A **continuous-time dynamical system** is an initial-value problem

$$\dot{x} = f(t, x), \quad x(t_0) = x_0, \quad (t_0, x_0) \in J \times X.$$

The set  $X$  is the **state space**, also called the **phase space**. The system is **autonomous** if there is a function  $g : X \rightarrow \mathbb{R}^n$  such that  $f(t, x) = g(x)$  for every  $(t, x) \in J \times X$ .

## Notation: Time derivative

The variable  $t$  denotes time, and

$$\dot{x} = \frac{dx}{dt}.$$

## Recall: Solution and trajectory

A **solution** of  $\dot{x} = f(t, x)$  through  $(t_0, x_0)$  is a differentiable function  $x : I \rightarrow X$ , where  $I \subseteq J$  is an interval containing  $t_0$ , such that

$$x(t_0) = x_0, \quad \dot{x}(t) = f(t, x(t)) \quad (t \in I).$$

Its image  $\{x(t) : t \in I\}$  is its **trajectory**, or **orbit**.

**Definition: Periodic solution**

A solution  $x : \mathbb{R} \rightarrow X$  is **periodic** if some  $T > 0$  satisfies

$$x(t + T) = x(t)$$

for every  $t \in \mathbb{R}$ . If  $x$  is nonconstant, its trajectory is a **periodic orbit**.

**Definition: Equilibrium**

For an autonomous system  $\dot{x} = g(x)$ , a point  $x^* \in X$  is an **equilibrium**, also called a **fixed point**, if

$$g(x^*) = 0.$$

**Observation: Equilibria and constant solutions**

For the autonomous system  $\dot{x} = g(x)$ , the constant function

$$x(t) = x^*$$

is a solution if and only if  $x^*$  is an equilibrium.

**Definition: Local and global flow**

Let  $X \subseteq \mathbb{R}^n$  be open, and let  $g \in C^1(X, \mathbb{R}^n)$ . For each  $x_0 \in X$ , let  $I_{x_0}$  be the largest interval containing 0 on which the unique solution  $x(\cdot; x_0)$  of

$$\dot{x} = g(x), \quad x(0) = x_0,$$

is defined. Define

$$\mathcal{D} = \{(t, x_0) : x_0 \in X, t \in I_{x_0}\}.$$

The **local flow** is

$$\Phi : \mathcal{D} \rightarrow X, \quad \Phi(t, x_0) = x(t; x_0).$$

It is a **global flow** if  $I_{x_0} = \mathbb{R}$  for every  $x_0 \in X$ .

**Notation: Time-indexed flow map**

For each  $t \in \mathbb{R}$ , define

$$X_t = \{x \in X : (t, x) \in \mathcal{D}\}.$$

The corresponding **time- $t$  map** is

$$\Phi_t : X_t \rightarrow X, \quad \Phi_t(x) = \Phi(t, x).$$

**Recall: Action of the real numbers**

An **action of**  $(\mathbb{R}, +)$  **on**  $X$  is a map  $A : \mathbb{R} \times X \rightarrow X$  satisfying

$$A(0, x) = x, \quad A(t + s, x) = A(t, A(s, x))$$

for all  $s, t \in \mathbb{R}$  and  $x \in X$ .

**Observation: Autonomous flow law**

For the local flow of an autonomous system,

$$\Phi(0, x) = x, \quad x \in X.$$

If  $x \in X_s$  and  $\Phi_s(x) \in X_t$ , then

$$x \in X_{t+s}, \quad \Phi_{t+s}(x) = \Phi_t(\Phi_s(x)).$$

For a global flow,  $X_t = X$  for every  $t \in \mathbb{R}$ , so

$$\Phi_0 = \text{id}_X, \quad \Phi_{t+s} = \Phi_t \circ \Phi_s.$$

Thus a global flow is an action of  $(\mathbb{R}, +)$  on  $X$ .

**Recall: Reducing an equation to first order**

Let  $n \geq 1$  be an integer and let  $G : \mathbb{R}^n \rightarrow \mathbb{R}$ . For the  $n$ th-order scalar equation

$$q^{(n)} = G(q, \dot{q}, \dots, q^{(n-1)}),$$

define

$$x_1 = q, \quad x_2 = \dot{q}, \quad \dots, \quad x_n = q^{(n-1)}.$$

Then

$$\dot{x}_1 = x_2, \quad \dots, \quad \dot{x}_{n-1} = x_n, \quad \dot{x}_n = G(x_1, \dots, x_n).$$

The resulting first-order system has state  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ . Its Euclidean phase-space dimension is  $n$ .

**Question: What is the state of a damped oscillator?**

Rewrite

$$m\ddot{q} + b\dot{q} + kq = 0, \quad m > 0, \quad b \geq 0, \quad k > 0,$$

as a first-order autonomous system.

**Solution.** Define

$$x_1 = q, \quad x_2 = \dot{q}.$$

Then

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2,$$

or

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -k/m & -b/m \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

**Question: What is the state of a damped oscillator? (continued)**

The state is  $(q, \dot{q}) \in \mathbb{R}^2$ . Position alone is insufficient because the same position can occur with different velocities.

**Definition: Vector field**

Let  $X \subseteq \mathbb{R}^n$  be open. A **vector field** on  $X$  is a function

$$g : X \rightarrow \mathbb{R}^n.$$

In the autonomous system  $\dot{x} = g(x)$ , the vector  $g(x)$  gives the instantaneous velocity at the state  $x$ .

**Definition: Linear and nonlinear vector fields**

A vector field  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is **linear** if

$$f(au + bv) = af(u) + bf(v)$$

for all  $u, v \in \mathbb{R}^n$  and all  $a, b \in \mathbb{R}$ . Equivalently,

$$f(x) = Ax$$

for some constant  $n \times n$  matrix  $A$ . A vector field that is not linear is **nonlinear**.

**Question: Why is the pendulum nonlinear?**

Compare

$$\ddot{\theta} + \frac{g_0}{L}\theta = 0 \quad \text{and} \quad \ddot{\theta} + \frac{g_0}{L}\sin\theta = 0, \quad g_0 > 0, \quad L > 0,$$

where  $g_0$  is the magnitude of gravitational acceleration and  $L$  is the pendulum length.

**Solution.** With  $\omega = \dot{\theta}$ , the systems are

$$\begin{aligned} \dot{\theta} &= \omega, & \dot{\omega} &= -\frac{g_0}{L}\theta, \\ \dot{\theta} &= \omega, & \dot{\omega} &= -\frac{g_0}{L}\sin\theta. \end{aligned}$$

The first is linear. The second is nonlinear because, in general,

$$\sin(a\theta_1 + b\theta_2) \neq a\sin\theta_1 + b\sin\theta_2.$$

$$\lim_{\theta \rightarrow 0} \frac{\sin\theta}{\theta} = 1.$$

Thus  $\sin\theta$  is well approximated by  $\theta$  when  $|\theta|$  is small. Replacing  $\sin\theta$  by  $\theta$  gives the linear small-angle model.

**Visualization: Time graph versus phase trajectory**

For a state  $x = (q, v)$  with  $v = \dot{q}$ :

| time graph                | phase trajectory                  |
|---------------------------|-----------------------------------|
| $(t, q(t))$               | $(q(t), v(t))$                    |
| one component versus time | the complete state in phase space |

**Definition: Phase portrait**

A **phase portrait** is a diagram of the qualitatively distinct trajectories and equilibria in phase space, with the direction of increasing time marked on each non-equilibrium trajectory.

# Asymptotic Notation, Phase Lines, and Stability

Tuesday, September 1, 2026

## Notation: Big-O notation near a finite point

Let  $D \subseteq \mathbb{R}^m$ , let  $\mathbf{a} \in \mathbb{R}^m$ , and suppose that  $D$  contains points arbitrarily close to  $\mathbf{a}$  other than  $\mathbf{a}$ . Let  $g : D \rightarrow \mathbb{R}^p$  and  $h : D \rightarrow \mathbb{R}$ . The statement

$$g(\mathbf{x}) = O(h(\mathbf{x})) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

means that there are constants  $C > 0$  and  $\delta > 0$  such that

$$\|g(\mathbf{x})\| \leq C|h(\mathbf{x})|$$

whenever  $\mathbf{x} \in D$  and

$$0 < \|\mathbf{x} - \mathbf{a}\| < \delta.$$

The domain  $D$  specifies any one-sided restriction. For example, when  $m = 1$ , the choices  $D = (0, \infty)$  and  $a = 0$  specify  $x \downarrow 0$ .

## Notation: Big-O notation at infinity

Let  $D \subseteq \mathbb{R}^m$  be unbounded, let  $g : D \rightarrow \mathbb{R}^p$ , and let  $h : D \rightarrow \mathbb{R}$ . The statement

$$g(\mathbf{x}) = O(h(\mathbf{x})) \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D)$$

means that there are constants  $C > 0$  and  $M > 0$  such that

$$\|g(\mathbf{x})\| \leq C|h(\mathbf{x})|$$

whenever  $\mathbf{x} \in D$  and  $\|\mathbf{x}\| > M$ . For a scalar argument,  $x \rightarrow +\infty$  uses  $x > M$ , whereas  $x \rightarrow -\infty$  uses  $x < -M$ .

**Observation: An approach label does not assert a limit**

In a statement such as

$$g(x) = O(h(x)) \quad (x \rightarrow a),$$

the label  $x \rightarrow a$  specifies where the estimate must hold. It does not assert that  $g(x)$ ,  $h(x)$ , or their quotient has a limit as  $x \rightarrow a$ .

**Notation: Little-o notation near a finite point**

Let  $D \subseteq \mathbb{R}^m$ , let  $\mathbf{a} \in \mathbb{R}^m$ , and let  $g : D \rightarrow \mathbb{R}^p$  and  $h : D \rightarrow \mathbb{R}$ . Suppose  $D$  contains points arbitrarily close to  $\mathbf{a}$  other than  $\mathbf{a}$ . The statement

$$g(\mathbf{x}) = o(h(\mathbf{x})) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

means that, for every  $\eta > 0$ , there is a  $\delta > 0$  such that

$$\|g(\mathbf{x})\| \leq \eta|h(\mathbf{x})|$$

whenever  $\mathbf{x} \in D$  and  $0 < \|\mathbf{x} - \mathbf{a}\| < \delta$ .

**Observation: Ratio test for little-o notation**

Let  $D \subseteq \mathbb{R}^m$ , let  $\mathbf{a} \in \mathbb{R}^m$ , let  $g : D \rightarrow \mathbb{R}^p$ , and let  $h : D \rightarrow \mathbb{R}$ . Suppose  $D$  contains points arbitrarily close to  $\mathbf{a}$  other than  $\mathbf{a}$ , and suppose there is a  $\delta_0 > 0$  such that  $h(\mathbf{x}) \neq 0$  whenever  $\mathbf{x} \in D$  and  $0 < \|\mathbf{x} - \mathbf{a}\| < \delta_0$ . Then

$$g(\mathbf{x}) = o(h(\mathbf{x})) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

if and only if

$$\frac{\|g(\mathbf{x})\|}{|h(\mathbf{x})|} \rightarrow 0 \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D).$$

**Notation: Little-o notation at infinity**

Let  $D \subseteq \mathbb{R}^m$  be unbounded, let  $g : D \rightarrow \mathbb{R}^p$ , and let  $h : D \rightarrow \mathbb{R}$ . The statement

$$g(\mathbf{x}) = o(h(\mathbf{x})) \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D)$$

means that, for every  $\eta > 0$ , there is an  $M > 0$  such that

$$\|g(\mathbf{x})\| \leq \eta|h(\mathbf{x})|$$

whenever  $\mathbf{x} \in D$  and  $\|\mathbf{x}\| > M$ . For a scalar argument,  $x \rightarrow +\infty$  uses  $x > M$ , whereas  $x \rightarrow -\infty$  uses  $x < -M$ .

**Notation: Asymptotic equivalence near a finite point**

Let  $D \subseteq \mathbb{R}^m$ , let  $\mathbf{a} \in \mathbb{R}^m$ , and let  $g, h : D \rightarrow \mathbb{R}$ . Suppose  $D$  contains points arbitrarily close to  $\mathbf{a}$  other than  $\mathbf{a}$ , and suppose there is a  $\delta_0 > 0$  such that  $h(\mathbf{x}) \neq 0$  whenever  $\mathbf{x} \in D$  and  $0 < \|\mathbf{x} - \mathbf{a}\| < \delta_0$ . The statement

$$g(\mathbf{x}) \sim h(\mathbf{x}) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

means that

$$\lim_{\substack{\mathbf{x} \rightarrow \mathbf{a} \\ \mathbf{x} \in D}} \frac{g(\mathbf{x})}{h(\mathbf{x})} = 1.$$

Equivalently,  $g(\mathbf{x}) - h(\mathbf{x}) = o(h(\mathbf{x}))$  along the same approach.

**Notation: Asymptotic equivalence at infinity**

Let  $D \subseteq \mathbb{R}^m$  be unbounded, let  $g, h : D \rightarrow \mathbb{R}$ , and suppose  $h(\mathbf{x}) \neq 0$  for all  $\mathbf{x} \in D$  with sufficiently large norm. The statement

$$g(\mathbf{x}) \sim h(\mathbf{x}) \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D)$$

means that

$$\frac{g(\mathbf{x})}{h(\mathbf{x})} \rightarrow 1 \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D).$$

For a scalar argument,  $x \rightarrow +\infty$  and  $x \rightarrow -\infty$  use the corresponding one-sided tails.

**Definition: Phase line**

A scalar autonomous system on an interval  $I \subseteq \mathbb{R}$  has the form

$$\dot{x} = f(x), \quad x \in I.$$

Its **phase line** consists of the equilibria  $f(x^*) = 0$  together with right arrows where  $f(x) > 0$  and left arrows where  $f(x) < 0$ .

**Visualization: From the vector field to the phase line**

| condition at $x$ | $\dot{x}$     | motion        |
|------------------|---------------|---------------|
| $f(x) > 0$       | $\dot{x} > 0$ | $\rightarrow$ |
| $f(x) = 0$       | $\dot{x} = 0$ | $\bullet$     |
| $f(x) < 0$       | $\dot{x} < 0$ | $\leftarrow$  |

**Observation: Arrows between equilibria**

If  $f$  is continuous and an interval contains no equilibrium, then  $f$  has constant sign there. Every solution that remains in that interval is strictly monotone.

**Definition: Lyapunov stability**

Consider an autonomous system on an interval  $I \subseteq \mathbb{R}$  with unique forward solutions. An equilibrium  $x^* \in I$  is **Lyapunov stable** if, for every  $\varepsilon > 0$ , there is a  $\delta > 0$  such that every  $x_0 \in I$  with  $|x_0 - x^*| < \delta$  generates a solution defined for all  $t \geq 0$  and

$$|x(t; x_0) - x^*| < \varepsilon \quad \text{for every } t \geq 0.$$

It is **unstable** if it is not Lyapunov stable.

**Definition: Attracting equilibrium**

Consider an autonomous system on an interval  $I \subseteq \mathbb{R}$  with unique forward solutions. An equilibrium  $x^* \in I$  is **attracting** if some  $\eta > 0$  has the following property: whenever  $x_0 \in I$  and  $|x_0 - x^*| < \eta$ , the solution is defined for all  $t \geq 0$  and

$$\lim_{t \rightarrow \infty} x(t; x_0) = x^*.$$

**Definition: Basin of attraction**

For an autonomous system with unique solutions on an interval  $I \subseteq \mathbb{R}$ , the **basin of attraction** of an attracting equilibrium  $x^* \in I$  is

$$\mathcal{B}(x^*) = \left\{ x_0 \in I : \begin{array}{l} x(t; x_0) \text{ exists for every } t \geq 0, \\ \lim_{t \rightarrow \infty} x(t; x_0) = x^* \end{array} \right\}.$$

**Definition: Asymptotic stability**

An equilibrium is **asymptotically stable** if it is both Lyapunov stable and attracting.

**Question: What does the phase line of the sine flow determine?**

For

$$\dot{x} = \sin x,$$

find and classify every equilibrium. For  $x(0) = \pi/4$ , determine the direction of motion, the limiting state, and the concavity of  $x(t)$ .

**Solution.** The equilibria are

$$x^* = k\pi, \quad k \in \mathbb{Z}.$$

For  $j \in \mathbb{Z}$ ,

$$\sin x \begin{cases} > 0, & x \in (2j\pi, (2j+1)\pi), \\ < 0, & x \in ((2j-1)\pi, 2j\pi), \end{cases}$$

so the phase line is

$$\cdots \longrightarrow (-\pi) \longleftarrow (0) \longrightarrow (\pi) \longleftarrow (2\pi) \longrightarrow \cdots.$$

Thus  $x^* = (2j+1)\pi$  is asymptotically stable and  $x^* = 2j\pi$  is unstable.

The smooth, bounded vector field  $\sin x$  gives unique solutions for all time. For  $x(0) = \pi/4$ , the solution increases and cannot cross either equilibrium 0 or  $\pi$ . Hence it remains in  $(0, \pi)$  and is

**Question: What does the phase line of the sine flow determine? (continued)**

bounded above, so it has a limit  $L \in [\pi/4, \pi]$  as  $t \rightarrow \infty$ . If  $L < \pi$ , then  $\sin L > 0$ , so eventually  $\dot{x}(t) \geq \sin L/2 > 0$ . This would force unbounded growth, contradicting convergence to  $L$ . Therefore

$$\lim_{t \rightarrow \infty} x(t) = \pi.$$

Differentiating  $\dot{x} = \sin x$  with respect to  $t$  gives

$$\ddot{x} = \cos x \dot{x} = \sin x \cos x.$$

Therefore

$$\ddot{x} \begin{cases} > 0, & 0 < x < \pi/2, \\ = 0, & x = \pi/2, \\ < 0, & \pi/2 < x < \pi. \end{cases}$$

The time graph is increasing, initially concave up, and then concave down.

**Question: Can an implicit equilibrium be counted and classified?**

Analyze

$$\dot{x} = x - \cos x$$

without solving the equilibrium equation in closed form.

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**Solution.** An equilibrium satisfies  $x = \cos x$ , so it must lie in  $[-1, 1]$ . Let

$$g(x) = x - \cos x.$$

On  $[-1, 1]$ ,

$$g'(x) = 1 + \sin x > 0.$$

Also,

$$g(0) = -1 < 0, \quad g(1) = 1 - \cos 1 > 0.$$

The intermediate value theorem gives at least one zero in  $(0, 1)$ , and strict monotonicity gives at most one. Hence there is exactly one equilibrium  $x^* \in (0, 1)$ . Numerically,

$$g(0.739) < 0, \quad g(0.740) > 0, \quad x^* \approx 0.739085.$$

Because  $g(x) < 0$  immediately to the left of  $x^*$  and  $g(x) > 0$  immediately to the right, both adjacent arrows point away from  $x^*$ . Thus  $x^*$  is unstable.

# Logistic Growth, Linearization, and Critical Slowing

Thursday, September 3, 2026

## Definition: Half-stability

Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be continuous, and let  $x^* \in \text{int}(I)$  be an equilibrium of  $\dot{x} = f(x)$ . The equilibrium  $x^*$  is **half-stable** if  $f$  has the same sign on the two sides of  $x^*$  sufficiently near  $x^*$ . Precisely, there is a  $\delta > 0$  with  $[x^* - \delta, x^* + \delta] \subset I$  such that either

$$f(x) > 0 \quad \text{for every } x \in (x^* - \delta, x^*) \cup (x^*, x^* + \delta),$$

or the same statement holds with  $f(x) < 0$ . Equivalently, the two adjacent phase-line arrows point in the same direction: one toward  $x^*$  and one away from  $x^*$ .

## Observation: A half-stable equilibrium is unstable

For a scalar autonomous system with unique forward solutions, every half-stable equilibrium is Lyapunov unstable because initial conditions on its repelling side leave some fixed neighborhood of the equilibrium.

## Definition: Per-capita growth rate

For a population  $N(t) > 0$ , the **per-capita growth rate** is

$$\frac{\dot{N}}{N}.$$

## Definition: Carrying capacity

For a population model

$$\dot{N} = Ng(N), \quad N \geq 0,$$

a number  $K > 0$  is a **carrying capacity** if  $g(K) = 0$  and there is a  $\delta \in (0, K)$  such that

$$g(N) > 0 \quad (K - \delta < N < K), \quad g(N) < 0 \quad (K < N < K + \delta).$$

**Idea: The logistic model**

Assume that, for  $N > 0$ , the per-capita growth rate is

$$\frac{\dot{N}}{N} = r \left( 1 - \frac{N}{K} \right), \quad r > 0, \quad K > 0.$$

The function  $r(1 - N/K)$  extends continuously from  $N > 0$  to  $N \geq 0$ . Using that extension and multiplying by  $N$  gives the population model

$$\dot{N} = rN \left( 1 - \frac{N}{K} \right), \quad N \geq 0.$$

The factor  $N$  makes extinction an equilibrium; the factor  $1 - N/K$  models resource limitation.

**Definition: Scalar linearization**

Let  $f$  be differentiable at an equilibrium  $x^*$  of  $\dot{x} = f(x)$ . With

$$u = x - x^*,$$

the **linearization at  $x^*$**  is

$$\dot{u} = f'(x^*)u.$$

**Definition: Hyperbolic equilibrium**

Let  $f$  be differentiable at an equilibrium  $x^*$  of  $\dot{x} = f(x)$ . The equilibrium  $x^*$  is **hyperbolic** if

$$f'(x^*) \neq 0.$$

**Notation: Linearized rate**

For an equilibrium  $x^*$  of  $\dot{x} = f(x)$ , the **linearized rate** is

$$\lambda = f'(x^*).$$

**Notation: Characteristic time scale**

For a hyperbolic equilibrium with linearized rate  $\lambda \neq 0$ , the **characteristic time scale** is

$$\tau_c = \frac{1}{|\lambda|}.$$

**Observation: Critical slowing**

As a nonzero linearized rate  $\lambda$  approaches 0, its characteristic time scale satisfies

$$\tau_c = \frac{1}{|\lambda|} \rightarrow \infty.$$

This divergence of the characteristic time scale is called **critical slowing**.

**Recall: Local existence and uniqueness**

If  $I \subseteq \mathbb{R}$  is an open interval and  $f \in C^1(I)$ , each initial-value problem  $\dot{x} = f(x)$ ,  $x(0) = x_0 \in I$ , has a unique local solution.

**Theorem: Compact-interval continuation criterion**

Let  $I \subseteq \mathbb{R}$  be open, let  $f \in C^1(I)$ , and let  $x : [0, T_{\max}) \rightarrow I$  be a maximal forward solution of  $\dot{x} = f(x)$ , where  $0 < T_{\max} \leq \infty$ . Here **maximal forward** means that the solution has no extension to  $[0, T)$  for any  $T > T_{\max}$ . If there is a compact interval  $K_0 \subset I$  such that

$$x(t) \in K_0 \quad (0 \leq t < T_{\max}),$$

then  $T_{\max} = \infty$ .

**Theorem: Scalar linear stability test**

Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f \in C^1(I)$ , and let  $x^* \in I$  be an equilibrium of  $\dot{x} = f(x)$ . If  $f'(x^*) < 0$ , then  $x^*$  is locally asymptotically stable. If  $f'(x^*) > 0$ , then  $x^*$  is unstable.

**Proof.** Since  $f(x^*) = 0$ ,

$$\lim_{x \rightarrow x^*} \frac{f(x)}{x - x^*} = f'(x^*).$$

In either case under consideration,  $f'(x^*) \neq 0$ . Choose  $r > 0$  so that  $[x^* - r, x^* + r] \subset I$  and, for  $0 < |x - x^*| \leq r$ , the quotient has the same sign as  $f'(x^*)$ .

If  $f'(x^*) < 0$ , both phase-line arrows point toward  $x^*$ . Uniqueness prevents a solution from crossing the equilibrium. Thus a solution starting at  $|x_0 - x^*| < r$  stays between  $x_0$  and  $x^*$ ; compact-interval continuation makes it defined for every  $t \geq 0$ . In particular,

$$|x(t) - x^*| \leq |x_0 - x^*|,$$

which proves stability: starting closer keeps the solution at least as close. The solution is monotone and bounded, so it has a limit  $L$ . If  $f(L) \neq 0$ , continuity would eventually give  $|\dot{x}(t)| \geq |f(L)|/2 > 0$ , which prevents convergence to a finite limit. Hence  $f(L) = 0$ . The only equilibrium between  $x_0$  and  $x^*$  is  $x^*$ , so  $L = x^*$ , proving local asymptotic stability.

If  $f'(x^*) > 0$ , then  $f(x) > 0$  on  $(x^*, x^* + r]$ . For any  $x_0 \in (x^*, x^* + r)$ , set

$$m = \min_{x \in [x_0, x^* + r]} f(x) > 0.$$

Until the solution reaches  $x^* + r$ ,

$$\dot{x}(t) \geq m, \quad x(t) \geq x_0 + mt.$$

It must reach that endpoint in finite time: otherwise compact-interval continuation and this inequality contradict  $x(t) < x^* + r$  for all  $t \geq 0$ . Thus initial values arbitrarily close to  $x^*$  leave the fixed neighborhood  $(x^* - r, x^* + r)$ . This is instability.  $\square$

**Question: What does the logistic model predict?**

For  $r, K > 0$ , analyze

$$\dot{N} = rN \left(1 - \frac{N}{K}\right), \quad N \geq 0.$$

Find and classify the equilibria, compute their characteristic time scales, and solve the equation for  $N(0) = N_0 \geq 0$ .

**Solution.** Let

$$f(N) = rN \left(1 - \frac{N}{K}\right).$$

The equilibria are  $N^* = 0$  and  $N^* = K$ . For  $N > 0$ ,

|                  |  |          |  |               |
|------------------|--|----------|--|---------------|
| $\frac{N}{f(N)}$ |  | $(0, K)$ |  | $(K, \infty)$ |
|                  |  | +        |  | -             |

so 0 is unstable from the physical side and  $K$  attracts every positive initial population. Moreover,

$$f'(N) = r - \frac{2rN}{K},$$

and therefore

$$f'(0) = r > 0, \quad f'(K) = -r < 0.$$

Thus 0 is hyperbolic and unstable,  $K$  is hyperbolic and asymptotically stable, and both characteristic time scales equal  $1/r$ .

If  $N_0 = 0$  or  $N_0 = K$ , the solution is the corresponding constant solution. For  $N_0 > 0$  with  $N_0 \neq K$ , separation gives

$$\frac{dN}{N(1 - N/K)} = r dt, \quad \frac{1}{N(1 - N/K)} = \frac{1}{N} + \frac{1}{K - N}.$$

Integrating yields

$$\log N - \log |K - N| = rt + C.$$

Imposing  $N(0) = N_0$  and solving for  $N$  gives

$$\boxed{N(t) = \frac{K}{1 + \left(\frac{K}{N_0} - 1\right) e^{-rt}}}, \quad t \geq 0.$$

The same formula includes the constant solution  $N(t) = K$  when  $N_0 = K$ . For every  $N_0 > 0$ ,

$$\lim_{t \rightarrow \infty} N(t) = K.$$

**Question: What if the linearized rate vanishes?**

Classify the origin for

$$\dot{x} = -x^3, \quad \dot{x} = x^3, \quad \dot{x} = x^2, \quad \dot{x} = 0.$$

For  $\dot{x} = -x^3$ , find the solution with initial value  $x_0$  and determine its asymptotic decay.

**Solution.** Every equation has  $f'(0) = 0$ , so linearization is inconclusive. Their phase lines give

| equation         | behavior of $x^* = 0$                                             |
|------------------|-------------------------------------------------------------------|
| $\dot{x} = -x^3$ | asymptotically stable                                             |
| $\dot{x} = x^3$  | unstable                                                          |
| $\dot{x} = x^2$  | half-stable and Lyapunov unstable                                 |
| $\dot{x} = 0$    | Lyapunov stable but not attracting; every point is an equilibrium |

For  $\dot{x} = -x^3$ , the initial value  $x_0 = 0$  gives  $x(t) = 0$ . If  $x_0 \neq 0$ , then

$$x^{-3} dx = -dt.$$

Integrating from 0 to  $t$  gives

$$-\frac{1}{2x(t)^2} + \frac{1}{2x_0^2} = -t,$$

so

$$x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}}.$$

The maximal interval is

$$\left(-\frac{1}{2x_0^2}, \infty\right).$$

Consequently,

$$x(t) \sim \begin{cases} (2t)^{-1/2}, & x_0 > 0, \\ -(2t)^{-1/2}, & x_0 < 0 \end{cases} \quad (t \rightarrow \infty).$$

The convergence is algebraic rather than exponential.

# Existence, Uniqueness, Blow-Up, and Potentials

Tuesday, September 8, 2026

## Definition: Maximal solution

A **maximal solution** is a solution whose interval of definition is not properly contained in the interval of definition of any extension satisfying the same initial-value problem.

## Definition: Global forward solution

A solution through time  $t_0$  is **global forward in time** if it is defined for every  $t \geq t_0$ .

## Definition: Finite-time blow-up

Let  $x : (\alpha, t_*) \rightarrow \mathbb{R}$  be a maximal solution with  $t_* < \infty$ . It has **forward finite-time blow-up at  $t_*$**  if

$$\lim_{t \uparrow t_*} |x(t)| = \infty.$$

## Definition: Locally Lipschitz map

Let  $D \subseteq \mathbb{R}^n$ . A map  $F : D \rightarrow \mathbb{R}^m$  is **locally Lipschitz** if, for every  $p \in D$ , there are constants  $r > 0$  and  $L \geq 0$  such that

$$\|F(x) - F(y)\| \leq L\|x - y\|$$

for all  $x, y \in D$  with  $\|x - p\| < r$  and  $\|y - p\| < r$ .

## Theorem: Local existence and uniqueness

Consider

$$\dot{x} = f(t, x), \quad x(t_0) = x_0.$$

If  $f$  is continuous on a closed rectangle whose interior contains  $(t_0, x_0)$ , and if there is an  $L \geq 0$  such that

$$|f(t, x) - f(t, y)| \leq L|x - y|$$

whenever  $(t, x)$  and  $(t, y)$  lie in that rectangle, then there is a unique local solution.

**Proof sketch.** Continuity gives a bound  $|f(t, x)| \leq M$  on the rectangle. Choose  $b, h > 0$  so that the smaller rectangle  $|t - t_0| \leq h$ ,  $|x - x_0| \leq b$  lies inside it and

$$Mh < b, \quad Lh < 1.$$

**Theorem: Local existence and uniqueness (continued)**

On  $J = [t_0 - h, t_0 + h]$ , define the Picard iterates by

$$u_0(t) = x_0, \quad u_{n+1}(t) = x_0 + \int_{t_0}^t f(s, u_n(s)) ds.$$

If  $|u_n - x_0| \leq b$ , then  $|u_{n+1} - x_0| \leq Mh < b$ ; hence all iterates stay in the rectangle. For their successive differences, set

$$D_n = \max_{t \in J} |u_{n+1}(t) - u_n(t)|.$$

The integral formula and the Lipschitz bound give

$$D_0 \leq Mh, \quad D_{n+1} \leq LhD_n, \quad D_n \leq Mh(Lh)^n.$$

Because  $Lh < 1$ , the differences are bounded by a convergent geometric series, so  $u_n$  converges uniformly to a continuous function  $x$ . The Lipschitz bound also gives uniform convergence of  $f(s, u_n(s))$  to  $f(s, x(s))$ . Passing to the limit yields

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds,$$

which gives  $\dot{x} = f(t, x)$  and  $x(t_0) = x_0$ .

For any two solutions  $x, y$  through  $(t_0, x_0)$ , restrict to a common closed interval about  $t_0$  of half-length at most  $h$  on which both stay in the rectangle. Their maximum difference  $D$  satisfies

$$D \leq LhD.$$

Since  $Lh < 1$ , this forces  $D = 0$ , proving local uniqueness. □

**Corollary: Autonomous local existence and uniqueness**

Let  $I \subseteq \mathbb{R}$  be open, let  $f \in C^1(I)$ , and let  $x_0 \in I$ . Then

$$\dot{x} = f(x), \quad x(0) = x_0,$$

has a unique local solution.

**Proof.** On every compact interval contained in  $I$ , the derivative  $f'$  is bounded. The mean value theorem therefore makes  $f$  locally Lipschitz, so the local existence-and-uniqueness theorem applies. □

**Question: How can one initial condition have infinitely many futures?**

On the state space  $x \geq 0$ , solve

$$\dot{x} = x^{1/3}, \quad x(0) = 0,$$

and exhibit infinitely many solutions.

**Solution.** The constant function  $x(t) = 0$  is a solution. For every waiting time  $a \geq 0$ , define

$$x_a(t) = \begin{cases} 0, & t \leq a, \\ \left(\frac{2(t-a)}{3}\right)^{3/2}, & t \geq a. \end{cases}$$

For  $t > a$ ,

$$\dot{x}_a(t) = \left(\frac{2(t-a)}{3}\right)^{1/2} = x_a(t)^{1/3}.$$

At  $t = a$ , both  $\dot{x}_a(a)$  and  $x_a(a)^{1/3}$  equal 0, so  $x_a$  is differentiable and satisfies the equation there as well. Every  $x_a$  also satisfies  $x_a(0) = 0$ . Distinct waiting times give distinct solutions. The vector field is continuous but is not locally Lipschitz at 0, since

$$\frac{|x^{1/3} - 0|}{|x - 0|} = x^{-2/3} \rightarrow \infty \quad (x \downarrow 0).$$

**Question: Does a smooth vector field guarantee global existence?**

Solve

$$\dot{x} = 1 + x^2, \quad x(0) = 0,$$

and find its maximal interval.

**Solution.** The vector field is smooth, so the local solution is unique. Separation gives

$$\int_0^{x(t)} \frac{du}{1+u^2} = t, \quad \arctan x(t) = t.$$

Therefore

$$x(t) = \tan t.$$

Its maximal interval is

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Indeed,

$$\lim_{t \downarrow -\pi/2} x(t) = -\infty, \quad \lim_{t \uparrow \pi/2} x(t) = +\infty,$$

so no real-valued extension is possible across either endpoint. The solution has forward finite-time blow-up at  $t_* = \pi/2$  and is not global forward in time.

**Theorem: No nonconstant periodic solution on a line**

Let  $I \subseteq \mathbb{R}$  be an open interval and let  $f : I \rightarrow \mathbb{R}$  be continuous. Every periodic solution of

$$\dot{x} = f(x)$$

is constant.

**Proof.** Let  $F$  be an antiderivative of  $f$ , and suppose  $x(t+T) = x(t)$  for some  $T > 0$ . Then

$$\begin{aligned} 0 &= F(x(t+T)) - F(x(t)) \\ &= \int_t^{t+T} F'(x(s)) \dot{x}(s) ds \\ &= \int_t^{t+T} f(x(s)) \dot{x}(s) ds \\ &= \int_t^{t+T} \dot{x}(s)^2 ds. \end{aligned}$$

The integrand is continuous and nonnegative, so it is identically zero. Hence  $\dot{x}(t) = 0$  for every  $t$ , and  $x$  is constant.  $\square$

**Definition: Potential for a scalar autonomous equation**

Let  $I \subseteq \mathbb{R}$  be an open interval and let  $f : I \rightarrow \mathbb{R}$  be continuous. A **potential** for

$$\dot{x} = f(x)$$

is a differentiable function  $V : I \rightarrow \mathbb{R}$  satisfying

$$f(x) = -V'(x).$$

Equivalently,

$$V(x) = - \int_{x_{\text{ref}}}^x f(s) ds + C,$$

where  $x_{\text{ref}} \in I$  and  $C \in \mathbb{R}$  are arbitrary.

**Recall: Critical point**

A **critical point** of a differentiable function  $V$  is a point where  $V' = 0$ .

**Observation: Equilibria are critical points**

For  $\dot{x} = -V'(x)$ ,

$$x^* \text{ is an equilibrium} \iff V'(x^*) = 0.$$

**Theorem: Potential descent identity**

Let  $I \subseteq \mathbb{R}$  be an open interval, let  $V : I \rightarrow \mathbb{R}$  be a potential for  $\dot{x} = f(x)$ , and let  $x : J \rightarrow I$  be a solution. For every  $t \in J$ ,

$$\frac{d}{dt}V(x(t)) = -(f(x(t)))^2 = -(V'(x(t)))^2 \leq 0.$$

Equality holds at time  $t$  exactly when  $x(t)$  is an equilibrium.

**Proof.** By the chain rule and  $\dot{x} = f(x) = -V'(x)$ ,

$$\frac{d}{dt}V(x(t)) = V'(x(t))\dot{x}(t) = V'(x(t))(-V'(x(t))) = -(V'(x(t)))^2.$$

The square vanishes exactly when  $V'(x(t)) = 0$ , equivalently when  $f(x(t)) = 0$ . □

**Definition: Potential well**

A strict local minimum of a potential is a **well**.

**Definition: Potential barrier**

Let  $x_- < x_b < x_+$ , where  $x_-$  and  $x_+$  are wells of a scalar potential  $V$ . If  $x_b$  is a local maximum of  $V$ , then  $x_b$  is a **potential barrier** between those two wells.

**Definition: Bistability**

A system on a specified state space is **bistable** if it has exactly two distinct asymptotically stable equilibria in that state space.

**Corollary: Nondegenerate extrema and stability**

Let  $I \subseteq \mathbb{R}$  be an open interval, let  $V \in C^2(I)$  be a potential for  $\dot{x} = -V'(x)$ , and let  $x^* \in I$  satisfy  $V'(x^*) = 0$ .

1. If  $V''(x^*) > 0$ , then  $x^*$  is locally asymptotically stable.
2. If  $V''(x^*) < 0$ , then  $x^*$  is unstable.

**Proof.** Since  $f = -V'$ ,

$$f'(x^*) = -V''(x^*).$$

The two conclusions follow from the scalar linear stability test. □

**Question: How does a double-well potential encode bistability?**

For

$$\dot{x} = x - x^3,$$

find a potential, classify every equilibrium, and determine the two basins of attraction.

**Solution.** Since  $V'(x) = -f(x) = -x + x^3$ , integration gives

$$V(x) = -\frac{x^2}{2} + \frac{x^4}{4}.$$

The equilibria satisfy

$$x - x^3 = x(1 - x^2) = 0,$$

so they are  $-1$ ,  $0$ , and  $1$ . Also,

$$V''(x) = -1 + 3x^2,$$

and hence

$$V''(0) = -1 < 0, \quad V''(\pm 1) = 2 > 0.$$

Thus  $0$  is an unstable barrier, while  $-1$  and  $1$  are asymptotically stable wells. The phase-line sign table is

| $x$       | $(-\infty, -1)$ | $(-1, 0)$ | $(0, 1)$ | $(1, \infty)$ |
|-----------|-----------------|-----------|----------|---------------|
| $x - x^3$ | +               | -         | +        | -             |

For  $x_0 > 0$ , the arrows and uniqueness keep the solution between  $x_0$  and  $1$ . This is a compact interval, so the continuation criterion gives a solution for all  $t \geq 0$ . Unless  $x_0 = 1$ , the solution is monotone and has a limit. That limit must be an equilibrium: at a non-equilibrium limit, continuity would keep  $|\dot{x}|$  bounded away from zero, preventing convergence. The only equilibrium between  $x_0$  and  $1$  is  $1$ . Since  $f(-x) = -f(x)$ , reflection gives convergence to  $-1$  for every  $x_0 < 0$ . Thus

$$\mathcal{B}(-1) = (-\infty, 0), \quad \mathcal{B}(1) = (0, \infty).$$

The initial condition  $x_0 = 0$  gives the constant solution at the barrier. Thus the system is bistable on  $\mathbb{R}$ .

# Numerical Time Stepping and Error Evidence

Thursday, September 10, 2026

## Notation: Time grid and numerical states

Let  $J \subseteq \mathbb{R}$  be an interval, let  $X \subseteq \mathbb{R}$ , and consider

$$\dot{x} = f(t, x), \quad x(t_0) = x_0,$$

where  $f : J \times X \rightarrow \mathbb{R}$ . Let  $x : I_x \rightarrow X$  denote the exact solution through  $(t_0, x_0)$ , where  $I_x \subseteq J$  is its interval of definition. Choose a step size  $h > 0$  and define

$$t_n = t_0 + nh, \quad n = 0, 1, 2, \dots,$$

whenever  $t_n \in I_x$ . The symbol  $x_n \in X$  denotes the numerical approximation to  $x(t_n)$ .

## Definition: One-step method

For each  $h > 0$ , let  $D_h \subseteq J \times X$  and let  $\Psi_h : D_h \rightarrow X$ . A **one-step method** computes

$$x_{n+1} = \Psi_h(t_n, x_n)$$

whenever  $(t_n, x_n) \in D_h$ .

## Definition: Local defect

For a one-step method and an exact solution  $x(t)$ , the **local defect** at step  $n + 1$  is the unscaled exact-data error

$$d_{n+1}(h) = x(t_{n+1}) - \Psi_h(t_n, x(t_n)).$$

This quantity is defined whenever  $t_{n+1} \in I_x$  and  $(t_n, x(t_n)) \in D_h$ . Some texts call  $d_{n+1}(h)/h$  the local truncation error; here the term **local defect** always means the unscaled quantity above.

**Definition: Global error**

Let  $x(t)$  be the exact solution and let  $x_n$  be the numerical state obtained from the same initial value. The **global error** at  $t_n$  is

$$e_n = x(t_n) - x_n.$$

**Definition: Global convergence order**

Fix an initial-value problem whose exact solution exists on  $[t_0, T]$ , where  $T > t_0$ . A method has **global order**  $p$  on  $[t_0, T]$ , where  $p > 0$ , if there are constants  $C > 0$  and  $h_0 > 0$ , independent of  $h$ , such that

$$\max_{\{n: t_n \leq T\}} |e_n| \leq Ch^p$$

whenever  $0 < h < h_0$ . This definition also requires the numerical states in the displayed maximum to exist.

**Idea: Euler's method is left-endpoint quadrature**

The exact solution satisfies

$$x(t_{n+1}) = x(t_n) + \int_{t_n}^{t_{n+1}} f(t, x(t)) dt.$$

Replacing the integrand by its left-endpoint value gives

$$x_{n+1} = x_n + hf(t_n, x_n).$$

**Question: What error orders appear for Euler's method?**

Apply Euler's method to

$$\dot{x} = x, \quad x(0) = x_0 \neq 0.$$

Determine the order of the unscaled local defect and the order of the global error after a fixed time  $T > 0$ .

**Solution.** All estimates below are as  $h \downarrow 0$  with  $T$  fixed. The local-defect estimates are uniform over grid points in  $[0, T]$ . Starting from the exact value  $x(t_n)$ ,

$$x(t_n + h) = e^h x(t_n), \quad \Psi_h(t_n, x(t_n)) = (1 + h)x(t_n).$$

Thus the one-step defect is

$$e^h x(t_n) - (1 + h)x(t_n) = x(t_n) \left( \frac{h^2}{2} + O(h^3) \right),$$

which is  $O(h^2)$ . At any grid point  $t_n = nh \leq T$ , Euler gives

$$x_n = (1 + h)^n x_0.$$

Since

$$\log(1 + h) = h - \frac{h^2}{2} + O(h^3),$$

**Question: What error orders appear for Euler's method? (continued)**

we obtain

$$(1+h)^n = \exp\left(t_n - \frac{t_n}{2}h + O(h^2)\right).$$

Subtracting from  $x(t_n) = x_0 e^{t_n}$  gives

$$e_n = x_0 e^{t_n} \left( \frac{t_n}{2}h + O(h^2) \right).$$

These estimates are uniform for  $0 \leq t_n \leq T$ . Thus the local defect is  $O(h^2)$  and the global error on  $[0, T]$  is  $O(h)$ .

**Definition: Improved Euler method**

The **improved Euler method** first predicts

$$\tilde{x}_{n+1} = x_n + hf(t_n, x_n)$$

and then accepts the average-slope update

$$x_{n+1} = x_n + \frac{h}{2} [f(t_n, x_n) + f(t_n + h, \tilde{x}_{n+1})].$$

For an autonomous equation, the explicit time arguments are omitted.

**Question: What error orders appear for improved Euler?**

Apply improved Euler to

$$\dot{x} = x, \quad x(0) = x_0 \neq 0.$$

Determine the order of the unscaled local defect and the order of the global error after a fixed time  $T > 0$ .

**Solution.** All estimates below are as  $h \downarrow 0$  with  $T$  fixed. The local-defect estimates are uniform over grid points in  $[0, T]$ . The predictor is

$$\tilde{x}_{n+1} = (1+h)x_n.$$

The accepted update is

$$x_{n+1} = x_n + \frac{h}{2} (x_n + \tilde{x}_{n+1}) = \left(1 + h + \frac{h^2}{2}\right) x_n.$$

Since

$$e^h - \left(1 + h + \frac{h^2}{2}\right) = \frac{h^3}{6} + O(h^4),$$

the local defect is  $O(h^3)$ . Moreover,

$$\log\left(1 + h + \frac{h^2}{2}\right) = h - \frac{h^3}{6} + O(h^4),$$

so at any grid point  $t_n = nh \leq T$ ,

$$\left(1 + h + \frac{h^2}{2}\right)^n = \exp\left(t_n - \frac{t_n}{6}h^2 + O(h^3)\right).$$

**Question: What error orders appear for improved Euler? (continued)**

Subtracting from  $x(t_n) = x_0 e^{t_n}$  gives

$$e_n = x_0 e^{t_n} \left( \frac{t_n}{6} h^2 + O(h^3) \right).$$

These estimates are uniform for  $0 \leq t_n \leq T$ . Thus the local defect is  $O(h^3)$  and the global error on  $[0, T]$  is  $O(h^2)$ .

**Definition: Classical Runge–Kutta method**

For  $\dot{x} = f(t, x)$ , define

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f(t_n + h/2, x_n + hk_1/2), \\ k_3 &= f(t_n + h/2, x_n + hk_2/2), \\ k_4 &= f(t_n + h, x_n + hk_3). \end{aligned}$$

The **classical fourth-order Runge–Kutta method** is

$$x_{n+1} = x_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

**Theorem: Order of the classical Runge–Kutta method**

Let  $D \subseteq \mathbb{R}^2$  be open, let  $f \in C^4(D)$ , and suppose the graph of the exact solution of  $\dot{x} = f(t, x)$  on  $[t_0, T]$  lies in a compact subset of  $D$ . If  $f$  is Lipschitz in  $x$  on a neighborhood of that graph, then, for all sufficiently small  $h > 0$ , its numerical states exist through time  $T$  and there are constants  $C_1, C_2 > 0$  such that

$$\max_{\{n: t_{n+1} \leq T\}} |d_{n+1}(h)| \leq C_1 h^5, \quad \max_{\{n: t_n \leq T\}} |e_n| \leq C_2 h^4.$$

**Question: How do three methods compare on linear decay?**

For

$$\dot{x} = -2x, \quad x(0) = 1,$$

use  $h = 1/2$  to approximate  $x(1)$  by Euler, improved Euler, and classical Runge–Kutta.

**Solution.** The exact value is

$$x(1) = e^{-2} \approx 0.135335.$$

Euler gives

$$x_1 = 1 + \frac{1}{2}(-2) = 0, \quad x_2 = 0.$$

For improved Euler,

$$\tilde{x}_1 = 1 + \frac{1}{2}(-2) = 0, \quad x_1 = 1 + \frac{1}{4}(-2 + 0) = \frac{1}{2},$$

and

$$\tilde{x}_2 = \frac{1}{2} + \frac{1}{2}(-1) = 0, \quad x_2 = \frac{1}{2} + \frac{1}{4}(-1 + 0) = \frac{1}{4}.$$

**Question: How do three methods compare on linear decay? (continued)**

For the first Runge–Kutta step,

$$(k_1, k_2, k_3, k_4) = \left(-2, -1, -\frac{3}{2}, -\frac{1}{2}\right),$$

so

$$x_1 = 1 + \frac{1/2}{6} \left(-2 - 2 - 3 - \frac{1}{2}\right) = \frac{3}{8}.$$

For the second step,

$$(k_1, k_2, k_3, k_4) = \left(-\frac{3}{4}, -\frac{3}{8}, -\frac{9}{16}, -\frac{3}{16}\right),$$

and

$$x_2 = \frac{3}{8} + \frac{1/2}{6} \left(-\frac{3}{4} - \frac{3}{4} - \frac{9}{8} - \frac{3}{16}\right) = \frac{9}{64} = 0.140625.$$

**Visualization: Slope field for the logistic equation**

For

$$\dot{x} = x(1 - x),$$

the slope depends only on  $x$ , so it is constant along every horizontal line in the  $(t, x)$  plane:

| $x$         | $\dot{x}$ | time-graph direction                                                                |
|-------------|-----------|-------------------------------------------------------------------------------------|
| $x < 0$     | $-$       |   |
| $x = 0$     | $0$       | horizontal                                                                          |
| $0 < x < 1$ | $+$       |  |
| $x = 1$     | $0$       | horizontal                                                                          |
| $x > 1$     | $-$       |  |

The equilibrium solution graphs are  $x(t) = 0$  and  $x(t) = 1$ .

**Theorem: Step-halving signature**

Let  $Q \in \mathbb{R}$ , and suppose  $Q_h \in \mathbb{R}$  is defined for every sufficiently small  $h > 0$ . Suppose, as  $h \downarrow 0$ ,

$$Q_h = Q + Ch^p + o(h^p), \quad C \in \mathbb{R} \setminus \{0\}, \quad p > 0.$$

Then

$$\frac{Q_h - Q_{h/2}}{Q_{h/2} - Q_{h/4}} \rightarrow 2^p,$$

and therefore

$$p_{\text{obs}} = \log_2 \left| \frac{Q_h - Q_{h/2}}{Q_{h/2} - Q_{h/4}} \right| \rightarrow p.$$

**Proof.** The expansion gives

$$Q_h - Q_{h/2} = Ch^p(1 - 2^{-p}) + o(h^p)$$

and

$$Q_{h/2} - Q_{h/4} = Ch^p 2^{-p}(1 - 2^{-p}) + o(h^p).$$

The denominator has a nonzero leading coefficient because  $C \neq 0$  and  $p > 0$ , so it is nonzero for sufficiently small  $h$ . Division gives the limit  $2^p$ ; continuity of  $\log_2$  then gives  $p_{\text{obs}} \rightarrow p$ .  $\square$

**Corollary: Richardson extrapolation**

Under the hypotheses of the step-halving theorem,

$$Q = Q_{h/2} + \frac{Q_{h/2} - Q_h}{2^p - 1} + o(h^p).$$

**Proof.** Multiplying the finer approximation by  $2^p$  cancels the leading errors:

$$\begin{aligned} 2^p Q_{h/2} - Q_h &= (2^p - 1)Q + (2^p C(h/2)^p - Ch^p) + o(h^p) \\ &= (2^p - 1)Q + o(h^p). \end{aligned}$$

Divide by  $2^p - 1$  and rewrite the left side to obtain the stated formula. □

**Question: What order is visible in an error table?**

For  $h = 0.2, 0.1, 0.05$ , suppose the absolute errors are

$$1.6 \times 10^{-3}, \quad 4.1 \times 10^{-4}, \quad 1.0 \times 10^{-4}.$$

Estimate the observed order.

**Solution.** If an error has the leading form  $Ch^p$ , then halving  $h$  divides the error by approximately  $2^p$ . Therefore the base-2 logarithm of each consecutive error ratio estimates  $p$ . The two refinement estimates are

$$\log_2 \left( \frac{1.6 \times 10^{-3}}{4.1 \times 10^{-4}} \right) \approx 1.96$$

and

$$\log_2 \left( \frac{4.1 \times 10^{-4}}{1.0 \times 10^{-4}} \right) \approx 2.04.$$

The data are consistent with second-order convergence.

# Bifurcations and the Saddle-Node Mechanism

Tuesday, September 15, 2026

## Notation: State and parameter derivatives

In a scalar family

$$\dot{x} = f(x; r),$$

$x \in \mathbb{R}$  is the state and  $r \in \mathbb{R}$  is a parameter held fixed along each solution. Where the derivatives exist,

$$f_x = \frac{\partial f}{\partial x}, \quad f_r = \frac{\partial f}{\partial r}, \quad f_{xx} = \frac{\partial^2 f}{\partial x^2}.$$

## Definition: Equilibrium branch

For

$$\dot{x} = f(x; r),$$

an **equilibrium branch** is a curve of points  $(r, x^*)$  satisfying

$$f(x^*; r) = 0.$$

## Definition: Bifurcation value

A **bifurcation value** is a parameter value at which the qualitative phase portrait changes.

## Definition: Bifurcation diagram

A **bifurcation diagram** plots  $r$  horizontally and equilibrium values  $x^*$  vertically. Time is not an axis of a bifurcation diagram.

**Visualization: Branch-style convention**

| branch style | stability             |
|--------------|-----------------------|
| solid        | asymptotically stable |
| dashed       | unstable              |

**Definition: Saddle-node bifurcation**

A **saddle-node bifurcation**, also called a **fold**, is the local creation or destruction of two equilibria, one asymptotically stable and one unstable.

**Definition: Saddle-node bottleneck**

A **saddle-node bottleneck** is slow passage through a neighborhood of the state at which a saddle-node pair collided, after the parameter has moved to the side on which no nearby equilibrium exists.

**Definition: Normal form**

A **normal form** is a simple parameterized equation representing the leading local dynamics near a bifurcation.

**Question: How are equilibria destroyed in the saddle-node normal form?**

For

$$\dot{x} = r + x^2,$$

find all equilibrium branches and classify them for every  $r$ .

**Solution.** Equilibria satisfy

$$r + x^2 = 0.$$

Thus

$$\begin{array}{l|l} r < 0 & x_{\pm}^* = \pm\sqrt{-r} \\ r = 0 & x^* = 0 \text{ is a double equilibrium} \\ r > 0 & \text{no equilibrium} \end{array}$$

Since  $f_x = 2x$ ,

$$f_x(-\sqrt{-r}; r) < 0, \quad f_x(\sqrt{-r}; r) > 0.$$

The lower branch is asymptotically stable and the upper branch is unstable. At  $r = 0$ ,  $\dot{x} = x^2 \geq 0$ , so the origin attracts from the left and repels to the right; it is half-stable. The equilibrium set in the  $(r, x^*)$ -plane is the parabola

$$r = -(x^*)^2.$$

**Visualization: Saddle-node phase lines and branches**

For  $\dot{x} = r + x^2$ ,

| $r$     | equilibria              | phase line                                         | branch type                                 |
|---------|-------------------------|----------------------------------------------------|---------------------------------------------|
| $r < 0$ | $-\sqrt{-r}, \sqrt{-r}$ | $\rightarrow \bullet \leftarrow \circ \rightarrow$ | asymptotically stable lower, unstable upper |
| $r = 0$ | $0$                     | $\rightarrow \odot \rightarrow$                    | half-stable fold                            |
| $r > 0$ | $\emptyset$             | $\rightarrow$                                      | no branch                                   |

Here  $\bullet$  denotes asymptotically stable,  $\circ$  unstable, and  $\odot$  half-stable.

**Question: What changes when the quadratic sign is reversed?**

Analyze

$$\dot{x} = r - x^2.$$

**Solution.** There are no equilibria for  $r < 0$ . For  $r \geq 0$ ,

$$x_{\pm}^* = \pm\sqrt{r}.$$

For  $r > 0$ , the roots are distinct and  $f_x = -2x$  gives

$$f_x(-\sqrt{r}; r) > 0, \quad f_x(\sqrt{r}; r) < 0.$$

Thus the negative branch is unstable and the positive branch is asymptotically stable. At  $r = 0$ ,  $\dot{x} = -x^2 \leq 0$ , so the origin repels from the left and attracts from the right. The equilibrium set in the  $(r, x^*)$ -plane is the parabola

$$r = (x^*)^2.$$

**Question: How is an implicit saddle-node located?**

Analyze

$$\dot{x} = r - x - e^{-x}.$$

**Solution.** Let

$$f(x; r) = r - x - e^{-x}.$$

At a collision of equilibria,

$$f(x; r) = 0, \quad f_x(x; r) = 0.$$

These equations are

$$r - x = e^{-x}, \quad -1 + e^{-x} = 0.$$

Hence

$$x_c = 0, \quad r_c = 1.$$

Moreover,

$$f_{xx}(x; r) = -e^{-x} < 0,$$

so  $f$  has a unique global maximum at  $x = 0$ , with value  $r - 1$ . Also,

$$\lim_{x \rightarrow -\infty} f(x; r) = \lim_{x \rightarrow +\infty} f(x; r) = -\infty.$$

**Question: How is an implicit saddle-node located? (continued)**

Since  $f$  increases on  $(-\infty, 0)$  and decreases on  $(0, \infty)$ , the equilibrium count is

$$\begin{array}{l|l} r < 1 & \text{no equilibrium} \\ r = 1 & x^* = 0 \text{ is double and half-stable} \\ r > 1 & \text{two equilibria: left unstable, right asymptotically stable} \end{array}$$

Writing  $\rho = r - 1$  and using

$$e^{-x} = 1 - x + \frac{x^2}{2} + O(x^3) \quad (x \rightarrow 0)$$

gives

$$\dot{x} = \rho - \frac{x^2}{2} + O(x^3) \quad (x \rightarrow 0),$$

the saddle-node form to leading order.

**Proposition: Quadratic reduction at a generic fold**

Consider the parameterized equation  $\dot{x} = f(x; r)$ . Let  $f \in C^3$  near  $(x_c, r_c)$  and suppose

$$f(x_c; r_c) = 0, \quad f_x(x_c; r_c) = 0, \quad f_r(x_c; r_c) \neq 0, \quad f_{xx}(x_c; r_c) \neq 0.$$

With

$$u = x - x_c, \quad \rho = r - r_c,$$

the local equation is

$$\dot{u} = a\rho + bu^2 + O(|u|^3 + |u\rho| + \rho^2) \quad ((u, \rho) \rightarrow (0, 0)),$$

where

$$a = f_r(x_c; r_c), \quad b = \frac{1}{2}f_{xx}(x_c; r_c).$$

**Proof.** Taylor expansion gives

$$\begin{aligned} f(x_c + u; r_c + \rho) &= f(x_c; r_c) + f_x(x_c; r_c)u + f_r(x_c; r_c)\rho \\ &\quad + \frac{1}{2}f_{xx}(x_c; r_c)u^2 + O(|u|^3 + |u\rho| + \rho^2) \quad ((u, \rho) \rightarrow (0, 0)). \end{aligned}$$

The first two terms vanish, while  $a \neq 0$  and  $b \neq 0$ . Substituting the definitions of  $a$  and  $b$  gives the stated equation.  $\square$

**Corollary: Normal form at a generic fold**

Let  $a, b \in \mathbb{R} \setminus \{0\}$ . The equation  $\dot{u} = a\rho + bu^2$  can be transformed by nonzero rescalings and reversals of state and parameter into

$$\dot{X} = R + X^2 \quad \text{or} \quad \dot{X} = R - X^2.$$

Its equilibrium count is

$$\begin{array}{c|c} -a\rho/b > 0 & 2 \\ -a\rho/b = 0 & 1 \text{ double} \\ -a\rho/b < 0 & 0. \end{array}$$

**Proof.** The rescaling  $X = |b|u$ ,  $R = |b|a\rho$  gives

$$\dot{X} = \begin{cases} R + X^2, & b > 0, \\ R - X^2, & b < 0. \end{cases}$$

Equilibria satisfy  $u^2 = -a\rho/b$ , which has the stated root counts. □

**Question: How long does a saddle-node bottleneck last?**

For

$$\dot{x} = \varepsilon + x^2, \quad \varepsilon > 0,$$

find the time required to cross from  $-A$  to  $A$ , where  $A > 0$  is fixed, and determine its asymptotic behavior as  $\varepsilon \downarrow 0$ .

**Solution.** Separation gives

$$T_A(\varepsilon) = \int_{-A}^A \frac{dx}{\varepsilon + x^2} = \frac{2}{\sqrt{\varepsilon}} \arctan\left(\frac{A}{\sqrt{\varepsilon}}\right).$$

Since  $\arctan(A/\sqrt{\varepsilon}) \rightarrow \pi/2$ ,

$$T_A(\varepsilon) \sim \frac{\pi}{\sqrt{\varepsilon}} \quad (\varepsilon \downarrow 0).$$

The system has no equilibrium, but it moves slowly near the former collision point  $x = 0$ .

**Question: How is the saddle-node orientation reversed?**

For  $\dot{x} = r + x^2$ , set  $y = -x$  and  $\rho = -r$ . Find the equation for  $y$ .

**Solution.** Since  $x = -y$ ,

$$\dot{y} = -\dot{x} = -r - x^2 = \rho - y^2.$$

Thus reversing both the state and parameter axes converts  $\dot{x} = r + x^2$  into  $\dot{y} = \rho - y^2$ .

# Persistent Branches and Transcritical Exchange

Thursday, September 17, 2026

## Definition: Persistent equilibrium branch

Let  $R \subseteq \mathbb{R}$  be a specified parameter set. For  $\dot{x} = f(x; r)$ , a **persistent equilibrium branch on  $R$**  is a continuous function  $x^* : R \rightarrow \mathbb{R}$  satisfying

$$f(x^*(r); r) = 0 \quad \text{for every } r \in R.$$

## Definition: Transcritical bifurcation

A **transcritical bifurcation** occurs when two equilibrium branches cross and exchange stability.

## Definition: Transcritical normal form

The **transcritical normal form** is

$$\dot{x} = rx - x^2.$$

## Question: How does the transcritical normal form exchange stability?

Analyze

$$\dot{x} = rx - x^2 = x(r - x).$$

**Solution.** The equilibrium branches are

$$x_0^*(r) = 0, \quad x_1^*(r) = r.$$

Since

$$f_x(x; r) = r - 2x,$$

the branch rates are

$$\lambda_0(r) = f_x(0; r) = r, \quad \lambda_1(r) = f_x(r; r) = -r.$$

Therefore

|         | $x^* = 0$             | $x^* = r$             |
|---------|-----------------------|-----------------------|
| $r < 0$ | asymptotically stable | unstable              |
| $r > 0$ | unstable              | asymptotically stable |

At  $r = 0$ ,  $\dot{x} = -x^2 \leq 0$ , so the origin repels from the left and attracts from the right; it is half-stable.

**Visualization: Transcritical branch crossing**

| $r$     | ordered equilibria | phase line                                                    |
|---------|--------------------|---------------------------------------------------------------|
| $r < 0$ | $r < 0$            | $\leftarrow \circ_{x=r} \rightarrow \bullet_{x=0} \leftarrow$ |
| $r = 0$ | $0$                | $\leftarrow \odot_{x=0} \leftarrow$                           |
| $r > 0$ | $0 < r$            | $\leftarrow \circ_{x=0} \rightarrow \bullet_{x=r} \leftarrow$ |

The bifurcation diagram is the crossing of

$$x^* = 0 \quad \text{and} \quad x^* = r,$$

with solid and dashed segments exchanged at  $(r, x^*) = (0, 0)$ .

**Recall: Scalar implicit function theorem**

Let  $g$  be  $C^1$  near  $(0, 0)$ , with  $g(0; 0) = 0$  and  $g_x(0; 0) \neq 0$ . Near  $(0, 0)$ , the equation  $g(x; r) = 0$  defines a unique  $C^1$  function  $x = \chi(r)$  satisfying

$$\chi(0) = 0, \quad \chi'(0) = -\frac{g_r(0; 0)}{g_x(0; 0)}.$$

**Lemma: Factorization forced by a persistent zero branch**

Let  $f \in C^2$  near  $(0, 0)$  and suppose  $f(0; r) = 0$  for all sufficiently small  $r$ . Then

$$f(x; r) = xg(x; r)$$

for a  $C^1$  function  $g$ .

**Proof.** Define

$$g(x; r) = \int_0^1 f_x(sx; r) ds.$$

Since  $f \in C^2$ , differentiation under the integral shows that  $g \in C^1$ . The fundamental theorem of calculus gives

$$xg(x; r) = \int_0^x f_x(u; r) du = f(x; r) - f(0; r) = f(x; r).$$

□

**Proposition: Exchange of stability on two local branches**

For the parameterized system  $\dot{x} = f(x; r)$ , let  $f(x; r) = xg(x; r)$ , where  $g \in C^1$  near  $(0, 0)$ , and suppose

$$g(0; 0) = 0, \quad a = g_r(0; 0) \neq 0, \quad b = g_x(0; 0) \neq 0.$$

Then the local equilibrium branches and their linearized rates are

$$x_0(r) = 0, \quad x_1(r) = -\frac{a}{b}r + o(r) \quad (r \rightarrow 0),$$

$$\lambda_0(r) = ar + o(r), \quad \lambda_1(r) = -ar + o(r) \quad (r \rightarrow 0).$$

Hence the branches exchange stability at  $r = 0$ .

**Proposition: Exchange of stability on two local branches (continued)****Proof.** Since

$$g(x; r) = ar + bx + o(|r| + |x|) \quad ((x, r) \rightarrow (0, 0)),$$

the implicit function theorem gives

$$g(x_1(r); r) = 0, \quad x_1(r) = -\frac{a}{b}r + o(r) \quad (r \rightarrow 0).$$

On  $x_0(r) = 0$ ,

$$\lambda_0(r) = f_x(0; r) = g(0; r) = ar + o(r) \quad (r \rightarrow 0).$$

On the nonzero branch,  $g(x_1(r); r) = 0$ , so

$$\lambda_1(r) = x_1(r)g_x(x_1(r); r) = -ar + o(r) \quad (r \rightarrow 0).$$

The signs are opposite for all sufficiently small  $r \neq 0$ . □**Question: Where is the bifurcation curve in a two-parameter model?**For real parameters  $a, b$ , consider

$$\dot{x} = x(1 - x^2) - a(1 - e^{-bx}),$$

and find the transcritical bifurcation curve at  $x = 0$  and the leading nonzero equilibrium branch.**Solution.** The origin is an equilibrium for every  $(a, b)$ . Fix any  $b \neq 0$  and vary  $a$  near  $1/b$ . Taylor expansion as  $x \rightarrow 0$ , uniformly for these  $a$ , gives

$$1 - e^{-bx} = bx - \frac{1}{2}b^2x^2 + O(x^3),$$

so

$$\dot{x} = (1 - ab)x + \frac{1}{2}ab^2x^2 + O(x^3).$$

The linear coefficient vanishes on

$$\boxed{ab = 1}.$$

For  $x^* \neq 0$ , division by  $x^*$  gives

$$0 = 1 - ab + \frac{1}{2}ab^2x^* + O((x^*)^2).$$

At  $(x^*, a) = (0, 1/b)$ , the derivative of this expression with respect to  $x^*$  is  $b/2 \neq 0$ . The implicit function theorem therefore gives the second branch, and the displayed equation gives its expansion:

$$x^* = \frac{2(ab - 1)}{ab^2} + O((ab - 1)^2) \quad (a \rightarrow 1/b).$$

Since  $f_x(x; a, b) = 1 - ab + ab^2x + O(x^2)$ , substitution gives the branch rates

$$\lambda_0 = 1 - ab, \quad \lambda_1 = ab - 1 + O((ab - 1)^2) \quad (a \rightarrow 1/b).$$

These have opposite signs for all sufficiently nearby parameters with  $ab \neq 1$ .

**Question: How is a nonzero persistent branch moved to the origin?**

Analyze

$$\dot{x} = r \log x + x - 1, \quad x > 0,$$

near the persistent equilibrium  $x = 1$ .

**Solution.** Set

$$u = x - 1.$$

Taylor expansion as  $u \rightarrow 0$ , uniformly for  $r$  near  $-1$ , gives

$$\log(1 + u) = u - \frac{1}{2}u^2 + O(u^3),$$

so

$$\dot{u} = (r + 1)u - \frac{r}{2}u^2 + O(u^3).$$

The bifurcation value is

$$r_c = -1.$$

For a nonzero equilibrium, divide by  $u$ :

$$0 = (r + 1) - \frac{r}{2}u + O(u^2).$$

The derivative with respect to  $u$  at  $(u, r) = (0, -1)$  is  $1/2 \neq 0$ , so the implicit function theorem gives a second local branch. Solving to leading order and substituting into  $f_x(x; r) = r/x + 1$  gives

$$\begin{aligned} x_0^* &= 1, & \lambda_0 &= r + 1, \\ x_1^* &= 1 + \frac{2(r + 1)}{r} + O((r + 1)^2), & \lambda_1 &= -(r + 1) + O((r + 1)^2), \end{aligned} \quad (r \rightarrow -1).$$

For  $r < -1$  sufficiently close to  $-1$ ,  $x_0^*$  is asymptotically stable and  $x_1^*$  is unstable; for  $r > -1$ , their stability is reversed. At  $r = -1$ ,

$$\dot{u} = \frac{1}{2}u^2 + O(u^3) > 0 \quad (0 < |u| \text{ sufficiently small}),$$

so  $x = 1$  attracts from the left and repels to the right: it is half-stable.

**Question: Is a rational vector field still transcritical?**

Analyze

$$\dot{x} = \frac{x(r + x)}{1 + x^2}.$$

**Solution.** The equilibrium branches are

$$x_0^* = 0, \quad x_1^* = -r.$$

At either equilibrium, the numerator vanishes, so differentiation gives

$$f_x(x^*; r) = \frac{r + 2x^*}{1 + (x^*)^2}.$$

**Question: Is a rational vector field still transcritical? (continued)**

Therefore

$$f_x(0; r) = r, \quad f_x(-r; r) = -\frac{r}{1 + r^2}.$$

For  $r < 0$ ,  $x^* = 0$  is asymptotically stable and  $x^* = -r$  is unstable; for  $r > 0$ , their stability is reversed. At  $r = 0$ ,

$$\dot{x} = \frac{x^2}{1 + x^2} \geq 0,$$

so the origin attracts from the left and repels to the right: it is half-stable. Near  $(x, r) = (0, 0)$ ,

$$\dot{x} = rx + x^2 + O(|x|^3 + |r|x^2) \quad ((x, r) \rightarrow (0, 0)).$$

The state reversal  $y = -x$  gives the leading form

$$\dot{y} = ry - y^2.$$

**Idea: Mathematical and physical branches**

If  $x$  denotes a population, concentration, or intensity, the physical state space may be  $x \geq 0$ . A branch with  $x^* < 0$  remains part of the mathematical bifurcation diagram but is not a physically admissible state.

# Laser Thresholds and Symmetry-Breaking Pitchforks

Tuesday, September 22, 2026

## Notation: Laser variables and parameters

Let

|               |                                         |
|---------------|-----------------------------------------|
| $n(t) \geq 0$ | number of photons                       |
| $N(t) \geq 0$ | number of excited atoms                 |
| $G > 0$       | gain coefficient                        |
| $k > 0$       | photon-loss rate                        |
| $N_0 > 0$     | excited-atom level without laser action |
| $\alpha > 0$  | depletion coefficient                   |

## Idea: Photon gain and loss

Stimulated emission contributes  $GnN$ , while photon loss contributes  $-kn$ :

$$\dot{n} = GnN - kn.$$

## Idea: Excited-atom depletion

Approximate the number of excited atoms by

$$N = N_0 - \alpha n,$$

so a larger photon population leaves fewer excited atoms available. Because  $n \geq 0$  and  $N \geq 0$ , the physically admissible photon states satisfy

$$0 \leq n \leq \frac{N_0}{\alpha}.$$

**Question: Where is the laser threshold?**

Use  $\dot{n} = GnN - kn$  and  $N = N_0 - \alpha n$  to obtain a scalar equation. Find all equilibrium branches and classify the physically admissible states.

**Solution.** Substitution of  $N = N_0 - \alpha n$  gives

$$\dot{n} = Gn(N_0 - \alpha n) - kn = (GN_0 - k)n - \alpha Gn^2.$$

Define

$$\mu = GN_0 - k.$$

Then

$$\dot{n} = f(n; \mu), \quad f(n; \mu) = n(\mu - \alpha Gn).$$

At the endpoints of the physical interval,

$$f(0; \mu) = 0, \quad f\left(\frac{N_0}{\alpha}; \mu\right) = -\frac{kN_0}{\alpha} < 0.$$

Uniqueness prevents crossing the equilibrium  $n = 0$ , and the vector field points inward at the upper endpoint. Thus  $[0, N_0/\alpha]$  is forward invariant. The continuation criterion gives existence for all  $t \geq 0$  because this interval is compact and the vector field is smooth.

The mathematical equilibrium branches are

$$n_0^* = 0, \quad n_1^* = \frac{\mu}{\alpha G}.$$

The physical parameters imply  $\mu > -k$ . Within this admissible parameter range, the branch  $n_1^*$  is defined for every  $\mu$ , but it is a physically admissible photon state only when  $\mu \geq 0$ . Since

$$f_n(n; \mu) = \mu - 2\alpha Gn,$$

their rates are

$$f_n(0; \mu) = \mu, \quad f_n\left(\frac{\mu}{\alpha G}; \mu\right) = -\mu.$$

For  $\mu = 0$ , the equation on the physical state space is

$$\dot{n} = -\alpha Gn^2,$$

and its solution is

$$n(t) = \frac{n_0}{1 + \alpha Gn_0 t}.$$

Thus every  $n_0 > 0$  converges to 0, while  $0 \leq n(t) \leq n_0$  for every  $t \geq 0$ . Hence  $n^* = 0$  is both attracting and Lyapunov stable at the nonhyperbolic threshold. On the full mathematical state space  $n \in \mathbb{R}$ , the two branches exchange stability at  $\mu = 0$ ; the negative branch for  $\mu < 0$  is not physically admissible. On the physical state space  $0 \leq n \leq N_0/\alpha$ ,

|             |                                                             |
|-------------|-------------------------------------------------------------|
| $N_0 < k/G$ | $n^* = 0$ is asymptotically stable                          |
| $N_0 = k/G$ | $n^* = 0$ is nonhyperbolic and asymptotically stable        |
| $N_0 > k/G$ | $n^* = 0$ is unstable,                                      |
|             | $n^* = (GN_0 - k)/(\alpha G) > 0$ is asymptotically stable. |

The pump threshold is

$$N_{0,c} = \frac{k}{G}.$$

**Definition: Reflection symmetry**

Let  $I \subseteq \mathbb{R}$  be symmetric about 0, let  $R \subseteq \mathbb{R}$  be a parameter set, and let  $f : I \times R \rightarrow \mathbb{R}$ . The scalar vector field has **reflection symmetry** if

$$f(-x; r) = -f(x; r) \quad \text{for every } x \in I \text{ and } r \in R.$$

**Observation: Consequences of reflection symmetry**

If  $f(-x; r) = -f(x; r)$ , then  $x = 0$  is an equilibrium and every nonzero equilibrium occurs with its reflected partner. If  $f$  is five times continuously differentiable in  $x$  near  $x = 0$ , then, for each fixed  $r$ ,

$$f(x; r) = a_1(r)x + a_3(r)x^3 + a_5(r)x^5 + o(x^5) \quad (x \rightarrow 0),$$

where

$$a_1(r) = f_x(0; r), \quad a_3(r) = \frac{f_{xxx}(0; r)}{3!}, \quad a_5(r) = \frac{f_{xxxxx}(0; r)}{5!}.$$

**Definition: Pitchfork bifurcation**

A **pitchfork bifurcation** occurs when a persistent equilibrium branch changes stability while a reflected pair of nonzero equilibrium branches meets it.

**Definition: Supercritical pitchfork**

Orient the parameter so that the origin changes from asymptotically stable for  $r < 0$  to unstable for  $r > 0$ . The pitchfork is **supercritical** if asymptotically stable nonzero branches exist for  $r > 0$ .

**Definition: Subcritical pitchfork**

With the same parameter orientation, the pitchfork is **subcritical** if unstable nonzero branches exist for  $r < 0$ .

**Question: How does the supercritical normal form break symmetry?**

Analyze

$$\dot{x} = rx - x^3 = x(r - x^2).$$

**Solution.** The origin is an equilibrium for every  $r$ . For  $r > 0$ , there are also two equilibria

$$x_{\pm}^* = \pm\sqrt{r}.$$

Since

$$f_x(x; r) = r - 3x^2,$$

we have

$$f_x(0; r) = r, \quad f_x(\pm\sqrt{r}; r) = -2r.$$

Therefore

| $r$     | $x^* = 0$                                  | $x^* = \pm\sqrt{r}$   |
|---------|--------------------------------------------|-----------------------|
| $r < 0$ | asymptotically stable                      | absent                |
| $r = 0$ | asymptotically stable with algebraic decay | coincident at 0       |
| $r > 0$ | unstable                                   | asymptotically stable |

**Question: How does the supercritical normal form break symmetry? (continued)**

At  $r = 0$ , the solution of  $\dot{x} = -x^3$  is

$$x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}},$$

so convergence is algebraic.

**Visualization: Supercritical potential landscape**

For  $\dot{x} = rx - x^3$ , a potential is

$$V(x; r) = -\frac{r}{2}x^2 + \frac{1}{4}x^4.$$

| $r$     | critical points of $V$ | landscape                            |
|---------|------------------------|--------------------------------------|
| $r < 0$ | $x = 0$                | one quadratic well                   |
| $r = 0$ | $x = 0$                | one quartic well                     |
| $r > 0$ | $x = 0, \pm\sqrt{r}$   | barrier at 0, wells at $\pm\sqrt{r}$ |

**Question: Why does the hyperbolic-tangent model have a pitchfork?**

For  $\beta > 0$ , analyze

$$\dot{x} = -x + \beta \tanh x,$$

and locate and classify its bifurcation.

**Solution.** The equation is reflection symmetric, and the origin is always an equilibrium. Since

$$\tanh x = x - \frac{x^3}{3} + O(x^5) \quad (x \rightarrow 0),$$

we have

$$\dot{x} = (\beta - 1)x - \frac{\beta}{3}x^3 + O(x^5) \quad (x \rightarrow 0).$$

The origin has rate  $\beta - 1$ . It is asymptotically stable for  $0 < \beta < 1$ , nonhyperbolic and asymptotically stable for  $\beta = 1$ , and unstable for  $\beta > 1$ .

For a nonzero equilibrium,

$$x^* = \beta \tanh x^*, \quad \beta = \frac{x^*}{\tanh x^*}.$$

For  $x > 0$ ,

$$\frac{d}{dx} \left( \frac{x}{\tanh x} \right) = \frac{\sinh x \cosh x - x}{\sinh^2 x}.$$

The numerator vanishes at 0 and has derivative  $2 \sinh^2 x > 0$  for  $x > 0$ . Thus  $x/\tanh x$  increases strictly from 1 to  $\infty$ . Hence no nonzero equilibria exist for  $0 < \beta \leq 1$ , while exactly one reflected pair exists for every  $\beta > 1$ . Along this branch,

$$f_x(x^*; \beta) = -1 + \frac{\beta}{\cosh^2 x^*} = -1 + \frac{x^*}{\sinh x^* \cosh x^*} < 0$$

for  $x^* > 0$ , by the same inequality  $\sinh x^* \cosh x^* > x^*$ . The reflected equilibrium has the same derivative. Thus both nonzero branches are asymptotically stable, and the bifurcation at  $\beta = 1$  is supercritical.

**Question: What does the local subcritical normal form predict?**

Analyze

$$\dot{x} = rx + x^3.$$

**Solution.** The origin has rate  $r$ . Nonzero equilibria satisfy

$$x^2 = -r,$$

so they exist for  $r < 0$  and are

$$x_{\pm}^* = \pm\sqrt{-r}.$$

At either nonzero equilibrium,

$$f_x(x^*; r) = r + 3(x^*)^2 = -2r > 0.$$

Thus the nonzero branches are unstable. The origin is asymptotically stable for  $r < 0$  and unstable for  $r > 0$ . At  $r = 0$ ,

$$\dot{x} = x^3 \begin{cases} < 0, & x < 0, \\ > 0, & x > 0, \end{cases}$$

so the origin is nonhyperbolic and unstable. For  $r \geq 0$ , every nonzero solution moves away from the origin; the local cubic model contains no large-amplitude saturation.**Question: How does a quintic term saturate a subcritical pitchfork?**

Analyze

$$\dot{x} = rx + x^3 - x^5 = x(r + x^2 - x^4).$$

**Solution.** For a nonzero equilibrium, set

$$y = x^2 > 0.$$

Then

$$y^2 - y - r = 0, \quad y_{\pm} = \frac{1 \pm \sqrt{1 + 4r}}{2}.$$

The two amplitude levels meet at

$$r = -\frac{1}{4}, \quad y = \frac{1}{2}, \quad x = \pm\frac{1}{\sqrt{2}}.$$

The equilibrium counts are

$$\begin{array}{l|l} r < -1/4 & x^* = 0 \\ r = -1/4 & x^* = 0, \pm 1/\sqrt{2} \\ -1/4 < r < 0 & x^* = 0, \pm\sqrt{y_-}, \pm\sqrt{y_+} \\ r = 0 & x^* = 0, \pm 1 \\ r > 0 & x^* = 0, \pm\sqrt{y_+}. \end{array}$$

At a nonzero equilibrium,

$$\begin{aligned} f_x(x; r) &= (r + x^2 - x^4) + x(2x - 4x^3) \\ &= 2x^2(1 - 2x^2) = 2y(1 - 2y). \end{aligned}$$

**Question: How does a quintic term saturate a subcritical pitchfork? (continued)**

For  $-1/4 < r < 0$ ,  $0 < y_- < 1/2$ , so the  $y_-$  branches are unstable. For every  $r > -1/4$ ,  $y_+ > 1/2$ , so the  $y_+$  branches are asymptotically stable. The origin has rate  $r$ . At  $r = -1/4$ ,

$$f(x) = -x \left( x^2 - \frac{1}{2} \right)^2,$$

so  $x = \pm 1/\sqrt{2}$  are half-stable and Lyapunov unstable. At  $r = 0$ , the origin is unstable and  $x = \pm 1$  are asymptotically stable. Therefore

$$-\frac{1}{4} < r < 0$$

has three asymptotically stable equilibria: the origin and two distinct large-amplitude equilibria, one positive and one negative. Suppose  $r$  is varied slowly enough that the state follows its current asymptotically stable equilibrium branch until that branch loses stability or ends. As  $r$  increases, the origin loses stability at  $r = 0$ . As  $r$  decreases, the large-amplitude branches end at  $r = -1/4$ .

**Question: What bifurcation is encoded by each leading equation?**

For  $j = 1, 2$ , let  $R_j(x, \mu)$  be continuous in  $(x, \mu)$ , odd in  $x$ , and continuously differentiable in  $x$ , with

$$R_j(x, \mu) = O(x^5), \quad \partial_x R_j(x, \mu) = O(x^4)$$

as  $x \rightarrow 0$ , uniformly for  $\mu$  near 0. Classify

$$\dot{x} = \mu x - 4x^3 + R_1(x, \mu), \quad \dot{x} = -\mu x + 2x^3 + R_2(x, \mu).$$

**Solution.** For the first equation, division by  $x \neq 0$  gives

$$0 = \mu - 4x^2 + O(x^4).$$

The right side has derivative  $-8x + O(x^3) < 0$  for small  $x > 0$ . It starts at  $\mu$  and is negative at a fixed small positive endpoint when  $\mu$  is close to 0. Hence there is exactly one small positive root for  $\mu > 0$  and none for  $\mu \leq 0$ . Reflection gives the negative root. All branch expansions below are as  $\mu \downarrow 0$ . The equilibrium equation gives

$$x^2 = \frac{\mu}{4} + O(\mu^2), \quad x_{\pm}^* = \pm \frac{\sqrt{\mu}}{2} + O(\mu^{3/2}),$$

for  $\mu > 0$ . The rates are  $f_x(0; \mu) = \mu$  and  $f_x(x_{\pm}^*; \mu) = -2\mu + O(\mu^2) < 0$  for sufficiently small  $\mu > 0$ . The pitchfork is supercritical.

For the second equation, division gives

$$0 = -\mu + 2x^2 + O(x^4).$$

The right side has derivative  $4x + O(x^3) > 0$  for small  $x > 0$ . The same crossing argument gives a reflected pair exactly when  $\mu > 0$ . The equilibrium equation gives

$$x^2 = \frac{\mu}{2} + O(\mu^2), \quad x_{\pm}^* = \pm \sqrt{\frac{\mu}{2}} + O(\mu^{3/2}),$$

for  $\mu > 0$ . The rates are  $f_x(0; \mu) = -\mu$  and  $f_x(x_{\pm}^*; \mu) = 2\mu + O(\mu^2) > 0$  for sufficiently small  $\mu > 0$ . With the standard parameter  $R = -\mu$ , these unstable nonzero branches exist for  $R < 0$ , so the pitchfork is subcritical.

# The Rotating Hoop and a Singular Overdamped Limit

Thursday, September 24, 2026

## Visualization: Rotating-hoop variables

| symbol          | meaning                                         |
|-----------------|-------------------------------------------------|
| $m > 0$         | bead mass                                       |
| $a > 0$         | hoop radius                                     |
| $g > 0$         | gravitational acceleration                      |
| $\omega \geq 0$ | constant angular speed of the hoop              |
| $b > 0$         | viscous-damping coefficient                     |
| $\phi$          | angle from the downward vertical                |
| $a \sin \phi$   | signed radial coordinate from the rotation axis |

The angular coordinate satisfies

$$-\pi < \phi \leq \pi, \quad -\pi \equiv \pi.$$

## Question: What is the rotating-hoop equation?

A bead moves on a vertical circular hoop rotating about its vertical diameter with angular speed  $\omega$ . Resolve gravity, centrifugal force, and viscous drag in the direction of increasing  $\phi$ .

**Solution.** The tangential acceleration is  $a\ddot{\phi}$ . The tangential force components are

| force        | tangential component               |
|--------------|------------------------------------|
| gravity      | $-mg \sin \phi$                    |
| centrifugal  | $m\omega^2(a \sin \phi) \cos \phi$ |
| viscous drag | $-ba\dot{\phi}$ .                  |

Newton's law gives

$$ma\ddot{\phi} = -ba\dot{\phi} - mg \sin \phi + ma\omega^2 \sin \phi \cos \phi.$$

**Question: How is the hoop equation nondimensionalized?**

For

$$ma\ddot{\phi} = -ba\dot{\phi} - mg \sin \phi + ma\omega^2 \sin \phi \cos \phi,$$

choose the slow time scale

$$T = \frac{ba}{mg}, \quad \tau = \frac{t}{T},$$

and derive the dimensionless equation.

**Solution.** With primes denoting  $d/d\tau$ ,

$$\dot{\phi} = \frac{1}{T}\phi', \quad \ddot{\phi} = \frac{1}{T^2}\phi''.$$

After division by  $mg$ ,

$$\frac{a}{gT^2}\phi'' = -\frac{ba}{mgT}\phi' - \sin \phi + \frac{a\omega^2}{g} \sin \phi \cos \phi.$$

The choice of  $T$  makes  $ba/(mgT) = 1$ . Define

$$\varepsilon = \frac{m^2g}{b^2a}, \quad \gamma = \frac{a\omega^2}{g}.$$

Then

$$\varepsilon\phi'' = -\phi' - \sin \phi + \gamma \sin \phi \cos \phi.$$

The strong-damping regime is the limit  $\varepsilon \downarrow 0$  with the other dimensionless parameters fixed.

**Definition: Overdamped limit**

For

$$\varepsilon\phi'' = -\phi' - \sin \phi + \gamma \sin \phi \cos \phi,$$

the limit  $\varepsilon \rightarrow 0$  is the **overdamped limit** because damping dominates inertia.

**Definition: Singular limit**

A parameter limit that lowers the order of the differential equation is called a **singular limit**. The reduced equation then has fewer independent initial conditions than the original equation.

**Definition: Initial layer**

The **initial layer** is the short initial interval during which the inertial term discarded by the overdamped reduction remains important.

**Notation: Fast time**

For  $\varepsilon > 0$ , the **fast time** is

$$s = \frac{\tau}{\varepsilon}$$

so a fixed interval  $0 \leq s \leq S$  corresponds to  $0 \leq \tau \leq \varepsilon S$ .

**Question: What is the reduced overdamped flow?**

For

$$\varepsilon \phi'' = -\phi' - \sin \phi + \gamma \sin \phi \cos \phi,$$

set  $\varepsilon = 0$ , find all equilibria for  $\gamma \geq 0$ , and classify them.

**Solution.** The reduced equation is

$$\phi' = f(\phi) = \sin \phi (\gamma \cos \phi - 1).$$

The bottom  $\phi = 0$  and top  $\phi = \pi$  are always equilibria. Since

$$f'(\phi) = \cos \phi (\gamma \cos \phi - 1) - \gamma \sin^2 \phi,$$

we have

$$f'(0) = \gamma - 1, \quad f'(\pi) = \gamma + 1 > 0.$$

Thus the top is always unstable, while the bottom is asymptotically stable for  $\gamma < 1$  and unstable for  $\gamma > 1$ .

If  $\gamma > 1$ , there are two additional equilibria

$$\phi_{\pm}^* = \pm \arccos(1/\gamma).$$

At either one,

$$f'(\phi_{\pm}^*) = -\gamma \sin^2 \phi_{\pm}^* < 0,$$

so both are asymptotically stable.

Near  $(\phi, \gamma) = (0, 1)$ ,

$$f(\phi) = (\gamma - 1)\phi - \frac{1}{2}\phi^3 + O(|\gamma - 1||\phi|^3 + |\phi|^5) \quad ((\phi, \gamma) \rightarrow (0, 1)).$$

Hence  $\gamma = 1$  is a supercritical pitchfork. At  $\gamma = 1$ ,

$$\phi' = -\frac{1}{2}\phi^3 + O(\phi^5) \quad (\phi \rightarrow 0).$$

The flow points toward 0 on both sides, so the bottom is nonhyperbolic and asymptotically stable.

**Visualization: Reduced equilibria versus rotation rate**

| $\gamma$            | bottom and symmetry-broken states                                         | top                   |
|---------------------|---------------------------------------------------------------------------|-----------------------|
| $0 \leq \gamma < 1$ | $\phi = 0$ asymptotically stable                                          | $\phi = \pi$ unstable |
| $\gamma = 1$        | $\phi = 0$ nonhyperbolic asymptotically stable                            | $\phi = \pi$ unstable |
| $\gamma > 1$        | $\phi = 0$ unstable, $\phi = \pm \arccos(1/\gamma)$ asymptotically stable | $\phi = \pi$ unstable |

The two stable nonzero branches satisfy

$$\cos \phi_{\pm}^* = \frac{1}{\gamma}, \quad \phi_{\pm}^* \rightarrow \pm \frac{\pi}{2} \quad (\gamma \rightarrow \infty).$$

**Question: Where does the missing initial condition go?**

Consider

$$\varepsilon y'' + 2y' + y = 0, \quad 0 < \varepsilon < 1, \quad y(0) = y_0, \quad y'(0) = v_0,$$

where primes denote differentiation with respect to  $\tau$  and  $y_0, v_0$  are fixed. Find the two time scales as  $\varepsilon \downarrow 0$  and explain how the solution satisfies both initial conditions.

**Solution.** The characteristic roots are

$$\lambda_{\pm} = \frac{-1 \pm \sqrt{1 - \varepsilon}}{\varepsilon}.$$

Using

$$\sqrt{1 - \varepsilon} = 1 - \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8} + O(\varepsilon^3),$$

as  $\varepsilon \downarrow 0$ , we obtain

$$\lambda_+ = -\frac{1}{2} - \frac{\varepsilon}{8} + O(\varepsilon^2), \quad \lambda_- = -\frac{2}{\varepsilon} + \frac{1}{2} + O(\varepsilon).$$

Thus

$$y(\tau) = Ae^{\lambda_+ \tau} + Be^{\lambda_- \tau}.$$

The initial conditions give

$$A + B = y_0, \quad \lambda_+ A + \lambda_- B = v_0,$$

so

$$B = \frac{\lambda_+ y_0 - v_0}{\lambda_+ - \lambda_-}, \quad A = y_0 - B.$$

The corresponding characteristic decay times are

$$\frac{1}{|\lambda_+|} = 2 + O(\varepsilon), \quad \frac{1}{|\lambda_-|} = \frac{\varepsilon}{2} + O(\varepsilon^2) \quad (\varepsilon \downarrow 0).$$

The reduced equation  $2y' + y = 0$  forces  $y'(0) = -y_0/2$ . The fast  $B$  term allows the full solution to have any prescribed slope  $v_0$ , then decays on a time interval comparable to  $\varepsilon$ . The  $A$  term evolves on a time interval comparable to 1.

**Definition: Slow curve**

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be continuous. For the system

$$\phi' = \Omega, \quad \varepsilon \Omega' = f(\phi) - \Omega, \quad \varepsilon > 0,$$

the **slow curve of the singular problem** is the algebraic set  $\Omega = f(\phi)$  obtained by setting  $\varepsilon = 0$ .

**Proposition: Leading fast-layer relaxation**

Fix  $\gamma \geq 0$ . Write  $\Omega = \phi'$  and  $f(\phi) = \sin \phi(\gamma \cos \phi - 1)$ . Fix  $S > 0$  and initial data  $(\phi_0, \Omega_0)$  independent of  $\varepsilon$ . For each  $\varepsilon > 0$ , let  $(\phi, \Omega)$  solve

$$\phi' = \Omega, \quad \varepsilon \Omega' = f(\phi) - \Omega, \quad (\phi(0), \Omega(0)) = (\phi_0, \Omega_0).$$

On the fast time  $s = \tau/\varepsilon$ , write  $\phi_\varepsilon(s) = \phi(\varepsilon s)$  and  $\Omega_\varepsilon(s) = \Omega(\varepsilon s)$ . Uniformly for  $0 \leq s \leq S$  as  $\varepsilon \downarrow 0$ ,

$$\begin{aligned} \phi_\varepsilon(s) &= \phi_0 + O(\varepsilon), \\ \Omega_\varepsilon(s) &= f(\phi_0) + (\Omega_0 - f(\phi_0))e^{-s} + O(\varepsilon). \end{aligned}$$

Thus trajectories relax toward the slow curve  $\Omega = f(\phi)$  on the initial layer.

**Proof.** In fast time,

$$\frac{d\phi_\varepsilon}{ds} = \varepsilon \Omega_\varepsilon, \quad \frac{d\Omega_\varepsilon}{ds} = f(\phi_\varepsilon) - \Omega_\varepsilon.$$

Multiplying the second equation by  $e^s$  and integrating gives

$$\Omega_\varepsilon(s) = e^{-s}\Omega_0 + \int_0^s e^{-(s-v)} f(\phi_\varepsilon(v)) dv.$$

Since  $|f(\phi)| \leq 1 + \gamma$ ,

$$|\Omega_\varepsilon(s)| \leq \max\{|\Omega_0|, 1 + \gamma\}.$$

Integrating the first equation gives

$$|\phi_\varepsilon(s) - \phi_0| \leq \varepsilon S \max\{|\Omega_0|, 1 + \gamma\} \quad (0 \leq s \leq S).$$

Since  $|f'(\phi)| \leq 1 + \gamma$ , the mean-value theorem now gives  $f(\phi_\varepsilon(s)) = f(\phi_0) + O(\varepsilon)$  uniformly on  $[0, S]$ . Substitute this into the integral and use  $\int_0^s e^{-(s-v)} dv = 1 - e^{-s}$  to obtain the formula for  $\Omega_\varepsilon$ .  $\square$

**Question: What is the fast-layer duration in physical time?**

With  $s = \tau/\varepsilon$  and  $t = T\tau$ , what physical time corresponds to  $s = 1$ ?

**Solution.** Since  $t = T\tau$ ,

$$T\varepsilon = \frac{ba}{mg} \frac{m^2g}{b^2a} = \frac{m}{b}, \quad \frac{m/b}{ba/(mg)} = \frac{m^2g}{b^2a} = \varepsilon.$$

Thus  $m/b$  is the inertial damping time, smaller than the slow time  $T = ba/(mg)$  by the factor  $\varepsilon$ .

# Imperfect Pitchforks and the Cusp

Tuesday, September 29, 2026

## Definition: Imperfection parameter

For real parameters  $r, h$ , consider

$$\dot{x} = h + rx - x^3.$$

The parameter  $h$  is the **imperfection parameter**:  $h = 0$  preserves reflection symmetry, while  $h \neq 0$  breaks it.

## Definition: Stability diagram

A **stability diagram** partitions parameter space into regions with qualitatively different equilibrium configurations.

## Definition: Generic cusp

For a  $C^3$  scalar family  $\dot{x} = f(x; \mu_1, \mu_2)$ , a **generic cusp** at  $(x_c; \mu_{1,c}, \mu_{2,c})$  satisfies

$$f = f_x = f_{xx} = 0, \quad f_{xxx} \neq 0, \quad \det \begin{pmatrix} f_{\mu_1} & f_{\mu_2} \\ f_{x\mu_1} & f_{x\mu_2} \end{pmatrix} \neq 0$$

at that point.

## Definition: Codimension

The **codimension** of a generic bifurcation is the number of independent parameters that must be adjusted to reach it.

## Observation: Codimensions of folds and cusps

A generic fold has codimension one, while a generic cusp has codimension two.

**Question: How many equilibria survive a symmetry-breaking bias?**

For fixed  $(r, h)$ , count the equilibria of

$$\dot{x} = h + rx - x^3.$$

**Solution.** Equilibria are intersections of

$$y = rx - x^3 \quad \text{and} \quad y = -h.$$

If  $r \leq 0$ , then

$$\frac{d}{dx}(rx - x^3) = r - 3x^2 \leq 0.$$

The cubic is strictly decreasing even when  $r = 0$ : its derivative vanishes only at the isolated point  $x = 0$ . Its values range from  $+\infty$  to  $-\infty$ , so it crosses each horizontal line exactly once.

If  $r > 0$ , the cubic has turning points

$$x_{\pm} = \pm \sqrt{\frac{r}{3}}.$$

Its extremal values have magnitude

$$h_c(r) = 2 \left( \frac{r}{3} \right)^{3/2}.$$

Therefore

$$\begin{array}{l|l} |h| < h_c(r) & 3 \text{ distinct equilibria} \\ |h| = h_c(r) & 2 \text{ distinct equilibria, one double} \\ |h| > h_c(r) & 1 \text{ equilibrium.} \end{array}$$

**Proposition: Fold locus of the imperfect pitchfork**

The multiple-equilibrium locus of

$$\dot{x} = h + rx - x^3$$

has the parametrization

$$\boxed{r = 3s^2, \quad h = -2s^3}, \quad s \in \mathbb{R}.$$

Equivalently,

$$\boxed{27h^2 = 4r^3, \quad r \geq 0}.$$

**Proof.** At a multiple equilibrium  $x = s$ ,

$$0 = f(s; r, h) = h + rs - s^3$$

and

$$0 = f_x(s; r, h) = r - 3s^2.$$

Thus

$$r = 3s^2, \quad h = s^3 - rs = -2s^3.$$

Squaring the second equation and cubing the first gives

$$h^2 = 4s^6, \quad r^3 = 27s^6,$$

so  $27h^2 = 4r^3$ . □

**Corollary: Ordinary folds away from the cusp**

For the family

$$\dot{x} = h + rx - x^3,$$

every point  $(r, h) = (3s^2, -2s^3)$  with  $s \neq 0$  is an ordinary saddle-node at  $x = s$ .

**Proof.** At such a point,

$$f_{xx}(s; r, h) = -6s \neq 0, \quad f_h(s; r, h) = 1 \neq 0.$$

Together with  $f = f_x = 0$ , these are the generic fold conditions, with  $h$  as the varying parameter and  $r$  held fixed.  $\square$

**Proposition: Generic cusp at the origin**

The family

$$\dot{x} = h + rx - x^3$$

has a generic cusp at  $(x, r, h) = (0, 0, 0)$ .

**Proof.** At  $(x, r, h) = (0, 0, 0)$ ,

$$f = f_x = f_{xx} = 0, \quad f_{xxx} = -6,$$

and, for parameter order  $(r, h)$ ,

$$\det \begin{pmatrix} f_r & f_h \\ f_{xr} & f_{xh} \end{pmatrix} = \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1 \neq 0.$$

Hence the defining cusp conditions hold.  $\square$

**Visualization: Stability diagram in parameter space**

$$\mathcal{C} = \{(r, h) : 27h^2 = 4r^3, r \geq 0\}.$$

| parameter region                   | distinct equilibria | stability                                                             |
|------------------------------------|---------------------|-----------------------------------------------------------------------|
| $r > 0, 27h^2 < 4r^3$              | $x_1 < x_2 < x_3$   | $x_1, x_3$ asymptotically stable;<br>$x_2$ unstable                   |
| $\mathcal{C} \setminus \{(0, 0)\}$ | $x_1 < x_2$         | one asymptotically stable simple root;<br>one half-stable double root |
| $27h^2 > 4r^3$ or $r < 0$          | $x_1$               | $x_1$ asymptotically stable                                           |
| $(r, h) = (0, 0)$                  | $x^* = 0$           | $\dot{x} = -x^3$ : asymptotically stable                              |

At every equilibrium,

$$\lambda = f_x(x^*; r, h) = r - 3(x^*)^2.$$

**Visualization: Cusp catastrophe surface**

The equilibrium surface is

$$\mathcal{S} = \{(r, h, x^*) : h = (x^*)^3 - rx^*\}.$$

A parametrization is

$$(r, s) \mapsto (r, s^3 - rs, s).$$

At fixed  $r$ , the curve  $h = s^3 - rs$  has a horizontal tangent where

$$\frac{\partial h}{\partial s} = 3s^2 - r = 0.$$

These points are ordinary folds when  $s \neq 0$ ; when  $s = 0$ , the condition also gives  $r = 0$ , yielding the cusp point. Their coordinates satisfy

$$r = 3s^2, \quad h = -2s^3, \quad 27h^2 = 4r^3.$$

| operation on $\mathcal{S}$ | result                               |
|----------------------------|--------------------------------------|
| $h = \text{constant}$      | $x^*$ versus $r$ bifurcation diagram |
| $r = \text{constant}$      | $x^*$ versus $h$ bifurcation diagram |
| projection onto $(r, h)$   | stability diagram                    |

**Question: What happens at a specified exact fold point?**

For  $(r, h) = (3, 2)$ , factor the vector field

$$\dot{x} = 2 + 3x - x^3$$

and classify every equilibrium.

**Solution.**

$$f(x) = 2 + 3x - x^3 = -(x + 1)^2(x - 2).$$

Thus  $x = -1$  is a double equilibrium and  $x = 2$  is a simple equilibrium. Near  $x = -1$ , the factor  $x - 2$  is negative, so

$$f(x) > 0$$

on both sides. Hence  $x = -1$  attracts from the left and repels to the right; it is half-stable. At the simple root,

$$f'(2) = 3 - 3(2)^2 = -9 < 0,$$

so  $x = 2$  is asymptotically stable. The fold formula with  $s = -1$  gives

$$r = 3, \quad h = 2.$$

**Question: What parameters produce a prescribed double equilibrium?**

For  $\dot{x} = h + rx - x^3$ , suppose  $x = s \neq 0$  is a double equilibrium. Find  $(r, h)$ , factor the vector field, and classify the remaining root.

**Solution.** The equations  $f(s) = f_x(s) = 0$  give

$$r = 3s^2, \quad h = -2s^3.$$

With these parameters,

$$f(x) = -(x - s)^2(x + 2s).$$

The remaining equilibrium is  $x = -2s$ , and

$$f'(-2s) = 3s^2 - 3(-2s)^2 = -9s^2 < 0.$$

Thus  $x = -2s$  is asymptotically stable. If  $s > 0$ , then  $h < 0$  and the fold lies on the lower arm of the cusp; if  $s < 0$ , then  $h > 0$  and it lies on the upper arm. At the double root  $x = s$ , the vector field is negative on both sides when  $s > 0$  and positive on both sides when  $s < 0$ . Hence the double root is half-stable and Lyapunov unstable.

**Observation: Potential of the imperfect pitchfork**

A potential for  $\dot{x} = h + rx - x^3$  is

$$V(x; r, h) = -hx - \frac{r}{2}x^2 + \frac{1}{4}x^4, \quad -V_x = h + rx - x^3.$$

**Observation: Landscape inside the cusp**

For  $V(x; r, h) = -hx - rx^2/2 + x^4/4$ , parameters  $r > 0$  and  $27h^2 < 4r^3$  give two local minima at the asymptotically stable equilibria and an intervening local maximum at the unstable equilibrium:

$$V_{xx}(x^*; r, h) = -f_x(x^*; r, h).$$

**Definition: Quasistatic parameter sweep**

A **quasistatic parameter sweep** is the ideal limit of arbitrarily slow parameter variation in which the state tracks an asymptotically stable equilibrium branch for as long as that branch remains asymptotically stable.

**Definition: Hysteresis**

**Hysteresis** is path dependence in which the observed state–parameter response does not retrace the same path when the direction of parameter variation is reversed.

**Observation: Hysteresis in a scalar fold system**

In the present scalar fold setting, suppose a sweep begins on an asymptotically stable branch and passes through the bistable parameter region. In the quasistatic idealization, increasing and decreasing sweeps can then track different branches and jump at different fold values.

**Visualization: Hysteresis at a fixed growth parameter**

For  $r = 3$ , the folds occur at

$$(h, x^*) = (-2, 1) \quad \text{and} \quad (h, x^*) = (2, -1).$$

| quasistatic sweep | tracked asymptotically stable branch | jump                                             |
|-------------------|--------------------------------------|--------------------------------------------------|
| $h \nearrow$      | lower branch                         | $h = 2 : \text{lower} \rightarrow \text{upper}$  |
| $h \searrow$      | upper branch                         | $h = -2 : \text{upper} \rightarrow \text{lower}$ |

In the quasistatic limit, the tracked asymptotically stable branch ends at the fold even though the vector field remains smooth. For a finite-rate sweep, the actual jump need not occur exactly at the static fold value.

# Spruce Budworm Outbreaks and Hysteresis

Thursday, October 1, 2026

## Notation: Budworm variables and parameters

Let  $N(t) \geq 0$  be the budworm population, and let

|         |                            |
|---------|----------------------------|
| $R > 0$ | intrinsic growth rate      |
| $K > 0$ | carrying capacity          |
| $A > 0$ | half-saturation population |
| $B > 0$ | maximum predation rate     |

Here  $K$  is the carrying capacity of the logistic growth term; equivalently, it is the carrying capacity the population would have in the absence of predation.

## Idea: Growth minus predation

Model the population rate as logistic growth minus saturating predation:

$$\dot{N} = RN \left( 1 - \frac{N}{K} \right) - \frac{BN^2}{A^2 + N^2}.$$

## Observation: Saturating predation response

The predation term is quadratic near extinction, reaches half of its maximum at  $N = A$ , and saturates at  $B$ :

$$\frac{BN^2}{A^2 + N^2} = O(N^2) \quad (N \rightarrow 0), \quad \frac{BA^2}{A^2 + A^2} = \frac{B}{2},$$

$$\frac{BN^2}{A^2 + N^2} \rightarrow B \quad (N \rightarrow \infty).$$

**Question: How is the budworm model nondimensionalized?**

Define

$$x = \frac{N}{A}, \quad \tau = \frac{B}{A}t, \quad r = \frac{RA}{B}, \quad k = \frac{K}{A}.$$

Derive the dimensionless equation.

**Solution.** Since  $N = Ax$  and  $\tau = (B/A)t$ ,

$$\dot{N} = A \frac{dx}{dt} = A \frac{dx}{d\tau} \frac{d\tau}{dt} = B \frac{dx}{d\tau}.$$

Also,

$$RN \left(1 - \frac{N}{K}\right) = RAx \left(1 - \frac{x}{k}\right)$$

and

$$\frac{BN^2}{A^2 + N^2} = B \frac{x^2}{1 + x^2}.$$

Division by  $B$  gives

$$\frac{dx}{d\tau} = rx \left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2}.$$

**Observation: Extinction is unstable**

For

$$F(x) = rx \left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2},$$

$F(0) = 0$  and  $F'(0) = r > 0$ . Hence extinction is unstable from the physical side  $x \geq 0$ .

**Visualization: Positive equilibria as line–curve intersections**

On the physical half-line  $x \geq 0$ , define

$$L(x) = r \left(1 - \frac{x}{k}\right), \quad H(x) = \frac{x}{1 + x^2}.$$

For  $x > 0$ , the equilibrium equation is  $L(x) = H(x)$ .

|                        | $L(x)$         | $H(x)$                        |
|------------------------|----------------|-------------------------------|
| vertical intercept     | $r$            | 0                             |
| horizontal intercept   | $k$            | 0                             |
| derivative             | $-r/k$         | $\frac{1 - x^2}{(1 + x^2)^2}$ |
| maximum                | $r$ at $x = 0$ | $1/2$ at $x = 1$              |
| $x \rightarrow \infty$ | $-\infty$      | 0                             |

The intersections give every positive equilibrium; the equilibrium  $x = 0$  must be added separately.

**Proposition: Positive multiple-equilibrium locus**

Let  $r, k > 0$  and

$$F(x; r, k) = rx \left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2}.$$

A positive equilibrium  $x = s$  is multiple if and only if  $s > 1$  and

$$r = \frac{2s^3}{(1 + s^2)^2}, \quad k = \frac{2s^3}{s^2 - 1}.$$

**Proof.** For  $s > 0$ , write

$$F(s; r, k) = s \left[ r \left(1 - \frac{s}{k}\right) - \frac{s}{1 + s^2} \right].$$

The equations  $F(s; r, k) = F_x(s; r, k) = 0$  are equivalent to

$$r \left(1 - \frac{s}{k}\right) = \frac{s}{1 + s^2}, \quad \frac{r}{k} = \frac{s^2 - 1}{(1 + s^2)^2}.$$

Substituting the second equation into the first gives

$$r = \frac{s(s^2 - 1)}{(1 + s^2)^2} + \frac{s}{1 + s^2} = \frac{2s^3}{(1 + s^2)^2}.$$

Then  $k = r/(r/k) = 2s^3/(s^2 - 1)$ . Since  $s > 0$ , the condition  $k > 0$  is equivalent to  $s > 1$ .  $\square$

**Definition: Ordinary semicubical cusp of a plane curve**

Let  $I \subseteq \mathbb{R}$  be an open interval and let  $\Gamma : I \rightarrow \mathbb{R}^2$  be three times continuously differentiable. The curve  $\Gamma$  has an **ordinary semicubical cusp** at  $s_c \in I$  if

$$\Gamma'(s_c) = \mathbf{0}, \quad \det(\Gamma''(s_c), \Gamma'''(s_c)) \neq 0.$$

**Proposition: Cusp of the fold locus**

For  $s > 1$ , define

$$k(s) = \frac{2s^3}{s^2 - 1}, \quad r(s) = \frac{2s^3}{(1 + s^2)^2}.$$

The two regular arms of this curve meet in an ordinary semicubical cusp at

$$s_c = \sqrt{3}, \quad k_c = 3\sqrt{3}, \quad r_c = \frac{3\sqrt{3}}{8}.$$

**Proof.** Differentiation gives

$$k'(s) = \frac{2s^2(s^2 - 3)}{(s^2 - 1)^2}, \quad r'(s) = \frac{2s^2(3 - s^2)}{(1 + s^2)^3}.$$

**Proposition: Cusp of the fold locus (continued)**

Both derivatives vanish only at  $s = \sqrt{3}$ . At that point,

$$(k'', r'') = \left( 3\sqrt{3}, -\frac{3\sqrt{3}}{16} \right), \quad (k''', r''') = \left( -21, \frac{3}{4} \right),$$

and the determinant of these vectors is  $-27\sqrt{3}/16 \neq 0$ . □

**Proposition: Ordinary saddle-nodes on the budworm fold locus**

Let

$$F(x; r, k) = rx \left( 1 - \frac{x}{k} \right) - \frac{x^2}{1 + x^2}, \quad r, k > 0.$$

Suppose that  $s > 1$ ,  $s \neq \sqrt{3}$ , and

$$r = \frac{2s^3}{(1 + s^2)^2}, \quad k = \frac{2s^3}{s^2 - 1}.$$

Then  $x = s$  is an ordinary saddle-node.

**Proof.** Set

$$q(x) = r \left( 1 - \frac{x}{k} \right) - \frac{x}{1 + x^2}, \quad F(x; r, k) = xq(x).$$

On the multiple-equilibrium locus,  $q(s) = q'(s) = 0$ , and

$$q''(s) = -\frac{2s(s^2 - 3)}{(1 + s^2)^3}.$$

Thus  $F_{xx}(s; r, k) = sq''(s) \neq 0$ . Moreover,

$$F_r(s; r, k) = s \left( 1 - \frac{s}{k} \right) = \frac{1 + s^2}{2s} > 0.$$

These are the nondegeneracy and parameter-transversality conditions for an ordinary saddle-node. □

**Proposition: Generic attracting cusp equilibrium in the budworm model**

Let

$$F(x; r, k) = rx \left( 1 - \frac{x}{k} \right) - \frac{x^2}{1 + x^2}.$$

At

$$x_c = \sqrt{3}, \quad k_c = 3\sqrt{3}, \quad r_c = \frac{3\sqrt{3}}{8},$$

$x_c$  is a generic triple equilibrium that attracts from both sides.

**Proof.** At these parameter values, the vector field factors exactly as

$$F(x; r_c, k_c) = -\frac{x(x - x_c)^3}{8(1 + x^2)}.$$

**Proposition: Generic attracting cusp equilibrium in the budworm model (continued)**

At  $x = x_c$ ,

$$F(x_c; r_c, k_c) = F_x(x_c; r_c, k_c) = F_{xx}(x_c; r_c, k_c) = 0,$$

while

$$F_{xxx}(x_c; r_c, k_c) = -\frac{3\sqrt{3}}{16} \neq 0, \quad \det \begin{pmatrix} F_k & F_r \\ F_{xk} & F_{xr} \end{pmatrix} \bigg|_{(x_c; r_c, k_c)} = -\frac{\sqrt{3}}{24} \neq 0.$$

Hence the triple root satisfies the generic cusp conditions. The exact factorization gives  $F(x; r_c, k_c) > 0$  for  $0 < x < x_c$  and  $F(x; r_c, k_c) < 0$  for  $x > x_c$ . Thus  $x_c$  is asymptotically stable.  $\square$

**Visualization: Budworm stability diagram**

Let

$$\mathcal{F} = \{(k(x), r(x)) : x > 1\}$$

be the fold locus. Its two arms correspond to

$$1 < x < \sqrt{3} \quad \text{and} \quad x > \sqrt{3}.$$

The two arms bound a region in the positive  $(k, r)$ -parameter plane. In the table below, **inside the cusp** means strictly between these arms, and **outside the cusp** means in their complement away from the fold locus.

| parameter location                                            | positive equilibria            | equilibrium types                                                                        |
|---------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------|
| outside the cusp<br>on $\mathcal{F} \setminus \{(k_c, r_c)\}$ | 1<br>2 distinct,<br>one double | one asymptotically stable<br>one asymptotically stable simple;<br>one half-stable double |
| $(k_c, r_c)$                                                  | 1 triple                       | one nonhyperbolic attracting                                                             |
| inside the cusp                                               | 3                              | two asymptotically stable;<br>one unstable                                               |

For  $k > k_c$ , the parameter path  $k = \text{constant}$  intersects both fold arms, so varying  $r$  passes through refuge-only, bistable, and outbreak-only regions.

**Definition: Refuge level**

Suppose a scalar population model has three positive equilibria

$$0 < a < b < c.$$

If  $a$  is asymptotically stable, then  $a$  is the **refuge level**.

**Definition: Outbreak threshold**

Suppose a scalar population model has three positive equilibria

$$0 < a < b < c.$$

If  $b$  is unstable and separates the basins of  $a$  and  $c$ , then  $b$  is the **outbreak threshold**.

**Definition: Outbreak level**

Suppose a scalar population model has three positive equilibria

$$0 < a < b < c.$$

If  $c$  is asymptotically stable, then  $c$  is the **outbreak level**.

**Recall: Bistability**

On a specified state space, **bistability** means that the system has exactly two distinct asymptotically stable equilibria in that state space.

**Recall: Hysteresis**

**Hysteresis** is path dependence in which the observed state–parameter response does not retrace the same path when the direction of parameter variation is reversed.

**Observation: Fold-induced hysteresis in the budworm model**

Suppose a quasistatic parameter sweep begins on an asymptotically stable budworm equilibrium and crosses the bistable region. Increasing and decreasing sweeps can then track different branches and jump at different fold values.

**Proposition: Stability and basins in the three-root regime**

Assume  $r, k > 0$  and that the positive equilibria are  $0 < a < b < c$ . Then  $a$  and  $c$  are asymptotically stable,  $b$  is unstable, and

$$\begin{array}{l|l} 0 < x_0 < b & x(\tau) \rightarrow a \\ x_0 = b & x(\tau) = b \\ x_0 > b & x(\tau) \rightarrow c. \end{array}$$

The initial condition  $x_0 = 0$  remains at the unstable extinction equilibrium.

**Proof.** Clearing the denominator in  $F(x)/x$  gives a cubic with the three distinct roots  $a, b, c$ . Therefore

$$F(x) = rx \left(1 - \frac{x}{k}\right) - \frac{x^2}{1+x^2} = -\frac{rx(x-a)(x-b)(x-c)}{k(1+x^2)}.$$

The positive roots are simple, and the sign alternates:

|        |          |          |          |               |
|--------|----------|----------|----------|---------------|
| $x$    | $(0, a)$ | $(a, b)$ | $(b, c)$ | $(c, \infty)$ |
| $F(x)$ | +        | −        | +        | −.            |

Uniqueness prevents crossing an equilibrium. The sign table therefore keeps each trajectory between its initial value and the equilibrium toward which the arrows point. This compact interval gives existence for all forward times. The trajectory is monotone and bounded, so it has a limit. That limit must be an equilibrium: otherwise its speed would stay bounded away from zero near the limit. The sign table selects exactly the limits stated above.  $\square$

**Question: What is the second fold for a specified carrying capacity?**

Verify the fold point at  $x = 2$  and find the other fold with the same  $k$ .

**Solution.** At  $x = 2$ ,

$$k = \frac{2(2)^3}{2^2 - 1} = \frac{16}{3}, \quad r = \frac{2(2)^3}{(1 + 2^2)^2} = \frac{16}{25}.$$

The intersection and slope equations are

$$\frac{16}{25} \left( 1 - \frac{2}{16/3} \right) = \frac{2}{5} = \frac{2}{1 + 2^2}$$

and

$$-\frac{16/25}{16/3} = -\frac{3}{25} = \frac{1 - 2^2}{(1 + 2^2)^2}.$$

For the other tangency, solve

$$\frac{2x^3}{x^2 - 1} = \frac{16}{3}.$$

Then

$$3x^3 - 8x^2 + 8 = (x - 2)(3x^2 - 2x - 4) = 0,$$

so the other admissible root is

$$x = \frac{1 + \sqrt{13}}{3} \approx 1.53518.$$

Its parameter value is

$$r = \frac{2x^3}{(1 + x^2)^2} = \frac{304 + 416\sqrt{13}}{2809} \approx 0.64219.$$

**Idea: Slow forest growth gives a parameter path**

Let  $S$  be average tree size. Hold  $R$  and  $B$  fixed, and suppose

$$K = K'S, \quad A = A'S,$$

where  $K', A' > 0$  are constant. Then

$$k = \frac{K}{A} = \frac{K'}{A'}$$

is constant, while

$$r = \frac{RA}{B} = \frac{RA'}{B}S$$

increases with  $S$ . Forest maturation therefore follows the fixed- $k$  path  $k = K'/A'$  toward larger  $r$ . If the refuge branch reaches its fold, it disappears and the population moves to the outbreak branch.

**Visualization: Empirical parameter scale reported with the model**

| forest state | $k$           | $r$         |
|--------------|---------------|-------------|
| young        | $\approx 300$ | $< 1/2$     |
| mature       | $\approx 300$ | $\approx 1$ |

The reported recovery time of the fir forest after replacement by birch is approximately

50–100 years.

These are empirical estimates used to locate a parameter path, not deductions from the dimensionless ODE.

**Idea: Scope of the model**

The outbreak mechanism is conditional on

logistic growth + saturating predation + separation of insect and forest time scales.

Spatial dispersal, heterogeneous foliage, and the slow tree dynamics are not state variables in the scalar model.

# Exam 1 Review: Scalar Flows and Bifurcations

Tuesday, October 6, 2026

## Idea: Scalar-flow analysis

For a parameter-dependent scalar equation

$$\dot{x} = f(x; r),$$

use the chain

$$\boxed{\text{state space}} \longrightarrow \boxed{f(x; r) = 0} \longrightarrow \boxed{\text{sign } f} \longrightarrow \boxed{\text{stability and limits}} \longrightarrow \boxed{\text{parameter dependence}}.$$

## Recall: Hyperbolic equilibrium test

Let  $x^*$  be an equilibrium of  $\dot{x} = f(x; r)$ , where  $f$  is continuously differentiable with respect to  $x$ . If  $f_x(x^*; r) \neq 0$ , then  $x^*$  is **hyperbolic**, and

$$f_x(x^*; r) < 0 \Rightarrow x^* \text{ is asymptotically stable,} \quad f_x(x^*; r) > 0 \Rightarrow x^* \text{ is unstable.}$$

## Observation: Nonhyperbolic scalar equilibria

For  $\dot{x} = f(x; r)$ , if  $f_x(x^*; r) = 0$ , the derivative test is inconclusive. When  $x^*$  is isolated, use the signs of  $f(x; r)$  on the intervals immediately to its left and right to determine stability.

## Question: What is the state space?

Classify each equation as autonomous or nonautonomous and linear or nonlinear. Give an autonomous state space.

$$\dot{x} = x(1 - x), \quad \ddot{q} + q^3 = 0, \quad \dot{y} = t - y.$$

**Solution.**

| equation             | autonomous | linearity             | autonomous state          |
|----------------------|------------|-----------------------|---------------------------|
| $\dot{x} = x(1 - x)$ | yes        | nonlinear             | $x \in \mathbb{R}$        |
| $\ddot{q} + q^3 = 0$ | yes        | nonlinear             | $(q, v) \in \mathbb{R}^2$ |
| $\dot{y} = t - y$    | no         | linear nonhomogeneous | $(y, s) \in \mathbb{R}^2$ |

For the second and third equations,

$$\dot{q} = v, \quad \dot{v} = -q^3,$$

**Question: What is the state space? (continued)**

and

$$\dot{y} = s - y, \quad \dot{s} = 1, \quad s(0) = 0,$$

respectively. The scalar equation  $\dot{y} = t - y$  is linear nonhomogeneous, but its autonomous lift has the affine vector field

$$F(y, s) = (s - y, 1),$$

which is not linear because  $F(0, 0) = (0, 1) \neq (0, 0)$ .

**Question: How does a repeated equilibrium affect a phase line?**

Analyze

$$\dot{x} = (x + 1)^2(2 - x).$$

**Solution.** The equilibria are  $x = -1$  and  $x = 2$ . Since  $(x + 1)^2 > 0$  for  $x \neq -1$ ,

| $x$    | $(-\infty, -1)$ | $-1$ | $(-1, 2)$ | $2$ | $(2, \infty)$ |
|--------|-----------------|------|-----------|-----|---------------|
| $f(x)$ | +               | 0    | +         | 0   | −             |

and the phase line is

$$\longrightarrow \odot_{-1} \longrightarrow \bullet_2 \longleftarrow .$$

Thus  $-1$  attracts from the left and repels from the right; it is half-stable and Lyapunov unstable. The point  $2$  is asymptotically stable. Moreover,

$$\lim_{t \rightarrow \infty} x(t) = \begin{cases} -1, & x(0) \leq -1, \\ 2, & x(0) > -1. \end{cases}$$

Since  $f'(2) = -9$ , perturbations near  $2$  decay to leading order as  $e^{-9t}$ . Since  $f'(-1) = 0$ , linearization does not classify  $-1$ .

**Question: How do phase line, potential, and basin agree?**

For

$$\dot{x} = -x(x - 1)(x + 2),$$

find the equilibria, stability, basins, and a potential  $V$  satisfying  $\dot{x} = -V'(x)$ .

**Solution.** The sign table is

| $x$    | $(-\infty, -2)$ | $-2$ | $(-2, 0)$ | $0$ | $(0, 1)$ | $1$ | $(1, \infty)$ |
|--------|-----------------|------|-----------|-----|----------|-----|---------------|
| $f(x)$ | +               | 0    | −         | 0   | +        | 0   | −             |

Hence  $-2$  and  $1$  are asymptotically stable, while  $0$  is unstable. Their basins are

$$\mathcal{B}(-2) = (-\infty, 0), \quad \mathcal{B}(1) = (0, \infty).$$

Expanding  $-f(x)$  gives

$$-f(x) = x(x - 1)(x + 2) = x^3 + x^2 - 2x,$$

so one may take

$$V(x) = \frac{x^4}{4} + \frac{x^3}{3} - x^2.$$

**Question: How do phase line, potential, and basin agree? (continued)**

Along solutions,

$$\dot{V} = V'(x)\dot{x} = -f(x)^2 \leq 0.$$

The asymptotically stable equilibria are local minima of  $V$ ; the unstable equilibrium is a local maximum.

**Visualization: Three local bifurcation signatures**

With  $r$  as the parameter and  $x$  as the state,

| type          | normal form          | equilibrium branches                | structural condition                     |
|---------------|----------------------|-------------------------------------|------------------------------------------|
| saddle-node   | $\dot{x} = r - x^2$  | $x = \pm\sqrt{r} \ (r \geq 0)$      | no imposed symmetry or persistent branch |
| transcritical | $\dot{x} = rx - x^2$ | $x = 0, \ r$                        | persistent branch $x = 0$                |
| pitchfork     | $\dot{x} = rx - x^3$ | $x = 0, \ \pm\sqrt{r} \ (r \geq 0)$ | $x \mapsto -x$ symmetry                  |

**Observation: Structural requirements for the three signatures**

For unrestricted one-parameter scalar families, the generic equilibrium bifurcation is the saddle-node. Transcritical and pitchfork bifurcations are generic only within families that retain the indicated persistent branch or symmetry.

**Question: Which local bifurcation occurs?**

Find the equilibrium branches and their stability for

$$\dot{x} = \mu - (x - 1)^2, \quad \dot{x} = x(\mu - 2x), \quad \dot{x} = \mu x - 2x^3.$$

**Solution.** For  $f_1(x; \mu) = \mu - (x - 1)^2$ ,

$$x_{\pm} = 1 \pm \sqrt{\mu} \quad (\mu \geq 0), \quad (f_1)_x = -2(x - 1).$$

For  $\mu > 0$ ,  $x_-$  is unstable and  $x_+$  is asymptotically stable. They collide at  $(\mu, x) = (0, 1)$ : a saddle-node. At  $\mu = 0$ ,  $\dot{x} = -(x - 1)^2$ , so  $x = 1$  attracts from the right and repels from the left: it is half-stable and Lyapunov unstable. For  $\mu < 0$ , there is no equilibrium.

For  $f_2(x; \mu) = x(\mu - 2x)$ ,

$$x = 0, \quad x = \frac{\mu}{2}, \quad (f_2)_x = \mu - 4x.$$

The scalar linearized rates on the two branches are  $\mu$  and  $-\mu$ , so the branches exchange stability at the origin: a transcritical bifurcation. At  $\mu = 0$ ,  $\dot{x} = -2x^2$ , so the coincident equilibrium is half-stable and Lyapunov unstable, attracting from the right and repelling from the left.

For  $f_3(x; \mu) = \mu x - 2x^3$ ,

$$x = 0, \quad x_{\pm} = \pm\sqrt{\frac{\mu}{2}} \quad (\mu > 0).$$

At the origin  $(f_3)_x = \mu$ , while

$$(f_3)_x(x_{\pm}) = \mu - 6\frac{\mu}{2} = -2\mu < 0.$$

Two asymptotically stable branches emerge as the origin loses stability: a supercritical pitchfork. At  $\mu = 0$ ,  $\dot{x} = -2x^3$  points toward the origin from both sides, so the origin remains asymptotically stable at the bifurcation value.

**Question: What does an overdamped reduction forget?**

For fixed initial data  $y_0, v_0 \in \mathbb{R}$  and  $\varepsilon > 0$ , solve

$$\varepsilon y'' + y' = 1, \quad y(0) = y_0, \quad y'(0) = v_0.$$

Compare the solution with the reduced equation obtained by setting  $\varepsilon = 0$ .

**Solution.** Put  $v = y'$ . Solving  $\varepsilon v' + v = 1$  with  $v(0) = v_0$  and then integrating gives

$$\begin{aligned} v(t) &= 1 + (v_0 - 1)e^{-t/\varepsilon}, \\ y(t) &= y_0 + t + \varepsilon(v_0 - 1)(1 - e^{-t/\varepsilon}). \end{aligned}$$

The reduced solution is  $y_{\text{red}}(t) = y_0 + t$ . For every  $t \geq 0$ ,

$$|y(t) - y_{\text{red}}(t)| \leq \varepsilon |v_0 - 1|.$$

Thus the positions converge uniformly on  $[0, \infty)$  as  $\varepsilon \downarrow 0$ . For the velocities,

$$v(t) - y'_{\text{red}}(t) = (v_0 - 1)e^{-t/\varepsilon}.$$

At every fixed  $t > 0$  this tends to zero, but at  $t = 0$  it remains  $v_0 - 1$ . Unless  $v_0 = 1$ , the reduced equation therefore misses the initial velocity. The factor  $e^{-t/\varepsilon}$  reduces the velocity discrepancy by  $e^{-1}$  every  $\varepsilon$  units of time, identifying the fast initial layer.

**Question: How many decimal places does step refinement support?**

An unknown quantity  $Q$  is approximated with step sizes  $h, h/2, h/4$ :

$$Q_h = 1.20384, \quad Q_{h/2} = 1.20496, \quad Q_{h/4} = 1.20524.$$

Assuming  $p > 0$ ,  $C \neq 0$ , and

$$Q_h = Q + Ch^p + o(h^p) \quad (h \downarrow 0),$$

estimate  $p$ , the finest-grid error, and  $Q$ .

**Solution.** The ratio gives the estimated order:

$$\frac{Q_{h/2} - Q_h}{Q_{h/4} - Q_{h/2}} = \frac{0.00112}{0.00028} = 4 = 2^2 \Rightarrow p \approx 2.$$

The finest-grid error estimate and Richardson extrapolation are

$$\begin{aligned} |Q - Q_{h/4}| &\approx \frac{|Q_{h/4} - Q_{h/2}|}{2^2 - 1} = \frac{0.00028}{3} \approx 9.33 \times 10^{-5}, \\ Q &\approx Q_{h/4} + \frac{Q_{h/4} - Q_{h/2}}{3} = 1.205333 \dots \end{aligned}$$

The data support three decimal places, provided the asymptotic error model is valid and roundoff is negligible.

# Flows on the Circle: Oscillation and Bottlenecks

Tuesday, October 13, 2026

## Definition: Circle as a quotient

The **circle** is

$$S^1 = \mathbb{R}/(2\pi\mathbb{Z}),$$

so  $\theta$  and  $\theta + 2\pi k$  represent the same point for every  $\theta \in \mathbb{R}$  and  $k \in \mathbb{Z}$ . The equivalence class of  $\theta$  is denoted by  $[\theta] \in S^1$ .

## Definition: Vector field on the circle

A **continuously differentiable vector field on  $S^1$**  is represented by

$$\dot{\theta} = f(\theta), \quad f(\theta + 2\pi) = f(\theta).$$

Here  $f : \mathbb{R} \rightarrow \mathbb{R}$  is continuously differentiable and  $2\pi$ -periodic.

## Definition: Lift of a circle-valued solution

Let  $q : \mathbb{R} \rightarrow S^1$  be the quotient map  $q(\theta) = [\theta]$ , and let  $\gamma : I \rightarrow S^1$  be a continuous solution curve. A **lift of  $\gamma$**  is a continuous function  $\tilde{\theta} : I \rightarrow \mathbb{R}$  satisfying

$$q(\tilde{\theta}(t)) = \gamma(t) \quad \text{for every } t \in I.$$

For the circle vector field represented by  $\dot{\theta} = f(\theta)$ , the lift is differentiable and satisfies

$$\dot{\tilde{\theta}} = f(\tilde{\theta})$$

after its initial representative has been chosen.

**Definition: Fixed point of a circle flow**

A class  $[\theta^*] \in S^1$  is a **fixed point** of  $\dot{\theta} = f(\theta)$  if

$$f(\theta^*) = 0.$$

**Question: Which formulas define circle flows?**

Analyze

$$\dot{\theta} = \sin \theta \quad \text{and} \quad \dot{\theta} = \theta.$$

**Solution.** The first right-hand side is  $2\pi$ -periodic. In one period its fixed points are 0 and  $\pi$ , and

|               |            |       |               |                 |
|---------------|------------|-------|---------------|-----------------|
| $\theta$      | $(0, \pi)$ | $\pi$ | $(\pi, 2\pi)$ | $0 \equiv 2\pi$ |
| $\sin \theta$ | +          | 0     | −             | 0               |

Thus  $[\pi]$  is asymptotically stable and  $[0]$  is unstable.

The formula  $f(\theta) = \theta$  is not  $2\pi$ -periodic. The equivalent labels 0 and  $2\pi$  prescribe different velocities, so it does not define a vector field on  $S^1$ .

**Theorem: Fixed-point–rotation dichotomy**

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be continuously differentiable and  $2\pi$ -periodic.

If  $f$  has a zero, every nonfixed trajectory remains in one component of

$$S^1 \setminus \{[\theta] : f(\theta) = 0\}$$

and approaches a boundary fixed point in forward time.

If  $f$  has no zero, then every trajectory rotates forever in the same direction and is periodic on  $S^1$ .

**Proof.** Periodicity makes  $f$  bounded, so solutions exist for all time. If  $f$  has no zero, continuity gives one sign and a positive lower bound for  $|f|$ . Every lift therefore completes a turn in finite time. Periodicity of  $f$  and uniqueness make the motion repeat after each turn.

If zeros occur, uniqueness prevents a nonfixed trajectory from meeting a fixed point in finite time. On each component of the complement of the zero set,  $f$  has one sign. The lift is monotone and bounded by the component's endpoints, so it has a limit  $L$ . If  $f(L) \neq 0$ , continuity would keep the speed bounded away from zero near  $L$ , contradicting convergence. Hence  $f(L) = 0$ .

**Proposition: Single-valued potential criterion**

Let  $f$  be continuous and  $2\pi$ -periodic. There exists a differentiable,  $2\pi$ -periodic function  $V$  such that

$$f(\theta) = -V'(\theta)$$

if and only if

$$\int_0^{2\pi} f(\theta) \, d\theta = 0.$$

**Proposition: Single-valued potential criterion (continued)**

**Proof.** If  $f = -V'$  and  $V$  is periodic, then

$$\int_0^{2\pi} f(\theta) \, d\theta = -V(2\pi) + V(0) = 0.$$

Conversely, if the integral is zero, define

$$V(\theta) = - \int_0^\theta f(s) \, ds.$$

Then  $V' = -f$  and

$$V(\theta + 2\pi) - V(\theta) = - \int_\theta^{\theta+2\pi} f(s) \, ds = 0.$$

Thus  $V$  is a function on  $S^1$ .

**Definition: Uniform oscillator**

A **uniform oscillator** is a circle flow of the form

$$\dot{\theta} = \omega,$$

where  $\omega \in \mathbb{R}$  is constant. Its lifts are

$$\theta(t) = \theta_0 + \omega t.$$

**Notation: Angular speed**

For the uniform oscillator  $\dot{\theta} = \omega$ , its **angular speed** is  $|\omega|$ . The sign of  $\omega$  determines the direction of rotation.

**Observation: Period of a uniform oscillator**

If  $\omega \neq 0$ , the period of  $\dot{\theta} = \omega$  on  $S^1$  is

$$T = \frac{2\pi}{|\omega|}.$$

**Definition: Phase difference**

For two phases  $[\theta_1], [\theta_2] \in S^1$ , their **phase difference** is

$$\phi = [\theta_1 - \theta_2] \in S^1.$$

**Question: When does one oscillator lap another?**

Suppose

$$\dot{\theta}_1 = \omega_1, \quad \dot{\theta}_2 = \omega_2, \quad \omega_1 > \omega_2 > 0.$$

Find the lap time. Then evaluate it when the periods are  $T_1 = 3$  seconds and  $T_2 = 4$  seconds.

**Solution.** For the lifted phase difference  $\tilde{\phi} = \tilde{\theta}_1 - \tilde{\theta}_2$ ,

$$\dot{\tilde{\phi}} = \omega_1 - \omega_2.$$

One lap occurs when  $\tilde{\phi}$  increases by  $2\pi$ , so

$$T_{\text{lap}} = \frac{2\pi}{\omega_1 - \omega_2}.$$

Since  $\omega_j = 2\pi/T_j$ ,

$$T_{\text{lap}} = \left( \frac{1}{T_1} - \frac{1}{T_2} \right)^{-1} = \left( \frac{1}{3} - \frac{1}{4} \right)^{-1} = 12 \text{ seconds}.$$

**Question: How does a nonuniform oscillator change with its parameter?**

Let  $\omega > 0$ ,  $a \geq 0$ , and

$$\dot{\theta} = \omega - a \sin \theta.$$

Classify the dynamics for  $a < \omega$ ,  $a = \omega$ , and  $a > \omega$ .

**Solution.** For  $a > \omega$ , define

$$\alpha = \arcsin\left(\frac{\omega}{a}\right) \in \left(0, \frac{\pi}{2}\right).$$

| parameter           | fixed points           | dynamics                                                |
|---------------------|------------------------|---------------------------------------------------------|
| $0 \leq a < \omega$ | none                   | $\dot{\theta} > 0$ ; counterclockwise rotation          |
| $a = \omega$        | $\theta^* = \pi/2$     | half-stable saddle-node point                           |
| $a > \omega$        | $\alpha, \pi - \alpha$ | $\alpha$ asymptotically stable, $\pi - \alpha$ unstable |

Indeed,

$$f'(\theta) = -a \cos \theta,$$

so  $f'(\alpha) < 0$  and  $f'(\pi - \alpha) > 0$ . At  $a = \omega$ ,

$$f\left(\frac{\pi}{2} + u\right) = \omega(1 - \cos u) \geq 0.$$

More precisely,

$$f\left(\frac{\pi}{2} + u\right) = \frac{\omega}{2}u^2 + O(u^4) \quad (u \rightarrow 0).$$

Locally, the arrows point toward the fixed point from the left and away from it on the right. A point starting immediately to the right nevertheless travels counterclockwise around the remainder of the circle and approaches  $[\pi/2]$  from the left. Thus  $[\pi/2]$  attracts every initial state on  $S^1$ , so it is globally attracting, but it is Lyapunov unstable.

**Visualization: Nonuniform-oscillator phase portraits**

| $0 \leq a < \omega$                                               | $a = \omega$                                                                                                                       | $a > \omega$                                                                                         |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| counterclockwise rotation<br>$\min \dot{\theta} = \omega - a > 0$ | $\rightarrow \odot_{\pi/2} \rightarrow$<br>$\dot{\theta} \sim \frac{\omega}{2}(\theta - \pi/2)^2 \quad (\theta \rightarrow \pi/2)$ | $\rightarrow \bullet_{\alpha} \leftarrow \circ_{\pi-\alpha} \rightarrow$<br>$\sin \alpha = \omega/a$ |

Here  $\bullet$  denotes an attracting fixed point and  $\circ$  a repelling fixed point. The symbol  $\odot$  denotes the threshold point, whose local arrows attract from the left and repel to the right. On the circle, the right-side trajectories travel counterclockwise around the remainder of the circle and approach the point from the left, so this point is globally attracting but Lyapunov unstable.

**Theorem: Period of a rotating circle flow**

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be continuously differentiable,  $2\pi$ -periodic, and satisfy  $f(\theta) \neq 0$  for every  $\theta \in \mathbb{R}$ . Every orbit on  $S^1$  has the same least positive period

$$T = \int_0^{2\pi} \frac{d\theta}{|f(\theta)|}.$$

**Proof.** If  $f > 0$ , separation of variables gives

$$T = \int_{\theta_0}^{\theta_0+2\pi} \frac{d\theta}{f(\theta)} = \int_0^{2\pi} \frac{d\theta}{f(\theta)},$$

where periodicity makes the value independent of  $\theta_0$ . If  $f < 0$ , a lift decreases by  $2\pi$  in one turn; reversing the limits gives the formula with  $|f|$ . A return on  $S^1$  requires the monotone lift to change by  $2\pi k$  for some nonzero  $k \in \mathbb{Z}$ , so one turn gives the least positive return time.

**Question: What is the exact period of the nonuniform oscillator?**

Assume  $\omega > a > 0$  and evaluate

$$T = \int_{-\pi}^{\pi} \frac{d\theta}{\omega - a \sin \theta}.$$

**Solution.** Use

$$u = \tan \frac{\theta}{2}, \quad \sin \theta = \frac{2u}{1+u^2}, \quad d\theta = \frac{2 du}{1+u^2}.$$

Then

$$\begin{aligned} T &= 2 \int_{-\infty}^{\infty} \frac{du}{\omega u^2 - 2au + \omega} \\ &= \frac{2}{\omega} \int_{-\infty}^{\infty} \frac{du}{(u - a/\omega)^2 + (\omega^2 - a^2)/\omega^2} \\ &= \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \end{aligned}$$

With  $\omega > 0$  fixed, as  $a \rightarrow \omega^-$ ,

$$T = \frac{2\pi}{\sqrt{(\omega - a)(\omega + a)}} \sim \pi \sqrt{\frac{2}{\omega}} (\omega - a)^{-1/2}.$$

**Question: Why does a saddle-node bottleneck produce a square-root delay?**

Let  $r > 0$  and consider the bottleneck normal form

$$\dot{x} = r + x^2.$$

Define the passage time from  $x = -\infty$  to  $x = \infty$  as the limit, as  $L \rightarrow \infty$ , of the time required to travel from  $x = -L$  to  $x = L$ . Find this passage time and recover its scaling from a rescaling.

**Solution.** Separation gives

$$T_{\text{bottle}} = \int_{-\infty}^{\infty} \frac{dx}{r + x^2}.$$

With  $x = \sqrt{r} \tan u$ ,

$$T_{\text{bottle}} = \frac{1}{\sqrt{r}} \int_{-\pi/2}^{\pi/2} du = \frac{\pi}{\sqrt{r}}.$$

Alternatively, set

$$x = r^{1/2}X, \quad t = r^{-1/2}\tau.$$

Then

$$\frac{dX}{d\tau} = 1 + X^2,$$

which contains no  $r$ . Its passage time in  $\tau$  is therefore independent of  $r$ . Since  $t = r^{-1/2}\tau$ , the passage time in  $t$  is proportional to  $r^{-1/2}$ .

**Question: How does a flatter bottleneck change the delay?**

Let  $n \in \mathbb{N}$  satisfy  $n \geq 1$ , let  $r > 0$ , and

$$\dot{x} = r + x^{2n}.$$

Determine the dependence of the passage time from  $x = -\infty$  to  $x = \infty$  on  $r$ .

**Solution.** The rescaling

$$x = r^{1/(2n)}X, \quad t = r^{1/(2n)-1}\tau$$

gives

$$\frac{dX}{d\tau} = 1 + X^{2n}.$$

Consequently,

$$T_{\text{bottle}} = r^{1/(2n)-1} \int_{-\infty}^{\infty} \frac{dX}{1 + X^{2n}}.$$

The integral is a finite positive constant  $C_n$ : its integrand is bounded near  $X = 0$  and at most  $|X|^{-2n}$  for  $|X| \geq 1$ . Thus

$$T_{\text{bottle}} = C_n r^{-(2n-1)/(2n)}.$$

# One Circle Equation, Three Applications

Thursday, October 15, 2026

**Question:** How does a driven pendulum reduce to a circle flow?

Let  $m > 0$  be the mass,  $L > 0$  the length,  $b > 0$  the viscous damping coefficient,  $g_0 > 0$  gravitational acceleration, and  $\Gamma \geq 0$  a constant applied torque. Starting from

$$mL^2\ddot{\theta} + bL^2\dot{\theta} + mg_0L \sin \theta = \Gamma,$$

nondimensionalize the equation and identify the overdamped limit.

**Solution.** Define

$$\tau = \frac{mg_0}{bL}t, \quad \gamma = \frac{\Gamma}{mg_0L}, \quad \varepsilon = \frac{m^2g_0}{b^2L}.$$

Here primes denote differentiation with respect to  $\tau$ . Since

$$\dot{\theta} = \frac{mg_0}{bL}\theta', \quad \ddot{\theta} = \left(\frac{mg_0}{bL}\right)^2\theta'',$$

division by  $mg_0L$  gives

$$\varepsilon\theta'' + \theta' + \sin \theta = \gamma.$$

The fast and slow dimensional time scales are

$$t_f = \frac{m}{b}, \quad t_s = \frac{bL}{mg_0}, \quad \varepsilon = \frac{t_f}{t_s}.$$

Consider  $\varepsilon \downarrow 0$  while  $\gamma$  remains bounded. After a transient on the dimensional fast time scale  $t_f$ , the formal leading reduced equation is

$$\theta' = \gamma - \sin \theta.$$

To ensure that the initial layer does not produce an order-one change in  $\theta$ , also assume

$$\varepsilon|\theta'(0)| \longrightarrow 0,$$

which is  $m|\dot{\theta}(0)|/b \rightarrow 0$  in dimensional variables. In particular, this condition holds if  $\theta'(0)$  remains bounded as  $\varepsilon \downarrow 0$ . The reduced equation does not retain an independently prescribed initial angular velocity.

**Visualization: Driven-pendulum regimes**

| $0 \leq \gamma < 1$               | $\gamma = 1$                                                                  | $\gamma > 1$                                 |
|-----------------------------------|-------------------------------------------------------------------------------|----------------------------------------------|
| $\sin \theta^* = \gamma$          | $\theta^* = \pi/2$                                                            | no equilibrium                               |
| $\theta_s = \arcsin \gamma$       | half-stable threshold                                                         | $\theta' > 0$                                |
| $\theta_u = \pi - \arcsin \gamma$ | saddle-node collision                                                         | continual overturning                        |
| $\cos \theta_s > 0$               | $\theta' \sim \frac{1}{2}(\theta - \pi/2)^2 \quad (\theta \rightarrow \pi/2)$ | $T_\tau := \frac{2\pi}{\sqrt{\gamma^2 - 1}}$ |

Here  $\theta_s$  is asymptotically stable and  $\theta_u$  is unstable. The dimensional period for  $\gamma > 1$  is

$$T_t = \frac{bL}{mg_0} T_\tau.$$

**Question: What happens below the torque threshold?**

Classify the equilibria of the reduced pendulum when  $\gamma = 0.6$ .

**Solution.** For  $\gamma = 0.6$ ,

$$\theta_s = \arcsin(0.6) \approx 0.6435, \quad \theta_u = \pi - \arcsin(0.6) \approx 2.4981.$$

For  $f(\theta) = 0.6 - \sin \theta$ ,

$$f'(\theta_s) = -0.8 < 0, \quad f'(\theta_u) = 0.8 > 0.$$

Thus  $\theta_s$  is asymptotically stable and  $\theta_u$  is unstable.

**Question: What is the period above the torque threshold?**

Compute the dimensionless period of the reduced pendulum when  $\gamma = 1.25$ .

**Solution.** Since  $\gamma = 5/4$ ,

$$T_\tau = \frac{2\pi}{\sqrt{(5/4)^2 - 1}} = \frac{8\pi}{3}.$$

**Recall: Phase difference**

Let  $\Theta$  be the phase of a periodic stimulus and  $\theta$  the phase of an oscillator. Their **phase difference** is

$$\phi = [\Theta - \theta] \in S^1.$$

**Definition: Phase locking**

For a continuous autonomous phase-difference equation

$$\dot{\phi} = g(\phi),$$

the phases are **phase-locked** if  $\phi(t)$  approaches an equilibrium  $\phi^*$  of  $g$ .

**Definition: Long-time frequency**

Let  $\tilde{\theta}(t)$  be a lift of a phase  $\theta(t) \in S^1$ . If the limit exists, the **long-time frequency** is

$$\bar{\omega}_\theta = \lim_{T \rightarrow \infty} \frac{\tilde{\theta}(T) - \tilde{\theta}(0)}{T}.$$

**Definition: Entrainment**

Two oscillators are **entrained** if their long-time frequencies exist and are equal.

**Observation: Phase locking transfers a long-time frequency**

Suppose  $\phi(t) \rightarrow \phi^*$  and choose compatible lifts for which  $\tilde{\phi} = \tilde{\Theta} - \tilde{\theta}$  converges. If either  $\bar{\omega}_\Theta$  or  $\bar{\omega}_\theta$  exists, then both exist and

$$\bar{\omega}_\Theta - \bar{\omega}_\theta = \lim_{T \rightarrow \infty} \frac{\tilde{\phi}(T) - \tilde{\phi}(0)}{T} = 0.$$

Thus a phase-locked pair is entrained whenever one of its long-time frequencies exists.

**Definition: Phase drift**

The motion exhibits **phase drift** if a lift  $\tilde{\phi}(t)$  of the phase difference is unbounded as  $t \rightarrow \infty$ .

**Definition: Phase slip**

A **phase slip** occurs between times  $t_1 < t_2$  if a lift of the phase difference satisfies

$$\tilde{\phi}(t_2) - \tilde{\phi}(t_1) = \pm 2\pi.$$

**Question: When does a firefly lock to a stimulus?**

Let

$$\dot{\Theta} = \Omega, \quad \dot{\theta} = \omega + A \sin(\Theta - \theta), \quad A > 0, \quad \Omega, \omega \in \mathbb{R}.$$

Find the phase-difference equation, the locking range, and the stable locked phase.

**Solution.** Define

$$\phi = \Theta - \theta, \quad \Delta = \Omega - \omega.$$

Then

$$\dot{\phi} = g(\phi), \quad g(\phi) = \Delta - A \sin \phi.$$

A fixed point exists exactly when

$$|\Delta| \leq A,$$

equivalently,

$$\omega - A \leq \Omega \leq \omega + A.$$

If  $|\Delta| < A$ , the stable phase is the unique solution satisfying

$$\sin \phi_s = \frac{\Delta}{A}, \quad -\frac{\pi}{2} < \phi_s < \frac{\pi}{2},$$

**Question: When does a firefly lock to a stimulus? (continued)**

because

$$g'(\phi_s) = -A \cos \phi_s < 0.$$

The other fixed point is unstable. If  $|\Delta| = A$ , the two fixed points coalesce into one half-stable, Lyapunov-unstable locked phase.

**Question: How fast does the unlocked firefly drift?**

For the firefly model

$$\dot{\phi} = \Delta - A \sin \phi, \quad A > 0,$$

find the drift period and mean drift rate when  $|\Delta| > A$ .

**Solution.** If  $\Delta > A$ , then

$$T_{\text{drift}} = \int_0^{2\pi} \frac{d\phi}{\Delta - A \sin \phi} = \frac{2\pi}{\sqrt{\Delta^2 - A^2}},$$

and

$$\bar{\omega}_\phi = \frac{2\pi}{T_{\text{drift}}} = \sqrt{\Delta^2 - A^2}.$$

For  $\Delta < -A$ , the same period uses  $|\Delta|$  and the mean drift rate is  $-\sqrt{\Delta^2 - A^2}$ .

**Question: What does locking look like numerically?**

Let  $\omega = 1$ ,  $A = 0.4$ , and  $\Omega = 1.2$ . Find the stable and unstable locked phases.

**Solution.** Here

$$\Delta = 0.2, \quad \sin \phi_s = \frac{1}{2},$$

so

$$\phi_s = \frac{\pi}{6}, \quad \phi_u = \frac{5\pi}{6}.$$

Because  $\phi = \Theta - \theta$ , the value  $\phi_s = \pi/6$  means that the firefly phase lags the stimulus phase by  $\pi/6$ .

**Question: What does phase drift look like numerically?**

Let  $\omega = 1$ ,  $A = 0.4$ , and  $\Omega = 1.6$ . Find the drift period and mean drift rate.

**Solution.** Here  $\Delta = 0.6 > A$ , and

$$T_{\text{drift}} = \frac{2\pi}{\sqrt{0.6^2 - 0.4^2}} = \frac{2\pi}{\sqrt{0.2}} \approx 14.0496,$$

$$\bar{\omega}_\phi = \sqrt{0.2} \approx 0.4472.$$

**Proposition: Locking range for a general response function**

Let  $F : S^1 \rightarrow \mathbb{R}$  be continuous and let

$$\dot{\Theta} = \Omega, \quad \dot{\theta} = \omega + AF(\Theta - \theta), \quad A > 0, \quad \Omega, \omega \in \mathbb{R}.$$

Define

$$F_{\min} = \min_{S^1} F, \quad F_{\max} = \max_{S^1} F.$$

A locked phase exists if and only if

$$AF_{\min} \leq \Omega - \omega \leq AF_{\max}.$$

**Proof.** Since  $\phi = \Theta - \theta$ ,

$$\dot{\phi} = g(\phi) = \Omega - \omega - AF(\phi).$$

A fixed point exists exactly when  $(\Omega - \omega)/A$  belongs to the range  $[F_{\min}, F_{\max}]$ .

**Proposition: Stability of a locked phase**

Let  $F : S^1 \rightarrow \mathbb{R}$  be continuous, let  $A > 0$ , let  $\Omega, \omega \in \mathbb{R}$ , and define

$$g(\phi) = \Omega - \omega - AF(\phi).$$

Assume  $g$  is locally Lipschitz, and let  $g(\phi^*) = 0$ . If  $F$  is differentiable at  $\phi^*$ , then  $\phi^*$  is locally asymptotically stable when  $F'(\phi^*) > 0$  and unstable when  $F'(\phi^*) < 0$ .

Without differentiability,  $\phi^*$  is locally asymptotically stable if there is a  $\delta > 0$  such that

$$g(\phi) > 0 \quad (\phi^* - \delta < \phi < \phi^*), \quad g(\phi) < 0 \quad (\phi^* < \phi < \phi^* + \delta),$$

where the intervals are interpreted in a local angular coordinate on  $S^1$ .

**Proof.** At a differentiable fixed point,

$$g'(\phi^*) = -AF'(\phi^*),$$

so  $F'(\phi^*) > 0$  gives the displayed inward signs and  $F'(\phi^*) < 0$  gives outward signs. Inward signs and uniqueness keep each nearby trajectory between its initial point and  $\phi^*$ . It therefore stays close and converges monotonically to  $\phi^*$ , the only zero in this interval. Outward signs force arbitrarily close initial points to leave a fixed smaller arc, proving instability.

**Definition: Josephson relations**

Let  $\phi$  be the phase difference across a Josephson junction,  $I_c > 0$  its critical current,  $I_s$  its supercurrent,  $V$  its voltage,  $e$  the elementary charge, and  $\hbar$  the reduced Planck constant. The **Josephson relations** are

$$I_s = I_c \sin \phi, \quad V = \frac{\hbar}{2e} \dot{\phi}.$$

**Definition: Shunted-junction model**

For a capacitance  $C \geq 0$  and resistance  $R > 0$  in parallel with the junction, the **resistively and capacitively shunted-junction model** with constant bias current  $I$  is

$$C\dot{V} + \frac{V}{R} + I_c \sin \phi = I.$$

**Definition: Long-time mean**

For any continuous observable  $q : [0, \infty) \rightarrow \mathbb{R}$  whose long-time mean exists, write

$$\langle q \rangle = \lim_{S \rightarrow \infty} \frac{1}{S} \int_0^S q(t) dt.$$

This number is the **long-time mean** of  $q$ .

**Observation: Mean of a periodic observable**

If  $q : [0, \infty) \rightarrow \mathbb{R}$  is continuous and periodic with period  $T > 0$ , then

$$\langle q \rangle = \frac{1}{T} \int_0^T q(t) dt.$$

**Question: What is the dimensionless junction equation?**

Rewrite the shunted-junction model as an equation for  $\phi$  and nondimensionalize it.

**Solution.** Substitution of the voltage relation gives

$$\frac{\hbar C}{2e} \ddot{\phi} + \frac{\hbar}{2eR} \dot{\phi} + I_c \sin \phi = I.$$

Define

$$\tau = \frac{2eI_c R}{\hbar} t, \quad i = \frac{I}{I_c}, \quad \beta = \frac{2eI_c R^2 C}{\hbar}.$$

With primes denoting  $\tau$ -derivatives,

$$\beta \phi'' + \phi' + \sin \phi = i.$$

The McCumber parameter  $\beta$  is the dimensionless capacitance. Consider  $\beta \downarrow 0$  while  $i$  remains bounded. After the initial transient,

$$\phi' = i - \sin \phi.$$

To ensure that the initial layer does not produce an order-one change in  $\phi$ , also assume

$$\beta |\phi'(0)| \rightarrow 0,$$

which is  $RC|\dot{\phi}(0)| \rightarrow 0$  in dimensional variables. In particular, this condition holds if  $\phi'(0)$  remains bounded as  $\beta \downarrow 0$ .

**Theorem: Overdamped current–voltage curve**

Let  $I \geq 0$ ,  $I_c > 0$ , and  $R > 0$ , and define

$$i = \frac{I}{I_c}, \quad \tau = \frac{2eI_c R}{\hbar} t.$$

For the overdamped junction equation

$$\phi' = i - \sin \phi, \quad V = I_c R \phi',$$

where the prime denotes differentiation with respect to  $\tau$ , the long-time mean voltage is

$$\langle V \rangle = \begin{cases} 0, & 0 \leq I \leq I_c, \\ R\sqrt{I^2 - I_c^2}, & I > I_c. \end{cases}$$

**Proof.** Since

$$\langle V \rangle = \frac{\hbar}{2e} \langle \dot{\phi} \rangle = I_c R \langle \phi' \rangle,$$

it suffices to average  $\phi'$ . If  $0 \leq i \leq 1$ , the phase approaches a fixed point, so  $\langle \phi' \rangle = 0$ .

If  $i > 1$ , the dimensionless period is

$$T = \frac{2\pi}{\sqrt{i^2 - 1}}.$$

Therefore

$$\langle \phi' \rangle = \frac{1}{T} \int_0^T \phi' d\tau = \frac{1}{T} \int_{\phi(0)}^{\phi(0)+2\pi} d\phi = \frac{2\pi}{T} = \sqrt{i^2 - 1}.$$

Hence

$$\langle V \rangle = I_c R \sqrt{i^2 - 1} = R\sqrt{I^2 - I_c^2}.$$

**Question: What voltage occurs above threshold?**

Let  $I = 5I_c/4$ , so  $i = 5/4$ . Compute the dimensionless period, mean angular phase rate, phase-slip event rate, and mean voltage.

**Solution.**

$$\begin{aligned} i &= \frac{5}{4}, & T &= \frac{2\pi}{\sqrt{25/16 - 1}} = \frac{8\pi}{3}, \\ \langle \phi' \rangle &= \frac{2\pi}{T} = \frac{3}{4}, & \nu_{\text{slip}} &= \frac{1}{T} = \frac{3}{8\pi}, \\ \langle V \rangle &= \frac{3}{4} I_c R. \end{aligned}$$

Thus the mean angular phase rate is  $3/4$  radians per unit dimensionless time, while phase-slip events occur at rate  $3/(8\pi)$  per unit dimensionless time.

**Visualization: One equation, three interpretations**

|                | dimensionless phase model | pendulum         | Josephson junction       |
|----------------|---------------------------|------------------|--------------------------|
| phase variable | $\phi$                    | mechanical angle | quantum phase difference |
| phase rate     | $\phi'$                   | angular velocity | $V/(I_c R)$              |
| drive          | $i$                       | torque ratio     | current ratio $I/I_c$    |
| periodic term  | $\sin \phi$               | gravity          | supercurrent             |
| inertia        | $\beta$                   | mass             | capacitance              |

Fixed point  $\leftrightarrow$  rest  $\leftrightarrow$  zero mean voltage, while rotation  $\leftrightarrow$  overturning  $\leftrightarrow$  nonzero mean voltage.

# Linear Phase Planes and Stability

Tuesday, October 20, 2026

## Definition: Planar linear system

A **homogeneous planar linear system** has the form

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad A \in \mathbb{R}^{2 \times 2}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2.$$

Thus  $a, b, c, d \in \mathbb{R}$ .

## Definition: Phase plane

The **phase plane** of a planar system is its two-dimensional state space.

## Recall: Trajectory

A **trajectory** is the image of one solution in the phase plane.

## Recall: Phase portrait

A **phase portrait** is a diagram of the qualitatively distinct trajectories and equilibria in the phase plane, with the direction of increasing time marked on each non-equilibrium trajectory.

## Observation: Equilibrium at the origin

The origin is an equilibrium of every homogeneous linear system because

$$A\mathbf{0} = \mathbf{0}.$$

## Definition: Invariant set

Let  $\Phi$  be a possibly local flow. A set  $M$  is **invariant** if, for every  $\mathbf{x}_0 \in M$  and every  $t$  in the maximal existence interval of the solution through  $\mathbf{x}_0$ ,

$$\Phi(t, \mathbf{x}_0) \in M.$$

**Definition: Stable set**

For an equilibrium  $\mathbf{x}^*$ , its **stable set** is

$$W^s(\mathbf{x}^*) = \{\mathbf{x}_0 : \Phi_t(\mathbf{x}_0) \rightarrow \mathbf{x}^* \text{ as } t \rightarrow \infty\},$$

where the solution through  $\mathbf{x}_0$  is required to exist for every  $t \geq 0$ .

**Definition: Unstable set**

For an equilibrium  $\mathbf{x}^*$ , its **unstable set** is

$$W^u(\mathbf{x}^*) = \{\mathbf{x}_0 : \Phi_t(\mathbf{x}_0) \rightarrow \mathbf{x}^* \text{ as } t \rightarrow -\infty\}.$$

Here the solution through  $\mathbf{x}_0$  is required to exist for every  $t \leq 0$ .

**Definition: Stable and unstable manifolds**

When  $W^s(\mathbf{x}^*)$  is a single point or a smooth curve or surface, it is called the **stable manifold** of  $\mathbf{x}^*$ . When  $W^u(\mathbf{x}^*)$  has one of these forms, it is called the **unstable manifold** of  $\mathbf{x}^*$ .

**Proposition: Superposition**

Let  $A \in \mathbb{R}^{2 \times 2}$ . If  $\mathbf{x}_1$  and  $\mathbf{x}_2$  solve  $\dot{\mathbf{x}} = A\mathbf{x}$ , then

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2$$

is a solution for all  $c_1, c_2 \in \mathbb{R}$ .

**Proof.**

$$\frac{d}{dt}(c_1\mathbf{x}_1 + c_2\mathbf{x}_2) = c_1A\mathbf{x}_1 + c_2A\mathbf{x}_2 = A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2).$$

**Question: What is the phase portrait of a harmonic oscillator?**

Let  $m, k > 0$  and consider

$$m\ddot{x} + kx = 0.$$

Find a planar system, its conserved energy, its trajectories, and its period.

**Solution.** Define

$$v = \dot{x}, \quad \omega = \sqrt{\frac{k}{m}}.$$

Then

$$\dot{x} = v, \quad \dot{v} = -\omega^2 x.$$

The energy

$$E(x, v) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

satisfies

$$\dot{E} = mv\dot{v} + kx\dot{x} = mv(-\omega^2 x) + kxv = 0.$$

Thus each nonzero trajectory lies on an ellipse

$$mv^2 + kx^2 = 2E_0, \quad E_0 > 0.$$

**Question: What is the phase portrait of a harmonic oscillator? (continued)**

The explicit solution with initial state  $(x_0, v_0)$  is

$$x(t) = x_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t,$$

$$v(t) = -\omega x_0 \sin \omega t + v_0 \cos \omega t.$$

Every nonzero solution has period

$$T = \frac{2\pi}{\omega}.$$

At  $(x, 0)$  with  $x > 0, \dot{v} < 0$ ; therefore the ellipses are traversed clockwise in the  $(x, v)$  plane.

**Question: What is one complete physical oscillation?**

Let  $m = 2, k = 8, E_0 = 4$ , and  $(x_0, v_0) = (1, 0)$ . Find the orbit and mark the states after each quarter-period.

**Solution.** Since  $\omega = 2$ ,

$$v^2 + 4x^2 = 4, \quad x(t) = \cos 2t, \quad v(t) = -2 \sin 2t, \quad T = \pi.$$

Hence

| $t$      | $x(t)$ | $v(t)$ |
|----------|--------|--------|
| 0        | 1      | 0      |
| $\pi/4$  | 0      | -2     |
| $\pi/2$  | -1     | 0      |
| $3\pi/4$ | 0      | 2      |
| $\pi$    | 1      | 0      |

and the state moves clockwise around the ellipse.

**Recall: Attracting equilibrium**

Let  $\mathbf{x}^*$  be an equilibrium of  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ . It is **attracting** if there exists  $\delta > 0$  such that every solution with  $\|\mathbf{x}_0 - \mathbf{x}^*\| < \delta$  exists for all  $t \geq 0$  and

$$\Phi_t(\mathbf{x}_0) \rightarrow \mathbf{x}^* \quad (t \rightarrow \infty).$$

**Recall: Lyapunov stability**

An equilibrium  $\mathbf{x}^*$  is **Lyapunov stable** if, for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that every solution with  $\|\mathbf{x}_0 - \mathbf{x}^*\| < \delta$  exists for all  $t \geq 0$  and

$$\|\Phi_t(\mathbf{x}_0) - \mathbf{x}^*\| < \varepsilon \quad \text{for all } t \geq 0.$$

**Recall: Asymptotic stability**

An equilibrium is **asymptotically stable** if it is both attracting and Lyapunov stable.

**Definition: Neutral stability**

An equilibrium is **neutrally stable** if it is Lyapunov stable but not attracting.

**Recall: Instability**

An equilibrium is **unstable** if it is not Lyapunov stable.

**Visualization: Convergence and confinement**

| property              | geometric content                            |
|-----------------------|----------------------------------------------|
| attracting            | nearby solutions converge to the equilibrium |
| Lyapunov stable       | nearby solutions remain nearby               |
| asymptotically stable | both convergence and confinement             |

For the harmonic oscillator, nearby energy ellipses remain nearby but never shrink:

$$\boxed{\text{Lyapunov stable}} + \boxed{\text{not attracting}} = \boxed{\text{neutrally stable}}.$$

**Question: How does one diagonal family change its phase portrait?**

For

$$\dot{x} = ax, \quad \dot{y} = -y,$$

determine the stability of the origin and the generic forward asymptotic behavior for every  $a \in \mathbb{R}$ .

**Solution.**

$$x(t) = x_0 e^{at}, \quad y(t) = y_0 e^{-t}.$$

Therefore

| $a$          | stability and geometry                                               | generic forward behavior<br>$x_0 y_0 \neq 0$ |
|--------------|----------------------------------------------------------------------|----------------------------------------------|
| $a < -1$     | asymptotically stable                                                | approach tangent to the $y$ -axis            |
| $a = -1$     | asymptotically stable;<br>each nonzero trajectory is a ray           | approach along the initial ray               |
| $-1 < a < 0$ | asymptotically stable                                                | approach tangent to the $x$ -axis            |
| $a = 0$      | $y = 0$ is a line of equilibria;<br>$\mathbf{0}$ is neutrally stable | vertical approach to $(x_0, 0)$              |
| $a > 0$      | $\mathbf{0}$ is unstable                                             | escape tangent to the $x$ -axis              |

For  $a > 0$ ,

$$W^s(\mathbf{0}) = \{(0, y) : y \in \mathbb{R}\}, \quad W^u(\mathbf{0}) = \{(x, 0) : x \in \mathbb{R}\}.$$

**Question: How can one prove stability directly?**

For

$$\dot{x} = -2x, \quad \dot{y} = -y,$$

prove Lyapunov stability and attraction by estimating the Euclidean norm.

**Solution.** The solution satisfies

$$\|(x(t), y(t))\|^2 = x_0^2 e^{-4t} + y_0^2 e^{-2t} \leq e^{-2t} \|(x_0, y_0)\|^2.$$

Taking square roots gives

$$\|(x(t), y(t))\| \leq e^{-t} \|(x_0, y_0)\|.$$

For every  $\varepsilon > 0$ , an initial norm less than  $\varepsilon$  stays less than  $\varepsilon$  for every  $t \geq 0$ , proving Lyapunov stability. The same bound tends to zero, proving attraction.

**Question: Can attraction occur without Lyapunov stability?**

On  $S^1$ , analyze

$$\dot{\theta} = 1 - \cos \theta.$$

**Solution.** The unique fixed point is  $[0]$ . Since

$$1 - \cos \theta > 0$$

for  $\theta \notin 2\pi\mathbb{Z}$ , every nonfixed trajectory moves counterclockwise and passes through every intermediate angle. In the representative interval  $[0, 2\pi)$ , its angle approaches  $2\pi$ , which represents  $[0]$ . Hence  $[0]$  is attracting.

Initial points with  $\theta_0 > 0$  arbitrarily small must pass through the opposite point  $[\pi]$ . Thus arbitrarily close initial states leave the fixed semicircle centered at  $[0]$ , disproving Lyapunov stability.

**Definition: Involution**

Let  $U \subseteq \mathbb{R}^2$ . A map  $R : U \rightarrow U$  is an **involution** if

$$R(R(\mathbf{z})) = \mathbf{z}$$

for every  $\mathbf{z} \in U$ .

**Definition: Reversing symmetry**

Let  $U \subseteq \mathbb{R}^2$  be open, and let  $R : U \rightarrow U$  be a differentiable involution. For a vector field  $\mathbf{F} : U \rightarrow \mathbb{R}^2$ , the map  $R$  is a **reversing symmetry** of

$$\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$$

if

$$\mathbf{F}(R(\mathbf{z})) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z}).$$

Here  $DR(\mathbf{z})$  is the derivative matrix of  $R$ . For  $\mathbf{z} = (x, y)$  and  $R = (R_1, R_2)$ ,

$$DR = \begin{pmatrix} \partial R_1 / \partial x & \partial R_1 / \partial y \\ \partial R_2 / \partial x & \partial R_2 / \partial y \end{pmatrix}.$$

**Observation: Time reversal of a solution**

If  $R$  is a reversing symmetry and  $\mathbf{z}(t)$  is a solution, then  $R(\mathbf{z}(-t))$  is also a solution. Indeed, the chain rule gives

$$\frac{d}{dt}R(\mathbf{z}(-t)) = -DR(\mathbf{z}(-t))\mathbf{F}(\mathbf{z}(-t)) = \mathbf{F}(R(\mathbf{z}(-t))).$$

**Proposition: Closure of the harmonic-oscillator orbit by symmetry**

For

$$\dot{x} = v, \quad \dot{v} = -x,$$

define

$$S_x(x, v) = (x, -v), \quad S_v(x, v) = (-x, v).$$

Both maps are reversing symmetries. Suppose a trajectory begins at

$$\mathbf{z}(0) = (0, -v_0), \quad v_0 > 0,$$

and first meets the  $x$ -axis at

$$\mathbf{z}(t_1) = (x_1, 0), \quad x_1 < 0.$$

Then  $\mathbf{z}(4t_1) = \mathbf{z}(0)$ .

**Proof.** Direct calculation gives

$$\mathbf{F}(S_x \mathbf{z}) = -S_x \mathbf{F}(\mathbf{z}), \quad \mathbf{F}(S_v \mathbf{z}) = -S_v \mathbf{F}(\mathbf{z}).$$

Uniqueness at the point fixed by  $S_x$  yields

$$\mathbf{z}(t_1 + s) = S_x \mathbf{z}(t_1 - s).$$

Thus

$$\mathbf{z}(2t_1) = S_x \mathbf{z}(0) = (0, v_0).$$

Applying the same reflection argument at  $t = 2t_1$  with  $S_v$  gives

$$\mathbf{z}(3t_1) = (-x_1, 0), \quad \mathbf{z}(4t_1) = (0, -v_0) = \mathbf{z}(0).$$

Uniqueness makes the trajectory periodic.

# Linear Classification I: Real Eigenvalues

Thursday, October 22, 2026

## Definition: Eigenvalue

Let  $A$  be a real  $2 \times 2$  matrix. A scalar  $\lambda \in \mathbb{C}$  is an **eigenvalue** of  $A$  if there is a nonzero vector  $\mathbf{v} \in \mathbb{C}^2$  such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

## Definition: Eigenvector

Let  $A$  be a real  $2 \times 2$  matrix, and let  $\lambda$  be an eigenvalue of  $A$ . A vector  $\mathbf{v} \in \mathbb{C}^2 \setminus \{\mathbf{0}\}$  is an **eigenvector associated with**  $\lambda$  if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

## Recall: Eigenspace and kernel notation

Let  $I$  denote the  $2 \times 2$  identity matrix. The **eigenspace** associated with an eigenvalue  $\lambda$  of  $A$  is

$$E_\lambda = \ker(A - \lambda I) = \{\mathbf{v} : (A - \lambda I)\mathbf{v} = \mathbf{0}\}.$$

Here  $\ker$  denotes the **kernel** or **nullspace**. For real  $\lambda$ , we use vectors in  $\mathbb{R}^2$ ; for nonreal  $\lambda$ , vectors in  $\mathbb{C}^2$ . The nonzero vectors in  $E_\lambda$  are the eigenvectors.

## Recall: Characteristic polynomial

For a real matrix  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , the **characteristic polynomial** is

$$p(s) = \det(A - sI) = (a - s)(d - s) - bc.$$

Its roots are the eigenvalues of  $A$ . A root that occurs twice is a **repeated eigenvalue**.

**Definition: Eigendirection**

Let  $A$  be a real  $2 \times 2$  matrix. If an eigenvalue  $\lambda$  of  $A$  and an associated eigenvector  $\mathbf{v}$  are real, the **eigendirection associated with  $\lambda$**  is the line

$$\text{span}\{\mathbf{v}\} = \{c\mathbf{v} : c \in \mathbb{R}\}.$$

**Proposition: Eigensolutions**

Let  $A \in \mathbb{R}^{2 \times 2}$ . If  $A\mathbf{v} = \lambda\mathbf{v}$ , where  $\lambda \in \mathbb{C}$  and  $\mathbf{v} \in \mathbb{C}^2 \setminus \{\mathbf{0}\}$ , then

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}$$

is a complex-valued solution of the complexified system  $\dot{\mathbf{x}} = A\mathbf{x}$ , meaning the same differential equation interpreted on  $\mathbb{C}^2$ .

**Proof.**

$$\dot{\mathbf{x}}(t) = \lambda e^{\lambda t}\mathbf{v} = e^{\lambda t}A\mathbf{v} = A\mathbf{x}(t).$$

**Theorem: General solution for two distinct real eigenvalues**

Let  $A \in \mathbb{R}^{2 \times 2}$  have distinct real eigenvalues  $\lambda_1, \lambda_2$  with associated real eigenvectors  $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^2$ . Then  $\mathbf{v}_1, \mathbf{v}_2$  are linearly independent, and every real solution of  $\dot{\mathbf{x}} = A\mathbf{x}$  has the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2, \quad c_1, c_2 \in \mathbb{R}.$$

**Proof.** If  $a, b \in \mathbb{R}$  satisfy  $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{0}$ , applying  $A - \lambda_1 I$  gives  $b(\lambda_2 - \lambda_1)\mathbf{v}_2 = \mathbf{0}$ . Thus  $b = 0$  and then  $a = 0$ . The eigenvectors are therefore linearly independent, so every initial state has a unique representation

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$$

Superposition shows that the displayed function is a solution, and at  $t = 0$  it equals  $\mathbf{x}_0$ . Uniqueness of the linear initial-value problem makes it the only solution.

**Definition: Linear saddle**

The origin of  $\dot{\mathbf{x}} = A\mathbf{x}$ , with  $A \in \mathbb{R}^{2 \times 2}$ , is a **saddle** if  $A$  has one negative and one positive real eigenvalue.

**Definition: Linear node and its repeated-eigenvalue cases**

The origin of  $\dot{\mathbf{x}} = A\mathbf{x}$ , with  $A \in \mathbb{R}^{2 \times 2}$ , is a **node** if both eigenvalues are real, nonzero, and have the same sign. A node is a **star node** if  $A = \lambda I$  for some  $\lambda \in \mathbb{R} \setminus \{0\}$ ; it is a **degenerate node** if its eigenvectors span only one line.

**Theorem: Classification for two distinct real eigenvalues**

Let  $A$  be a real  $2 \times 2$  matrix with distinct real eigenvalues  $\lambda_1$  and  $\lambda_2$ , labeled so that  $\lambda_1 < \lambda_2$ .

| eigenvalue signs            | equilibrium type   | stability                       |
|-----------------------------|--------------------|---------------------------------|
| $\lambda_1 < 0 < \lambda_2$ | saddle             | unstable                        |
| $\lambda_1 < \lambda_2 < 0$ | node               | asymptotically stable           |
| $0 < \lambda_1 < \lambda_2$ | node               | unstable                        |
| $\lambda_1 < \lambda_2 = 0$ | line of equilibria | Lyapunov stable, not attracting |
| $\lambda_1 = 0 < \lambda_2$ | line of equilibria | unstable                        |

**Proof.** In the eigenvector coordinates, each component is multiplied by  $e^{\lambda_j t}$ . If both eigenvalues are nonpositive, both components stay small whenever they start small; the invertible change of coordinates gives Lyapunov stability in the phase plane. If both eigenvalues are negative, both components also tend to zero, giving attraction.

A positive eigenvalue gives arbitrarily small initial states on its eigenline that grow without bound, proving instability. A zero eigenvalue gives a line of equilibria, so the origin is not attracting.

**Proposition: Dominant eigendirection**

Let  $A \in \mathbb{R}^{2 \times 2}$  have distinct real eigenvalues  $\lambda_1 < \lambda_2$  with associated real eigenvectors  $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^2$ , and let

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2.$$

If  $\lambda_1 < \lambda_2 < 0$  and  $c_2 \neq 0$ , then the trajectory approaches the origin tangent to  $\text{span}\{\mathbf{v}_2\}$  as  $t \rightarrow \infty$ . If  $0 < \lambda_1 < \lambda_2$  and  $c_1 \neq 0$ , then the trajectory approaches the origin tangent to  $\text{span}\{\mathbf{v}_1\}$  as  $t \rightarrow -\infty$ .

**Proof.** In the stable case, factoring out the slower exponential gives

$$\mathbf{x}(t) = e^{\lambda_2 t} (c_2 \mathbf{v}_2 + c_1 e^{(\lambda_1 - \lambda_2)t} \mathbf{v}_1),$$

and the parenthesized direction tends to  $c_2 \mathbf{v}_2$ . The unstable case is obtained by factoring out  $e^{\lambda_1 t}$  and letting  $t \rightarrow -\infty$ .

**Corollary: Stable and unstable sets of a linear saddle**

Let  $A \in \mathbb{R}^{2 \times 2}$  have eigenvalues  $\lambda_s < 0 < \lambda_u$ . Choose real eigenvectors  $\mathbf{v}_s, \mathbf{v}_u \in \mathbb{R}^2 \setminus \{\mathbf{0}\}$  such that  $A\mathbf{v}_s = \lambda_s \mathbf{v}_s$  and  $A\mathbf{v}_u = \lambda_u \mathbf{v}_u$ . Then

$$W^s(\mathbf{0}) = \text{span}\{\mathbf{v}_s\}, \quad W^u(\mathbf{0}) = \text{span}\{\mathbf{v}_u\}.$$

**Proof.** A solution converges to the origin as  $t \rightarrow \infty$  exactly when its  $\mathbf{v}_u$  coefficient is zero. It converges to the origin as  $t \rightarrow -\infty$  exactly when its  $\mathbf{v}_s$  coefficient is zero.

**Question: How do eigenvalues solve an initial-value problem?**

Solve

$$\dot{x} = x + y, \quad \dot{y} = 4x - 2y, \quad (x(0), y(0)) = (2, -3),$$

and classify the origin.

**Solution.**

$$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}, \quad \det(A - \lambda I) = \lambda^2 + \lambda - 6 = (\lambda - 2)(\lambda + 3).$$

Thus

$$\lambda_1 = -3, \quad \lambda_2 = 2.$$

Solving  $(A - \lambda I)\mathbf{v} = \mathbf{0}$  gives

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Hence

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = c_1 e^{-3t} \begin{pmatrix} 1 \\ -4 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The initial condition requires

$$c_1 + c_2 = 2, \quad -4c_1 + c_2 = -3,$$

so  $c_1 = c_2 = 1$ . Therefore

$$x(t) = e^{2t} + e^{-3t}, \quad y(t) = e^{2t} - 4e^{-3t}.$$

The origin is a saddle with

$$W^s(\mathbf{0}) = \text{span}\{(1, -4)\}, \quad W^u(\mathbf{0}) = \text{span}\{(1, 1)\}.$$

**Visualization: Real-eigenvalue geometry**

The following table describes trajectories whose coefficients along both displayed eigendirections are nonzero.

| portrait                    | $t \rightarrow \infty$     | $t \rightarrow -\infty$    |
|-----------------------------|----------------------------|----------------------------|
| $\lambda_1 < \lambda_2 < 0$ | tangent to $\mathbf{v}_2$  | parallel to $\mathbf{v}_1$ |
| $0 < \lambda_1 < \lambda_2$ | parallel to $\mathbf{v}_2$ | tangent to $\mathbf{v}_1$  |
| $\lambda_s < 0 < \lambda_u$ | parallel to $\mathbf{v}_u$ | parallel to $\mathbf{v}_s$ |

Here “parallel” describes the limiting direction of an unbounded trajectory; it does not assert that the distance to the eigenline tends to zero. The table does not apply to the eigendirection trajectories: each eigendirection is itself an invariant straight line, and a solution that starts on one remains on it. A saddle trajectory with nonzero coefficients along both eigendirections approaches the unstable eigendirection as  $t \rightarrow \infty$ ; it does not approach the stable manifold in forward time.

**Theorem: Repeated eigenvalue with two eigendirections**

Suppose the characteristic polynomial of a real  $2 \times 2$  matrix  $A$  is  $(s - \lambda)^2$  and the eigenspace for  $\lambda$  contains two linearly independent eigenvectors. Then

$$A = \lambda I.$$

If  $\lambda < 0$ , the origin is an asymptotically stable star node; if  $\lambda > 0$ , it is an unstable star node; if  $\lambda = 0$ , every point is an equilibrium.

**Proof.** Every vector  $\mathbf{x}$  is a linear combination of the two eigenvectors, so  $A\mathbf{x} = \lambda\mathbf{x}$  for every  $\mathbf{x}$ . Hence  $A = \lambda I$  and

$$\mathbf{x}(t) = e^{\lambda t} \mathbf{x}_0.$$

The classification follows.

**Definition: Defective eigenvalue**

Let  $A$  be a real  $2 \times 2$  matrix. A repeated eigenvalue  $\lambda$  of  $A$  is **defective** if

$$\dim \ker(A - \lambda I) = 1.$$

**Definition: Generalized eigenvector**

Let  $n \geq 1$  be an integer, let  $A \in \mathbb{C}^{n \times n}$ , let  $\lambda \in \mathbb{C}$  be an eigenvalue, and let  $\mathbf{v} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$  satisfy

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

Here  $I$  is the  $n \times n$  identity matrix. A vector  $\mathbf{w} \in \mathbb{C}^n$  satisfying

$$(A - \lambda I)\mathbf{w} = \mathbf{v}$$

is a **generalized eigenvector associated with  $\mathbf{v}$** . The pair  $(\mathbf{v}, \mathbf{w})$  is a **length-two Jordan chain**.

**Theorem: Solution from a length-two Jordan chain**

Let  $A$  be a real  $2 \times 2$  matrix and let  $\lambda \in \mathbb{R}$ . Suppose  $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ ,  $\mathbf{v} \neq \mathbf{0}$ , and

$$A\mathbf{v} = \lambda\mathbf{v}, \quad (A - \lambda I)\mathbf{w} = \mathbf{v}.$$

Then  $\mathbf{v}$  and  $\mathbf{w}$  are linearly independent, and every real solution of  $\dot{\mathbf{x}} = A\mathbf{x}$  is

$$\mathbf{x}(t) = e^{\lambda t} [(c_1 + c_2 t)\mathbf{v} + c_2 \mathbf{w}], \quad c_1, c_2 \in \mathbb{R}.$$

**Proof.** The two functions

$$e^{\lambda t} \mathbf{v}, \quad e^{\lambda t} (t\mathbf{v} + \mathbf{w})$$

solve the system. For the second,

$$\frac{d}{dt} [e^{\lambda t} (t\mathbf{v} + \mathbf{w})] = e^{\lambda t} [(\lambda t + 1)\mathbf{v} + \lambda \mathbf{w}],$$

**Theorem: Solution from a length-two Jordan chain (continued)**

while

$$Ae^{\lambda t}(t\mathbf{v} + \mathbf{w}) = e^{\lambda t}[\lambda t\mathbf{v} + A\mathbf{w}] = e^{\lambda t}[(\lambda t + 1)\mathbf{v} + \lambda\mathbf{w}].$$

If  $\mathbf{w}$  were a multiple of  $\mathbf{v}$ , then  $(A - \lambda I)\mathbf{w} = \mathbf{0}$ , contradicting  $\mathbf{v} \neq \mathbf{0}$ . Thus  $\mathbf{v}, \mathbf{w}$  are independent, and the two displayed solutions span the two-dimensional real solution space.

**Question: What does a defective node look like?**

Let

$$A = \begin{pmatrix} \lambda & b \\ 0 & \lambda \end{pmatrix}, \quad \lambda \in \mathbb{R}, \quad b \in \mathbb{R} \setminus \{0\}.$$

Find the general solution and classify the origin.

**Solution.** The eigenspace is

$$\ker(A - \lambda I) = \text{span}\{(1, 0)\}.$$

The system is

$$\dot{x} = \lambda x + by, \quad \dot{y} = \lambda y.$$

Thus

$$y(t) = y_0 e^{\lambda t}, \quad x(t) = e^{\lambda t}(x_0 + by_0 t).$$

For  $t \geq 0$ , the triangle inequality gives

$$\|(x(t), y(t))\| \leq e^{\lambda t}(1 + |b|t)\|(x_0, y_0)\|.$$

If  $\lambda < 0$ , this multiplier is bounded and tends to zero, proving Lyapunov stability and attraction. The origin is therefore an asymptotically stable degenerate node. If  $\lambda > 0$ , arbitrarily small nonzero initial states on the  $x$ -axis grow without bound, so the node is unstable. In both time directions, generic trajectories become parallel to the unique eigendirection. If  $\lambda = 0$ , every point on  $y = 0$  is an equilibrium and every solution with  $y_0 \neq 0$  has  $x(t) = x_0 + by_0 t$ . Thus the origin belongs to a nonisolated line of equilibria and is unstable.

**Question: Can a repeated eigenvalue alone determine the portrait?**

Classify

$$A_1 = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix}.$$

**Solution.** Both matrices have the repeated eigenvalue  $-2$ . For  $A_1$ , every nonzero vector is an eigenvector, so the origin is an asymptotically stable star node. For  $A_2$ ,

$$\ker(A_2 + 2I) = \text{span}\{(1, 0)\},$$

so the origin is an asymptotically stable degenerate node. The eigenvalue list does not distinguish these portraits; the eigenspace dimension does.

# Complex Eigenvalues and the Trace–Determinant Plane

Tuesday, October 27, 2026

## Definition: Linear spiral

The origin of  $\dot{\mathbf{x}} = A\mathbf{x}$ , with  $A \in \mathbb{R}^{2 \times 2}$ , is a **spiral** if  $A$  has nonreal eigenvalues with nonzero real part.

## Definition: Linear center

The origin of  $\dot{\mathbf{x}} = A\mathbf{x}$ , with  $A \in \mathbb{R}^{2 \times 2}$ , is a **center** if every nonzero trajectory is a closed curve surrounding the origin.

## Question: What do complex eigenvalues mean in a real system?

Let  $\alpha \in \mathbb{R}$ ,  $\beta \in \mathbb{R} \setminus \{0\}$ , and

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Find the eigenvalues and the polar equations.

**Solution.** The characteristic polynomial is

$$(\alpha - \lambda)^2 + \beta^2,$$

so

$$\lambda = \alpha \pm i\beta.$$

Write  $x = r \cos \theta$ ,  $y = r \sin \theta$ , with  $r > 0$  and a continuous choice of angle  $\theta$ . Then

$$r\dot{r} = x\dot{x} + y\dot{y} = \alpha(x^2 + y^2) = \alpha r^2,$$

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = \beta.$$

Therefore

$$r(t) = r_0 e^{\alpha t}, \quad \theta(t) = \theta_0 + \beta t.$$

Here  $r_0 = r(0) > 0$  and  $\theta_0 = \theta(0)$ . The origin is an asymptotically stable spiral if  $\alpha < 0$ , an unstable spiral if  $\alpha > 0$ , and a center if  $\alpha = 0$ . The angular revolution time is  $2\pi/|\beta|$ , and the sign of  $\beta$  gives the orientation. When  $\alpha \neq 0$ , this is the time for one revolution, not a period of the nonclosed trajectory.

**Notation: Trace**

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

write

$$\tau = \operatorname{tr} A = a + d.$$

**Notation: Determinant**For  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , write

$$\Delta = \det A = ad - bc.$$

**Notation: Discriminant**For a real  $2 \times 2$  matrix with trace  $\tau$  and determinant  $\Delta$ , write

$$D = \tau^2 - 4\Delta.$$

**Definition: Matrix exponential**For a real  $2 \times 2$  matrix  $A$  and  $t \in \mathbb{R}$ , the **matrix exponential** is

$$e^{At} := \sum_{k=0}^{\infty} \frac{(At)^k}{k!}.$$

**Observation: Matrix exponential and the initial-value problem**For a constant real  $2 \times 2$  matrix  $A$ ,

$$\frac{d}{dt} e^{At} = A e^{At}, \quad e^{A0} = I,$$

where  $I$  is the identity matrix. Thus the solution of  $\dot{\mathbf{x}} = A\mathbf{x}$ ,  $\mathbf{x}(0) = \mathbf{x}_0$ , is

$$\mathbf{x}(t) = e^{At} \mathbf{x}_0.$$

**Observation: Eigenvalues from trace and determinant**

The characteristic polynomial and eigenvalues are

$$p(\lambda) = \lambda^2 - \tau\lambda + \Delta, \quad \lambda_{1,2} = \frac{\tau \pm \sqrt{D}}{2}.$$

Thus

$$\lambda_1 + \lambda_2 = \tau, \quad \lambda_1 \lambda_2 = \Delta.$$

**Theorem: Trace–determinant classification for an isolated origin**

Let  $A$  be a real  $2 \times 2$  matrix with  $\Delta = \det A \neq 0$ . Then the origin is the unique equilibrium of  $\dot{\mathbf{x}} = A\mathbf{x}$ , and

| condition                        | type at the origin      | stability                                                 |
|----------------------------------|-------------------------|-----------------------------------------------------------|
| $\Delta < 0$                     | saddle                  | unstable                                                  |
| $\Delta > 0, D > 0$              | node                    | $\tau < 0$ : asymptotically stable; $\tau > 0$ : unstable |
| $\Delta > 0, D < 0, \tau \neq 0$ | spiral                  | $\tau < 0$ : asymptotically stable; $\tau > 0$ : unstable |
| $\Delta > 0, \tau = 0$           | center                  | neutrally stable                                          |
| $\Delta > 0, D = 0$              | star or degenerate node | $\tau < 0$ : asymptotically stable; $\tau > 0$ : unstable |

**Proof.** If  $\Delta < 0$ , the eigenvalues are real with opposite signs. If  $\Delta > 0$  and  $D > 0$ , they are real with the same sign, determined by their sum  $\tau$ .

For  $D < 0$ , put  $\beta = \sqrt{-D}/2 > 0$  and  $B = A - (\tau/2)I$ . Direct multiplication gives  $B^2 = -\beta^2 I$ , so the exponential series gives

$$e^{At} = e^{\tau t/2} \left( I \cos \beta t + \frac{B}{\beta} \sin \beta t \right).$$

For  $\mathbf{x}_0 \neq \mathbf{0}$ , the vectors  $\mathbf{x}_0$  and  $B\mathbf{x}_0$  are independent: a real eigenvalue of  $B$  would contradict  $B^2 = -\beta^2 I$ . Thus the parenthesized factor traces an ellipse when applied to  $\mathbf{x}_0$ ; the exponential shrinks or expands it when  $\tau \neq 0$ . The parenthesized matrix is bounded and periodic. If  $\tau < 0$ , its exponentially decaying prefactor proves both stability and attraction. If  $\tau > 0$ , successive times  $t = 2\pi k/\beta$  multiply each nonzero initial state by  $e^{\tau\pi k/\beta}$  for  $k = 1, 2, \dots$ , proving instability. If  $\tau = 0$ , every nonzero solution is periodic and stays close whenever it starts close: the origin is a neutrally stable center. Its period is  $2\pi/\beta = 2\pi/\sqrt{\Delta}$ .

If  $D = 0$  and  $\Delta > 0$ , the repeated eigenvalue is  $\tau/2 \neq 0$ ; the eigenspace dimension distinguishes a star from a degenerate node.

**Proposition: Zero-determinant boundary**

Let  $A$  be a real  $2 \times 2$  matrix with  $\Delta = \det A = 0$ , and let  $\tau = \text{tr } A$ .

| condition            | equilibria         | stability of the origin         |
|----------------------|--------------------|---------------------------------|
| $\tau < 0$           | $\ker A$ is a line | Lyapunov stable, not attracting |
| $\tau > 0$           | $\ker A$ is a line | unstable                        |
| $\tau = 0, A = 0$    | $\mathbb{R}^2$     | Lyapunov stable, not attracting |
| $\tau = 0, A \neq 0$ | $\ker A$ is a line | unstable                        |

**Proof.** If  $\tau \neq 0$ , the distinct eigenvalues are 0 and  $\tau$ . The zero eigendirection consists of equilibria, while the other component decays for  $\tau < 0$  and grows for  $\tau > 0$ .

Direct multiplication gives  $A^2 - \tau A + \Delta I = 0$ . If  $\tau = 0$ , this gives  $A^2 = 0$ . When  $A = 0$ , every solution is constant. When  $A \neq 0$ ,

$$e^{At} = I + tA,$$

so arbitrarily small initial states outside  $\ker A$  produce solutions that leave any fixed neighborhood of the origin.

**Visualization: Trace–determinant plane**

With horizontal coordinate  $\tau$  and vertical coordinate  $\Delta$ ,

| region or curve               | portrait                           |
|-------------------------------|------------------------------------|
| $\Delta < 0$                  | saddle                             |
| $0 < \Delta < \tau^2/4$       | node                               |
| $\Delta > \tau^2/4, \tau < 0$ | asymptotically stable spiral       |
| $\Delta > \tau^2/4, \tau > 0$ | unstable spiral                    |
| $\tau = 0, \Delta > 0$        | center                             |
| $\Delta = \tau^2/4 > 0$       | star or degenerate node            |
| $\Delta = 0, \tau < 0$        | Lyapunov-stable line of equilibria |
| $\Delta = 0, \tau > 0$        | unstable line of equilibria        |
| $\Delta = \tau = 0$           | zero matrix or unstable case       |

Within the half-plane  $\Delta > 0$ , the parabola  $\Delta = \tau^2/4$  separates real-eigenvalue nodes from complex-eigenvalue portraits; the latter are spirals except for centers on  $\tau = 0$ . If  $\tau < 0$  and  $\Delta > 0$ , the origin is asymptotically stable; if  $\tau < 0$  and  $\Delta = 0$ , it is Lyapunov stable but not attracting. If  $\tau > 0$ , the origin is unstable, and every point with  $\Delta < 0$  is a saddle.

**Question: Can trace and determinant classify without solving?**

Classify

$$A_1 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}.$$

**Solution.** For  $A_1$ ,

$$\tau = 5, \quad \Delta = 4 - 6 = -2 < 0,$$

so the origin is a saddle.

For  $A_2$ ,

$$\tau = 6, \quad \Delta = 8 - 3 = 5, \quad D = 36 - 20 = 16 > 0.$$

The origin is an unstable node. Its eigenvalues are

$$\lambda = \frac{6 \pm 4}{2} = 5, 1.$$

**Question: How do we solve a stable spiral?**

For

$$A = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix},$$

classify the origin, determine the orientation, and solve the initial-value problem  $\mathbf{x}(0) = (2, 0)$ .

**Solution.**

$$\tau = -2, \quad \Delta = 5, \quad D = -16,$$

so  $\lambda = -1 \pm 2i$  and the origin is an asymptotically stable spiral. At  $(1, 0)$ ,

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix},$$

**Question: How do we solve a stable spiral? (continued)**

so the rotation is counterclockwise.

Let

$$B = A + I = \begin{pmatrix} 0 & -4 \\ 1 & 0 \end{pmatrix}.$$

Since  $B^2 = -4I$ ,

$$e^{At} = e^{-t} \left( I \cos 2t + \frac{B}{2} \sin 2t \right) = e^{-t} \begin{pmatrix} \cos 2t & -2 \sin 2t \\ \frac{1}{2} \sin 2t & \cos 2t \end{pmatrix}.$$

Thus

$$x(t) = 2e^{-t} \cos 2t, \quad y(t) = e^{-t} \sin 2t.$$

One revolution takes time  $\pi$ , and

$$\mathbf{x}(t + \pi) = e^{-\pi} \mathbf{x}(t).$$

Thus the distance from the origin shrinks by  $e^{-\pi}$  per turn. The spiral trajectory itself is not periodic.

**Question: Why do boundary portraits require more than eigenvalue multiplicity?**

Classify

$$A_1 = -I, \quad A_2 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

**Solution.** Both  $A_1$  and  $A_2$  satisfy

$$\tau = -2, \quad \Delta = 1, \quad D = 0.$$

For  $A_1$ , every nonzero vector is an eigenvector, so the origin is an asymptotically stable star node.

For  $A_2$ ,

$$\ker(A_2 + I) = \text{span}\{(1, 0)\},$$

so the origin is an asymptotically stable degenerate node.

For  $A_3$ ,  $\Delta = 0$ ,  $\tau = -1$ , and

$$x(t) = x_0 e^{-t}, \quad y(t) = y_0.$$

Every point on the  $y$ -axis is an equilibrium. The origin is Lyapunov stable but not attracting.

**Definition: Linear love-affair model**

Let  $R(t)$  denote Romeo's feeling toward Juliet and  $J(t)$  Juliet's feeling toward Romeo, with positive values representing love and negative values representing hate. For constants  $a, b, c, d \in \mathbb{R}$ , a linear response model is

$$\dot{R} = aR + bJ, \quad \dot{J} = cR + dJ.$$

This is the **linear love-affair model**.

**Observation: Interpretation of the response coefficients**

In the linear love-affair model, the diagonal coefficients  $a, d$  measure responses to one's own feelings, while the off-diagonal coefficients  $b, c$  measure responses to the other person's feelings.

**Question: What happens to responsive Romeo and fickle Juliet?**

Let

$$\dot{R} = aJ, \quad \dot{J} = -bR, \quad a, b > 0.$$

Classify the origin and find the trajectories and period.

**Solution.**

$$A = \begin{pmatrix} 0 & a \\ -b & 0 \end{pmatrix}, \quad \tau = 0, \quad \Delta = ab > 0.$$

Thus the origin is a center with eigenvalues  $\pm i\sqrt{ab}$ . Moreover,

$$H(R, J) = bR^2 + aJ^2$$

is conserved because

$$\dot{H} = 2bR(aJ) + 2aJ(-bR) = 0.$$

For  $C > 0$ , the level sets  $H = C$  are the nonzero trajectories and are ellipses. The level set  $H = 0$  is the equilibrium at the origin. Every nonzero trajectory has period

$$T = \frac{2\pi}{\sqrt{ab}}.$$

At  $(R, 0)$  with  $R > 0$ ,  $\dot{J} = -bR < 0$ , so the motion is clockwise.

**Question: What happens to two identically cautious lovers?**

Let

$$\dot{R} = aR + bJ, \quad \dot{J} = bR + aJ, \quad a < 0 < b.$$

Give the complete classification as  $b$  varies.

**Solution.** The eigenpairs are

$$\lambda_+ = a + b, \quad \mathbf{v}_+ = (1, 1), \quad \lambda_- = a - b, \quad \mathbf{v}_- = (1, -1).$$

Since  $a - b < 0$ ,

| condition    | type                       | long-time behavior                  |
|--------------|----------------------------|-------------------------------------|
| $0 < b < -a$ | asymptotically stable node | $(R, J) \rightarrow (0, 0)$         |
| $b = -a$     | line $R = J$ of equilibria | $(R, J) \rightarrow (c, c)$         |
| $b > -a$     | saddle                     | $W^u = \{R = J\}, W^s = \{R = -J\}$ |

In the middle case,  $c = (R(0) + J(0))/2$ .

# Nonlinear Phase Portraits and Uniqueness Barriers

Thursday, October 29, 2026

## Definition: Planar autonomous system

Let  $U \subseteq \mathbb{R}^2$  be open, and let  $f, g : U \rightarrow \mathbb{R}$ . A **planar autonomous system** is

$$\dot{x} = f(x, y), \quad \dot{y} = g(x, y),$$

where  $(x, y) \in U$ .

## Notation: Vector form of a planar system

Write

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x} = (x, y), \quad \mathbf{F} : U \rightarrow \mathbb{R}^2, \quad \mathbf{F}(x, y) = (f(x, y), g(x, y)).$$

## Recall: Equilibrium of a planar system

A point  $(x^*, y^*) \in U$  is an **equilibrium** if

$$f(x^*, y^*) = g(x^*, y^*) = 0.$$

## Definition: Nullclines

The  **$x$ -nullcline** is

$$\{(x, y) \in U : f(x, y) = 0\}.$$

The  **$y$ -nullcline** is

$$\{(x, y) \in U : g(x, y) = 0\}.$$

## Observation: Directions along nullclines

Away from an equilibrium, the vector field is vertical on the  $x$ -nullcline and horizontal on the  $y$ -nullcline.

**Observation: A nullcline need not be invariant**

The condition  $f(x, y) = 0$  makes the vector field vertical on the  $x$ -nullcline, but it does not guarantee that the vector field is tangent to that nullcline. Thus an  $x$ -nullcline need not be invariant. The analogous statement holds for a  $y$ -nullcline.

**Recall: Invariant set**

Let  $\Phi$  be a possibly local flow. A set  $M$  is **invariant** if every solution starting in  $M$  remains in  $M$  throughout its maximal existence interval.

**Definition: Positively invariant set**

A set  $M$  is **positively invariant** if every solution starting at  $\mathbf{x}_0 \in M$  satisfies

$$\Phi(t, \mathbf{x}_0) \in M$$

for every  $t \geq 0$  in its maximal existence interval.

**Question: What can nullclines determine before integration?**

Analyze

$$\dot{x} = x + e^{-y}, \quad \dot{y} = -y.$$

**Solution.** The nullclines are

$$x = -e^{-y}, \quad y = 0.$$

Their unique intersection is the equilibrium  $(-1, 0)$ . The component signs are

| region               | sign $\dot{x}$ | sign $\dot{y}$ |
|----------------------|----------------|----------------|
| $x > -e^{-y}, y > 0$ | +              | −              |
| $x < -e^{-y}, y > 0$ | −              | −              |
| $x > -e^{-y}, y < 0$ | +              | +              |
| $x < -e^{-y}, y < 0$ | −              | +              |

The line  $y = 0$  is invariant because  $\dot{y} = 0$  there. On that line,

$$\dot{x} = x + 1,$$

so solutions move away from  $(-1, 0)$ . Also,

$$y(t) = y_0 e^{-t} \rightarrow 0 \quad (t \rightarrow \infty).$$

Thus  $(-1, 0)$  repels trajectories along the invariant line  $y = 0$ . On the other hand,  $y(t) \rightarrow 0$  as  $t \rightarrow \infty$  for every initial value, so trajectories are attracted toward the line  $y = 0$  in the  $y$ -coordinate.

**Visualization: Nullcline sign map for the worked nonlinear system**

|         | $x < -e^{-y}$                                                                     | $x > -e^{-y}$                                                                     |
|---------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| $y > 0$ |  |  |
| $y < 0$ |  |  |

On  $x = -e^{-y}$ ,

$$y > 0 : \downarrow, \quad y < 0 : \uparrow.$$

On  $y = 0$ ,

$$x < -1 : \leftarrow, \quad x > -1 : \rightarrow.$$

Away from  $x = -e^{-y}$ , trajectory slopes satisfy

$$\frac{dy}{dx} = \frac{-y}{x + e^{-y}}.$$

**Definition: Direction field**

A **direction field** records the direction of  $\mathbf{F}(\mathbf{x})$  at each state where  $\mathbf{F}(\mathbf{x}) \neq \mathbf{0}$ , but suppresses its magnitude.

**Recall: Classical fourth-order Runge–Kutta method**

Let  $\mathbf{x}_n$  approximate a solution of  $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$  at time  $t_n$ , and let  $h > 0$ . The **classical fourth-order Runge–Kutta method** defines

$$\mathbf{k}_1 = \mathbf{F}(\mathbf{x}_n),$$

$$\mathbf{k}_2 = \mathbf{F}(\mathbf{x}_n + \frac{h}{2}\mathbf{k}_1),$$

$$\mathbf{k}_3 = \mathbf{F}(\mathbf{x}_n + \frac{h}{2}\mathbf{k}_2),$$

$$\mathbf{k}_4 = \mathbf{F}(\mathbf{x}_n + h\mathbf{k}_3),$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).$$

**Theorem: Local existence and uniqueness**

Let  $D \subseteq \mathbb{R}^n$  be open and let  $\mathbf{F} : D \rightarrow \mathbb{R}^n$  be continuously differentiable. For every  $\mathbf{x}_0 \in D$ , the initial-value problem

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{x}_0,$$

has a unique solution on some open interval containing 0.

**Corollary: Distinct maximal trajectories do not cross**

Under the hypotheses of the local existence-and-uniqueness theorem, let  $\mathbf{x}_j : I_j \rightarrow D$ ,  $j = 1, 2$ , be maximal solutions, meaning that neither solution extends to a larger time interval. Their trajectory images are either disjoint or identical.

**Proof.** Suppose

$$\mathbf{x}_1(t_1) = \mathbf{x}_2(t_2) = \mathbf{p}.$$

Define

$$\mathbf{y}_1(s) = \mathbf{x}_1(t_1 + s), \quad \mathbf{y}_2(s) = \mathbf{x}_2(t_2 + s).$$

Autonomy makes both solutions pass through  $\mathbf{p}$  at time 0. Uniqueness makes them agree wherever both are defined, so they join to a solution on the union of their shifted time intervals. Maximality forces those intervals to be equal. Thus the original solutions trace the same trajectory, with different time origins.

**Recall: Gradient**

For a continuously differentiable function  $h(x, y)$ , its **gradient** is

$$\nabla h = (h_x, h_y).$$

**Recall: Derivative along a solution**

Let  $h$  be continuously differentiable on an open set containing a solution of  $\dot{x} = f(x, y)$ ,  $\dot{y} = g(x, y)$ . The chain rule gives

$$\frac{d}{dt}h(x(t), y(t)) = h_x f + h_y g = \nabla h \cdot (f, g).$$

**Proposition: Regular invariant-curve test**

Let  $D \subseteq \mathbb{R}^2$  be open, let  $\mathbf{F} : D \rightarrow \mathbb{R}^2$  be locally Lipschitz, and let

$$C = \{(x, y) \in D : h(x, y) = 0\},$$

where  $h \in C^1(D)$  and  $\nabla h \neq \mathbf{0}$  on  $C$ . Suppose

$$\nabla h(x, y) \cdot \mathbf{F}(x, y) = 0 \quad \text{for every } (x, y) \in C.$$

Then  $C$  is invariant under  $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ .

**Proof.** Near any point of  $C$ , the implicit-function theorem writes the curve as  $y = k(x)$ , after interchanging coordinates if needed. Write  $\mathbf{F} = (f, g)$ . Tangency gives

$$k'(x)f(x, k(x)) = g(x, k(x)).$$

Since  $k$  is  $C^1$ , the function  $x \mapsto f(x, k(x))$  is locally Lipschitz. Solve the scalar equation  $\dot{x} = f(x, k(x))$  and set  $y = k(x)$ . The displayed identity gives  $\dot{y} = g(x, y)$ , so this is a solution of the full system lying on  $C$ . Uniqueness makes it the solution through the given point. This argument applies at every point of  $C$ , so a solution cannot leave  $C$  while it remains in  $D$ .

**Question: Can a trajectory cross an invariant circle?**

Consider

$$\dot{x} = -y + x(1 - x^2 - y^2), \quad \dot{y} = x + y(1 - x^2 - y^2).$$

Find the polar equations and classify the unit circle.

**Solution.** Let  $r^2 = x^2 + y^2$ . Then

$$\begin{aligned} \frac{d}{dt}r^2 &= 2x\dot{x} + 2y\dot{y} \\ &= 2(x^2 + y^2)(1 - x^2 - y^2) \\ &= 2r^2(1 - r^2). \end{aligned}$$

For  $r > 0$ ,

$$\dot{r} = r(1 - r^2).$$

Also,

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = 1.$$

Hence

$$0 < r < 1 \Rightarrow \dot{r} > 0, \quad r = 1 \Rightarrow \dot{r} = 0, \quad r > 1 \Rightarrow \dot{r} < 0.$$

The unit circle is an invariant periodic orbit of period  $2\pi$ . For  $r_0 > 0$ , uniqueness keeps  $r(t)$  between  $r_0$  and 1; compact containment gives existence for all  $t \geq 0$ . The radius is monotone, and its limit must be the only positive radial equilibrium,  $r = 1$ . Thus trajectories spiral outward toward the circle from inside and inward toward it from outside, without crossing it.

**Proposition: A closed orbit is a barrier**

Let  $C$  be a simple closed trajectory of a planar system with unique solutions. A trajectory beginning in the bounded component of  $\mathbb{R}^2 \setminus C$  cannot enter the unbounded component while it exists.

**Proof.** By the Jordan curve theorem, every continuous path from the bounded component to the unbounded component meets  $C$ . Such an intersection would make the new trajectory coincide with  $C$  by uniqueness, which is impossible for a trajectory that began off  $C$ .

**Question: Does local existence imply existence for all future time?**

Solve

$$\dot{x} = x^2, \quad x(0) = x_0 > 0.$$

**Solution.** Separation gives

$$-\frac{1}{x(t)} = t + C.$$

The initial condition yields  $C = -1/x_0$ , so

$$x(t) = \frac{x_0}{1 - x_0 t}.$$

The right-hand side of the ODE is smooth on  $\mathbb{R}$ , but the solution becomes unbounded as  $t \uparrow 1/x_0$ . Thus local uniqueness does not imply global existence.

**Theorem: Continuation from compact containment**

Let  $\mathbf{F} : D \rightarrow \mathbb{R}^n$  be locally Lipschitz on an open set  $D$ , and let  $\mathbf{x} : [0, T_{\max}) \rightarrow D$  be a maximal solution. If  $\mathbf{x}([0, T_{\max}))$  lies in a compact subset of  $D$ , then  $T_{\max} = \infty$ .

Equivalently, if  $T_{\max} < \infty$ , then for every compact set  $K \subseteq D$  there exists  $t_K < T_{\max}$  such that

$$\mathbf{x}(t) \notin K \quad \text{whenever } t_K < t < T_{\max}.$$

**Idea: Analytical and numerical phase portraits**

Analytically justified phase-portrait information often comes from

$$\boxed{\text{equilibria}} + \boxed{\text{nullclines and signs}} + \boxed{\text{invariant sets}} + \boxed{\text{uniqueness barriers}}.$$

**Observation: A numerical trajectory is not a proof**

A numerical trajectory is an approximation depending on its initial state, time interval, method, and step size. It can reveal a candidate invariant boundary or closed orbit, but it does not prove invariance, nonintersection, periodicity, or an asymptotic limit.

# Fixed Points, Linearization, and Its Limits

Tuesday, November 3, 2026

## Definition: Jacobian matrix

Let  $\mathbf{F} = (f, g)$  be continuously differentiable near  $\mathbf{z}^* = (x^*, y^*)$ . The **Jacobian matrix of  $\mathbf{F}$  at  $\mathbf{z}^*$**  is

$$D\mathbf{F}(\mathbf{z}^*) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(x^*, y^*)}.$$

## Notation: Displacement from an equilibrium

Let  $\mathbf{z}^*$  be an equilibrium. For a nearby state  $\mathbf{z}$ , write

$$\mathbf{u} = \mathbf{z} - \mathbf{z}^*.$$

## Observation: Taylor expansion at an equilibrium

Let  $\mathbf{F}$  be twice continuously differentiable near an equilibrium  $\mathbf{z}^*$ . Taylor's theorem gives

$$\dot{\mathbf{u}} = J\mathbf{u} + \mathbf{R}(\mathbf{u}), \quad J = D\mathbf{F}(\mathbf{z}^*),$$

where there are constants  $C, \rho > 0$  such that

$$\|\mathbf{R}(\mathbf{u})\| \leq C\|\mathbf{u}\|^2 \quad \text{whenever } \|\mathbf{u}\| < \rho.$$

Equivalently,

$$\mathbf{R}(\mathbf{u}) = O(\|\mathbf{u}\|^2) \quad (\mathbf{u} \rightarrow \mathbf{0}).$$

## Definition: Linearized system

Let  $\mathbf{F}$  be continuously differentiable near an equilibrium  $\mathbf{z}^*$  of  $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$ , and let  $\mathbf{u} = \mathbf{z} - \mathbf{z}^*$ . The **linearized system at  $\mathbf{z}^*$**  is

$$\dot{\mathbf{u}} = D\mathbf{F}(\mathbf{z}^*)\mathbf{u}.$$

**Recall: Hyperbolic equilibrium**

An equilibrium is **hyperbolic** if every eigenvalue  $\lambda$  of its Jacobian satisfies

$$\operatorname{Re} \lambda \neq 0.$$

**Definition: Hyperbolic sink and source**

A hyperbolic equilibrium is a **sink** if every eigenvalue of its Jacobian has negative real part. It is a **source** if every eigenvalue of its Jacobian has positive real part.

**Recall: Homeomorphism**

A **homeomorphism** is a continuous bijection with a continuous inverse.

**Definition: Topological equivalence of phase portraits**

Two local phase portraits are **topologically equivalent** if a homeomorphism maps trajectories of one portrait to trajectories of the other and preserves the direction of time.

**Theorem: Hartman–Grobman theorem**

Let  $\mathbf{F}$  be continuously differentiable near a hyperbolic equilibrium  $\mathbf{z}^*$ . There are neighborhoods of  $\mathbf{z}^*$  and  $\mathbf{0}$  and a homeomorphism between them that maps  $\mathbf{z}^*$  to  $\mathbf{0}$  and maps local trajectory arcs of

$$\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$$

to local trajectory arcs of

$$\dot{\mathbf{u}} = D\mathbf{F}(\mathbf{z}^*)\mathbf{u}.$$

The map preserves the direction along each orbit while the arcs remain in the chosen neighborhoods. Consequently, a hyperbolic sink, source, or saddle has the same local orbit type as its linearization.

**Idea: What linearization preserves**

Hartman–Grobman preserves

local orbit structure

time orientation

sink/source/saddle type.

It does not assert equality of trajectories, preservation of distances, exact exponential rates, or any global basin statement.

**Question: What does linearization say at three equilibria?**

For

$$\dot{x} = -x + x^3, \quad \dot{y} = -2y,$$

find and classify every equilibrium.

**Solution.** Equilibria satisfy

$$x(x^2 - 1) = 0, \quad y = 0,$$

so they are

$$(0, 0), \quad (1, 0), \quad (-1, 0).$$

The Jacobian is

$$J(x, y) = \begin{pmatrix} -1 + 3x^2 & 0 \\ 0 & -2 \end{pmatrix}.$$

At  $(0, 0)$ , the eigenvalues are  $-1, -2$ , so it is an asymptotically stable node. At each of  $(\pm 1, 0)$ , the eigenvalues are  $2, -2$ , so it is a saddle. All three are hyperbolic; therefore the local classifications are valid for the nonlinear system.

**Visualization: Exact product structure of the three-equilibrium system**

The scalar  $x$ -flow is

$$\longleftarrow \circ_{-1} \longrightarrow \bullet_0 \longleftarrow \circ_1 \longrightarrow,$$

and

$$y(t) = y_0 e^{-2t}.$$

The lines

$$x = -1, \quad x = 0, \quad x = 1, \quad y = 0$$

are invariant. In each vertical strip, the horizontal direction is the arrow from the scalar phase line and the vertical direction points toward  $y = 0$ .

**Question: Can cubic terms change a linear center?**

Let  $a \in \mathbb{R}$  and consider

$$\dot{x} = -y + ax(x^2 + y^2), \quad \dot{y} = x + ay(x^2 + y^2).$$

Classify the origin for every  $a$ .

**Solution.** The Jacobian at the origin is

$$J(0, 0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

with eigenvalues  $\pm i$ . The origin is nonhyperbolic, so linearization is inconclusive.

Let  $r^2 = x^2 + y^2$ . For  $r > 0$ ,

$$\begin{aligned} r\dot{r} &= x\dot{x} + y\dot{y} \\ &= a(x^2 + y^2)^2 = ar^4, \end{aligned}$$

and

$$\begin{aligned} r^2\dot{\theta} &= x\dot{y} - y\dot{x} \\ &= x^2 + y^2 = r^2. \end{aligned}$$

**Question: Can cubic terms change a linear center? (continued)**

Thus

$$\dot{r} = ar^3, \quad \dot{\theta} = 1.$$

Therefore

$$\begin{array}{l|l} a < 0 & \text{asymptotically stable spiral} \\ a = 0 & \text{center} \\ a > 0 & \text{unstable spiral} \end{array}$$

even though the linearization is the same in all three cases.

**Question: What is the exact radial behavior?**

Solve

$$\dot{r} = ar^3, \quad r(0) = r_0 > 0.$$

**Solution.** Separation gives

$$\int_{r_0}^{r(t)} r^{-3} dr = \int_0^t a ds,$$

so

$$-\frac{1}{2r(t)^2} + \frac{1}{2r_0^2} = at.$$

Hence

$$r(t) = \frac{r_0}{\sqrt{1 - 2ar_0^2t}}.$$

The maximal interval is the interval containing 0 on which  $1 - 2ar_0^2t > 0$ . If  $a = 0$ , then  $r(t) = r_0$  for every  $t \in \mathbb{R}$ . If  $a < 0$ , then  $r(t) \rightarrow 0$  algebraically as  $t \rightarrow \infty$ . If  $a > 0$ , the solution becomes unbounded at

$$T_{\max} = \frac{1}{2ar_0^2}.$$

Thus the phrase “unstable spiral” is local; for  $a > 0$  the nonzero solution does not exist for all  $t \geq 0$ .

**Visualization: When is linearization decisive?**

| spectrum                                         | status        | next step                                                          |
|--------------------------------------------------|---------------|--------------------------------------------------------------------|
| $\operatorname{Re} \lambda_j \neq 0$ for all $j$ | hyperbolic    | use Hartman–Grobman                                                |
| $\lambda = \pm i\omega$ , $\omega > 0$           | nonhyperbolic | linearization is inconclusive;<br>analyze the nonlinear system     |
| 0 is an eigenvalue                               | nonhyperbolic | analyze the nonlinear dynamics<br>in the zero-eigenvalue direction |

**Observation: Repeated real eigenvalues can be hyperbolic**

A planar equilibrium whose Jacobian has the repeated real eigenvalue  $\lambda \neq 0$  is hyperbolic. Whether the eigenspace is one- or two-dimensional changes the linear geometry, but the sign of  $\lambda$  still determines whether the equilibrium is a sink or a source.

**Definition: Locally invariant graph**

For  $\dot{u} = F(u, v)$ ,  $\dot{v} = G(u, v)$  and a differentiable function  $h : I \rightarrow \mathbb{R}$  on an interval  $I \subseteq \mathbb{R}$ , the graph

$$\Gamma = \{(u, h(u)) : u \in I\}$$

is **locally invariant near a point**  $p \in \Gamma$  if there is an open neighborhood  $N$  of  $p$  such that every solution starting in  $\Gamma \cap N$  remains in  $\Gamma$  for as long as it remains in  $N$ .

**Proposition: Invariant graph equation**

If the differentiable graph  $v = h(u)$  is locally invariant near every point of the graph for  $\dot{u} = F(u, v)$ ,  $\dot{v} = G(u, v)$ , then

$$h'(u)F(u, h(u)) = G(u, h(u)) \quad (u \in I).$$

**Proof.** Along the graph, the chain rule gives  $\dot{v} = h'(u)\dot{u}$ . Substitute the differential equations.

**Question: How does a nonlinear term bend an invariant direction?**

Consider

$$\dot{x} = x - xy, \quad \dot{y} = -2y + x^2.$$

Classify the origin. Assuming its local unstable invariant curve is  $C^3$  and tangent to the  $x$ -axis, find its quadratic term.

**Solution.**

$$J(x, y) = \begin{pmatrix} 1-y & -x \\ 2x & -2 \end{pmatrix}, \quad J(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}.$$

The eigenvalues are 1 and  $-2$ , so the origin is a hyperbolic saddle. Since  $x = 0$  implies  $\dot{x} = 0$ , the  $y$ -axis is the stable invariant curve.

The assumed  $C^3$  regularity and tangency give

$$\begin{aligned} y &= h(x) = cx^2 + O(x^3), \\ h'(x) &= 2cx + O(x^2) \end{aligned} \quad (x \rightarrow 0).$$

Substituting into the invariance equation gives

$$\begin{aligned} h'(x)x(1-h(x)) &= -2h(x) + x^2, \\ 2cx^2 + O(x^3) &= (1-2c)x^2 + O(x^3) \end{aligned} \quad (x \rightarrow 0).$$

Comparing quadratic coefficients,

$$2c = 1 - 2c, \quad c = \frac{1}{4}, \quad y = \frac{1}{4}x^2 + O(x^3) \quad (x \rightarrow 0).$$

# Competition, Separatrices, and Basins

Thursday, November 5, 2026

## Definition: Two-species competition model

Let  $x(t) \geq 0$  and  $y(t) \geq 0$  denote two competing populations. The model

$$\dot{x} = x(3 - x - 2y), \quad \dot{y} = y(2 - x - y)$$

is a **two-species competition model**: each species has self-limited growth and inhibits the growth of the other species.

## Notation: Population state space

The state space for the competition model is the closed first quadrant

$$Q = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}.$$

## Recall: Positive invariance

A set is **positively invariant** if every solution starting in it remains in it for all later times for which the solution exists.

## Proposition: Nonnegative populations remain nonnegative

The first quadrant  $Q$  is positively invariant.

**Proof.** On  $x = 0$ ,  $\dot{x} = 0$ , so the  $y$ -axis is invariant. On  $y = 0$ ,  $\dot{y} = 0$ , so the  $x$ -axis is invariant. By uniqueness, a trajectory that starts in  $Q$  cannot cross either invariant axis.

**Question: What are the nullclines and equilibria?**

Find every nullcline and equilibrium in  $Q$ .

**Solution.**

$$\dot{x} = 0 \iff x = 0 \text{ or } x + 2y = 3,$$

$$\dot{y} = 0 \iff y = 0 \text{ or } x + y = 2.$$

The intersections in  $Q$  are

$$(0, 0), \quad (3, 0), \quad (0, 2), \quad (1, 1).$$

Moreover,

$$\text{sign } \dot{x} = \text{sign}(3 - x - 2y) \quad (x > 0),$$

$$\text{sign } \dot{y} = \text{sign}(2 - x - y) \quad (y > 0).$$

**Visualization: Nullcline sign regions**

In the open first quadrant,

| inequalities            | $(\text{sign } \dot{x}, \text{sign } \dot{y})$ | direction                                                                             |
|-------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| $x + 2y < 3, x + y < 2$ | $(+, +)$                                       |    |
| $x + 2y > 3, x + y < 2$ | $(-, +)$                                       |    |
| $x + 2y < 3, x + y > 2$ | $(+, -)$                                       |   |
| $x + 2y > 3, x + y > 2$ | $(-, -)$                                       |  |

The oblique nullclines have intercepts

$$x + 2y = 3 : (3, 0), (0, 3/2),$$

$$x + y = 2 : (2, 0), (0, 2),$$

and cross at  $(1, 1)$ .

**Question: How is each equilibrium classified?**

Compute the Jacobian, eigenvalues, and eigendirections at all four equilibria.

**Solution.**

$$J(x, y) = \begin{pmatrix} 3 - 2x - 2y & -2x \\ -y & 2 - x - 2y \end{pmatrix}.$$

At  $(0, 0)$ ,

$$J(0, 0) = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix},$$

with eigenpairs

$$\lambda = 3, \mathbf{v} = (1, 0), \quad \lambda = 2, \mathbf{v} = (0, 1).$$

Thus the equilibrium is an unstable node.

At  $(0, 2)$ ,

$$J(0, 2) = \begin{pmatrix} -1 & 0 \\ -2 & -2 \end{pmatrix},$$

**Question: How is each equilibrium classified? (continued)**

with eigenpairs

$$\lambda = -1, \mathbf{v} = (1, -2), \quad \lambda = -2, \mathbf{v} = (0, 1).$$

Thus  $(0, 2)$  is an asymptotically stable node.

At  $(3, 0)$ ,

$$J(3, 0) = \begin{pmatrix} -3 & -6 \\ 0 & -1 \end{pmatrix},$$

with eigenpairs

$$\lambda = -1, \mathbf{v} = (3, -1), \quad \lambda = -3, \mathbf{v} = (1, 0).$$

Thus  $(3, 0)$  is an asymptotically stable node.

At  $(1, 1)$ ,

$$J(1, 1) = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix},$$

with eigenpairs

$$\lambda_s = -1 - \sqrt{2}, \quad \mathbf{v}_s = (\sqrt{2}, 1),$$

$$\lambda_u = -1 + \sqrt{2}, \quad \mathbf{v}_u = (\sqrt{2}, -1).$$

Thus  $(1, 1)$  is a saddle. Its stable direction has slope  $1/\sqrt{2}$  and its unstable direction has slope  $-1/\sqrt{2}$ .

**Theorem: Local invariant manifolds of a hyperbolic saddle**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $\mathbf{F} : U \rightarrow \mathbb{R}^2$  be continuously differentiable, and let  $p \in U$  be a hyperbolic saddle. There is a neighborhood  $N \subseteq U$  of  $p$  and unique continuously differentiable curves

$$W_{\text{loc}}^s(p), \quad W_{\text{loc}}^u(p)$$

through  $p$ , tangent at  $p$  to the stable and unstable eigenspaces of  $D\mathbf{F}(p)$ , respectively. They are characterized by

$$W_{\text{loc}}^s(p) = \{\mathbf{z}_0 \in N : \Phi_t(\mathbf{z}_0) \text{ is defined and belongs to } N \text{ for every } t \geq 0, \Phi_t(\mathbf{z}_0) \rightarrow p \text{ as } t \rightarrow \infty\},$$

$$W_{\text{loc}}^u(p) = \{\mathbf{z}_0 \in N : \Phi_t(\mathbf{z}_0) \text{ is defined and belongs to } N \text{ for every } t \leq 0, \Phi_t(\mathbf{z}_0) \rightarrow p \text{ as } t \rightarrow -\infty\}.$$

**Recall: Basin of attraction**

Let  $p$  be an attracting equilibrium in a state space  $D$ . Its basin of **attraction** relative to  $D$  is

$$\mathcal{B}_D(p) = \left\{ \mathbf{z}_0 \in D : \begin{array}{l} \Phi_t(\mathbf{z}_0) \text{ exists for every } t \geq 0, \\ \Phi_t(\mathbf{z}_0) \rightarrow p \text{ as } t \rightarrow \infty \end{array} \right\}.$$

**Definition: Relative closure**

Let  $A \subseteq D \subseteq \mathbb{R}^2$ . The **closure of  $A$  relative to  $D$**  is

$$\overline{A}^D = \left\{ p \in D : \begin{array}{l} \text{for every } \varepsilon > 0 \text{ there is a } q \in A \\ \text{such that } \|q - p\| < \varepsilon \end{array} \right\}.$$

**Definition: Relative boundary**

Let  $A \subseteq D \subseteq \mathbb{R}^2$ . The **boundary of  $A$  relative to  $D$**  is

$$\partial_D A = \overline{A}^D \cap \overline{D \setminus A}^D,$$

where both closures are taken relative to  $D$ .

**Definition: Basin boundary**

Let  $\mathcal{B}_D(p)$  be a basin of attraction in a state space  $D$ . Its **basin boundary** is its relative topological boundary

$$\partial_D \mathcal{B}_D(p).$$

**Definition: Basin separatrix**

Let  $p$  and  $q$  be distinct attracting equilibria in  $D$ . An invariant curve  $S \subseteq D$  contained in

$$\partial_D \mathcal{B}_D(p) \cap \partial_D \mathcal{B}_D(q)$$

is a **basin separatrix** for  $p$  and  $q$ .

**Proposition: A compact forward-invariant rectangle**

Let  $(x_0, y_0) \in Q$  and define

$$M_x = \max\{x_0, 3\}, \quad M_y = \max\{y_0, 2\}.$$

Then

$$R = [0, M_x] \times [0, M_y]$$

is positively invariant. Consequently, the solution exists for all  $t \geq 0$  and remains bounded.

**Proof.** On the right edge  $x = M_x$ ,

$$\dot{x} = M_x(3 - M_x - 2y) \leq 0.$$

On the top edge  $y = M_y$ ,

$$\dot{y} = M_y(2 - x - M_y) \leq 0.$$

The coordinate axes are invariant, so the vector field never points out through the remaining edges. Hence  $R$  is positively invariant. The vector field is smooth and the trajectory remains in the compact set  $R$ , so the continuation theorem gives existence for all forward time.

**Recall: Divergence of a planar vector field**

For a continuously differentiable vector field  $\mathbf{G} = (P, Q)$ , its **divergence** is

$$\operatorname{div} \mathbf{G} = \nabla \cdot \mathbf{G} = P_x + Q_y.$$

**Recall: Green's flux formula**

Let a simple closed, piecewise smooth curve  $C$  bound a region  $U$ , and let  $\mathbf{G}$  be continuously differentiable on a neighborhood of  $U \cup C$ . Then

$$\int_C \mathbf{G} \cdot \mathbf{n} \, ds = \iint_U \operatorname{div} \mathbf{G} \, dA,$$

where  $\mathbf{n}$  is the outward unit normal,  $ds$  is arc length, and  $dA$  is area. The left-hand side is the **outward flux**.

**Proposition: No periodic orbit lies in the open first quadrant**

The competition model has no nonconstant periodic trajectory contained in

$$Q^\circ = \{(x, y) : x > 0, y > 0\}.$$

**Proof.** Define

$$B(x, y) = \frac{1}{xy}$$

on  $Q^\circ$ . If  $\mathbf{F}$  is the competition vector field, then

$$B\mathbf{F} = \left( \frac{3 - x - 2y}{y}, \frac{2 - x - y}{x} \right)$$

and

$$\operatorname{div}(B\mathbf{F}) = -\frac{1}{y} - \frac{1}{x} < 0.$$

By uniqueness, a nonconstant periodic trajectory  $C \subset Q^\circ$  would be a simple closed curve. Its interior  $U$  also lies in the convex set  $Q^\circ$ . Since  $B\mathbf{F}$  is tangent to  $C$ , Green's flux formula gives

$$0 = \int_C B\mathbf{F} \cdot \mathbf{n} \, ds = \iint_U \operatorname{div}(B\mathbf{F}) \, dA < 0,$$

a contradiction.

**Observation: What the trapping calculations do not prove**

The compact forward-invariant rectangle and the exclusion of periodic orbits do not, by themselves, prove that every solution converges to an equilibrium or determine the global shape of the saddle's stable manifold.

**Visualization: Schematic separatrix and basin portrait**

Let

$$\Sigma = W^s((1, 1)) \cap Q, \quad S = \bar{\Sigma}^Q.$$

At  $(1, 1)$ ,  $\Sigma$  is tangent to  $(\sqrt{2}, 1)$  and has positive slope. The schematic global portrait is

| schematic initial position                       | schematic forward limit |
|--------------------------------------------------|-------------------------|
| component of $Q \setminus S$ containing $(3, 0)$ | $(3, 0)$                |
| on $\Sigma$                                      | $(1, 1)$                |
| $(0, 0)$                                         | $(0, 0)$                |
| component of $Q \setminus S$ containing $(0, 2)$ | $(0, 2)$                |

Here  $\Sigma$  is the proposed basin separatrix and  $S$  the proposed common basin boundary. These global claims require more than the local, forward-invariance, and no-cycle calculations above.

**Observation: Dynamics on the coordinate axes**

The axes contain the logistic flows

$$y = 0 : \dot{x} = x(3 - x), \quad x = 0 : \dot{y} = y(2 - y).$$

**Question: Does one local velocity determine the surviving species?**

Compute the vector field at  $(x, y) = (1/2, 1/2)$ .

**Solution.**

$$\dot{x} = \frac{1}{2} \left( 3 - \frac{1}{2} - 1 \right) = \frac{3}{4}, \quad \dot{y} = \frac{1}{2} \left( 2 - \frac{1}{2} - \frac{1}{2} \right) = \frac{1}{2}.$$

The initial motion is northeast. Its forward limit requires global information, such as its position relative to  $\Sigma = W^s((1, 1)) \cap Q$ , not just this one velocity.

**Visualization: Local data and global data**

| local data                      | global data         |
|---------------------------------|---------------------|
| $J(p)$                          | $\mathcal{B}_Q(p)$  |
| eigenvalues and eigendirections | $W^s((1, 1))$       |
| nullcline crossing direction    | separatrix location |

The Jacobian proves local attraction to each sink; it does not determine which sink attracts a distant state.

# Conservation, Energy Geometry, and Reversibility

Tuesday, November 10, 2026

## Definition: First integral

Let  $U \subseteq \mathbb{R}^2$  be open and let  $\mathbf{F} : U \rightarrow \mathbb{R}^2$  be continuously differentiable. A continuously differentiable function  $E : U \rightarrow \mathbb{R}$  is a **first integral** of  $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$  if

$$\nabla E(\mathbf{z}) \cdot \mathbf{F}(\mathbf{z}) = 0$$

for every  $\mathbf{z} \in U$ . Equivalently,  $E(\mathbf{z}(t))$  is constant along every solution.

## Definition: Conservative system

A system on an open set  $U$  is **conservative** if it has a first integral that is not constant on any nonempty open subset of  $U$ .

## Question: Why is mechanical energy conserved?

Let  $I \subseteq \mathbb{R}$  be an open interval, let  $V \in C^2(I)$ , let  $m > 0$ , and suppose  $x(t) \in I$  satisfies

$$m\ddot{x} = F(x), \quad F(x) = -V'(x).$$

Find a first integral of the planar system obtained with  $y = \dot{x}$ .

**Solution.** The planar system is

$$\dot{x} = y, \quad \dot{y} = -\frac{1}{m}V'(x).$$

Multiplying the second-order equation by  $\dot{x}$  gives

$$m\dot{x}\ddot{x} + V'(x)\dot{x} = \frac{d}{dt} \left( \frac{1}{2}m\dot{x}^2 + V(x) \right) = 0.$$

Thus

$$E(x, y) = \frac{1}{2}my^2 + V(x)$$

is conserved. Directly,

$$\mathbf{G}(x, y) = \left( y, -\frac{1}{m}V'(x) \right), \quad \nabla E \cdot \mathbf{G} = (V'(x), my) \cdot \left( y, -\frac{1}{m}V'(x) \right) = 0.$$

**Proposition: A conservative system has no attracting equilibrium**

A conservative system has no attracting equilibrium.

**Proof.** Let  $E$  be a first integral that is not constant on any nonempty open set. Suppose  $\mathbf{z}^*$  is attracting. Its basin contains a nonempty open neighborhood of  $\mathbf{z}^*$ . For any  $\mathbf{z}_0$  in the basin, conservation and continuity give

$$E(\mathbf{z}_0) = E(\Phi_t(\mathbf{z}_0)) \longrightarrow E(\mathbf{z}^*) \quad (t \rightarrow \infty).$$

Hence  $E(\mathbf{z}_0) = E(\mathbf{z}^*)$  for every  $\mathbf{z}_0$  in the basin. Thus  $E$  is constant on a nonempty open set, contradicting the definition of a conservative system.

**Visualization: Potential and phase-plane energy level**

Let  $I \subseteq \mathbb{R}$  be an interval, let  $m > 0$ , let  $V \in C^2(I)$ , and consider

$$\dot{x} = y, \quad \dot{y} = -\frac{1}{m}V'(x), \quad E(x, y) = \frac{1}{2}my^2 + V(x).$$

For

$$E(x, y) = \frac{1}{2}my^2 + V(x) = h,$$

the two velocity branches are

$$y = \pm \sqrt{\frac{2}{m}(h - V(x))}.$$

Therefore

| potential condition         | phase-plane meaning                     |
|-----------------------------|-----------------------------------------|
| $V(x) < h$                  | two states with velocities $\pm  y $    |
| $V(x) = h, V'(x) \neq 0$    | $y = 0$ ; non-equilibrium turning point |
| $V(x) = h, V'(x) = 0$       | equilibrium                             |
| $V(x) > h$                  | no state on the energy level            |
| $V'(x^*) = 0, V''(x^*) > 0$ | nearby oval levels about $(x^*, 0)$     |
| $V'(x^*) = 0, V''(x^*) < 0$ | saddle energy at $(x^*, 0)$             |

Since  $y = \dot{x}$ , the upper branch moves right and the lower branch moves left.

**Definition: Regular level**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $E \in C^1(U)$ , and let  $h \in \mathbb{R}$ . The level set

$$\{p \in U : E(p) = h\}$$

is **regular** if

$$\nabla E(p) \neq \mathbf{0}$$

at every point  $p$  in that level set.

**Recall: Hessian matrix**

For  $E \in C^2(U)$  on an open set  $U \subseteq \mathbb{R}^2$ , its **Hessian matrix** is

$$D^2E = \begin{pmatrix} E_{xx} & E_{xy} \\ E_{yx} & E_{yy} \end{pmatrix}.$$

The matrix is symmetric because  $E_{xy} = E_{yx}$ .

**Recall: Definite symmetric matrix**

A real symmetric matrix  $H$  is **positive definite** if  $\mathbf{v}^\top H \mathbf{v} > 0$  for every nonzero vector  $\mathbf{v}$ . It is **negative definite** if the inequality is reversed. Here  $\top$  denotes transpose.

**Lemma: Definite Hessian gives closed nearby levels**

Let  $E$  be twice continuously differentiable near  $\mathbf{z}^* \in \mathbb{R}^2$ , suppose

$$\nabla E(\mathbf{z}^*) = \mathbf{0},$$

and suppose  $D^2E(\mathbf{z}^*)$  is positive definite. Then  $\mathbf{z}^*$  is a strict local minimum, and for all sufficiently small  $c > 0$  the nearby component of

$$E(\mathbf{z}) = E(\mathbf{z}^*) + c$$

is a regular simple closed curve surrounding  $\mathbf{z}^*$ . For a negative definite Hessian, the same conclusion holds with  $c < 0$ .

**Proof.** Assume the Hessian is positive definite. By continuity, there are a radius  $R > 0$  and a constant  $m > 0$  such that

$$m\|\mathbf{v}\|^2 \leq \mathbf{v}^\top D^2E(\mathbf{z})\mathbf{v}$$

for every  $\mathbf{v} \in \mathbb{R}^2$  whenever  $\|\mathbf{z} - \mathbf{z}^*\| \leq R$ . For each unit vector  $\mathbf{e} \in \mathbb{R}^2$ , define

$$q_{\mathbf{e}}(r) = E(\mathbf{z}^* + r\mathbf{e}) - E(\mathbf{z}^*).$$

Because  $\nabla E(\mathbf{z}^*) = \mathbf{0}$ ,

$$q'_{\mathbf{e}}(r) = \int_0^r \mathbf{e}^\top D^2E(\mathbf{z}^* + s\mathbf{e})\mathbf{e} \, ds,$$

so, for  $0 < r \leq R$ ,

$$q'_{\mathbf{e}}(r) \geq mr > 0, \quad q_{\mathbf{e}}(r) \geq \frac{m}{2}r^2.$$

The second inequality makes  $\mathbf{z}^*$  a strict local minimum. The first makes the energy strictly increase along each ray. Thus, for every  $0 < c < mR^2/2$  and every unit vector  $\mathbf{e}$ , there is a unique radius  $r_c(\mathbf{e}) \in (0, R)$  satisfying

$$E(\mathbf{z}^* + r_c(\mathbf{e})\mathbf{e}) = E(\mathbf{z}^*) + c.$$

The implicit-function theorem makes  $r_c$  continuously differentiable in the direction  $\mathbf{e}$ . As  $\mathbf{e}$  runs around the unit circle,

$$\mathbf{e} \mapsto \mathbf{z}^* + r_c(\mathbf{e})\mathbf{e}$$

parametrizes a simple closed curve surrounding  $\mathbf{z}^*$ . Moreover,  $q'_{\mathbf{e}}(r_c(\mathbf{e})) > 0$ , so  $\nabla E$  does not vanish on this curve; the level is regular. Replacing  $E$  by  $-E$  proves the negative-definite case.  $\square$

**Proposition: Regular closed energy levels are periodic orbits**

Let  $E$  be a continuously differentiable first integral of a continuously differentiable planar vector field  $\mathbf{F}$ . Let  $C$  be a regular simple closed component of a level set of  $E$ . If  $C$  contains no equilibrium, then  $C$  is a periodic orbit.

**Proof.** Conservation keeps trajectories on  $C$ , and compactness permits continuation for all time. Since  $C$  contains no equilibrium, the speed  $\|\mathbf{F}\|$  has a positive minimum  $m$  on  $C$ . The direction of travel cannot reverse without the vector field vanishing. If  $L$  is the length of  $C$ , one circuit takes time

$$T = \int_C \frac{ds}{\|\mathbf{F}\|} \leq \frac{L}{m} < \infty.$$

Uniqueness makes the returning trajectory periodic.

**Definition: Nonlinear center**

An isolated equilibrium  $\mathbf{z}^*$  of a planar system is a **nonlinear center** if every sufficiently nearby non-equilibrium point lies on a periodic orbit that is a simple closed curve surrounding  $\mathbf{z}^*$ .

**Theorem: Definite-Hessian nonlinear-center criterion**

Let  $\mathbf{F}$  be continuously differentiable near an isolated equilibrium  $\mathbf{z}^*$  of a planar system, and let  $E$  be a twice continuously differentiable first integral. Suppose

$$\nabla E(\mathbf{z}^*) = \mathbf{0}$$

and  $D^2E(\mathbf{z}^*)$  is positive definite or negative definite. Then every non-equilibrium point sufficiently near  $\mathbf{z}^*$  lies on a periodic orbit that is a simple closed level curve of  $E$  surrounding  $\mathbf{z}^*$ . Hence  $\mathbf{z}^*$  is a nonlinear center.

**Proof.** Assume the Hessian is positive definite; otherwise replace  $E$  by  $-E$ . Since  $\mathbf{z}^*$  is isolated, apply the preceding lemma in a disk containing no other equilibrium. Every non-equilibrium point sufficiently near  $\mathbf{z}^*$  lies on one of the lemma's regular simple closed energy curves in that disk. The preceding proposition makes each curve a periodic orbit.

**Question: Why is a non-strict local minimum insufficient?**

Consider

$$\dot{x} = 0, \quad \dot{y} = x^2 + y^2, \quad E(x, y) = x^2.$$

Verify that the origin is an isolated equilibrium and a local minimum of a first integral that is not constant on any nonempty open set, but is not a center.

**Solution.** The equilibrium equations are

$$0 = 0, \quad x^2 + y^2 = 0,$$

so the origin is the unique equilibrium. Also,

$$\dot{E} = 2x\dot{x} = 0,$$

**Question: Why is a non-strict local minimum insufficient? (continued)**

and  $E = x^2 \geq 0 = E(0, 0)$ . The minimum is not strict, and the nearby levels  $x^2 = c$  are pairs of vertical lines rather than closed curves. At every non-equilibrium point,

$$\dot{y} = x^2 + y^2 > 0,$$

so no non-equilibrium trajectory is periodic. The missing hypothesis is the closed-contour condition; a definite Hessian would guarantee it.

Finally,  $E$  is not constant on any nonempty open set: every open disk contains a horizontal line segment on which  $x^2$  is not constant.

**Definition: Homoclinic trajectory**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $\mathbf{F} : U \rightarrow \mathbb{R}^2$ , and let  $p \in U$  be an equilibrium of  $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$ . A nonconstant solution  $\mathbf{z} : \mathbb{R} \rightarrow U$  is **homoclinic to**  $p$  if

$$\lim_{t \rightarrow -\infty} \mathbf{z}(t) = p = \lim_{t \rightarrow \infty} \mathbf{z}(t).$$

Its image is a **homoclinic trajectory** or **homoclinic orbit**.

**Question: What is the complete phase portrait of a double well?**

Consider

$$\ddot{x} = x - x^3.$$

Find the equilibria, conserved energy, local classifications, and energy-level regimes.

**Solution.** With  $y = \dot{x}$ ,

$$\dot{x} = y, \quad \dot{y} = x - x^3.$$

Choose

$$V(x) = -\frac{1}{2}x^2 + \frac{1}{4}x^4, \quad E(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

The equilibria are

$$(0, 0), \quad (1, 0), \quad (-1, 0).$$

The Jacobian is

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ 1 - 3x^2 & 0 \end{pmatrix}.$$

At  $(0, 0)$ , the eigenvalues are  $\pm 1$ , so it is a saddle. At  $(\pm 1, 0)$ , the eigenvalues are  $\pm i\sqrt{2}$ , and

$$D^2E(\pm 1, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive definite. Hence  $(\pm 1, 0)$  are nonlinear centers.

Their energy is  $-1/4$  and the saddle energy is 0. On  $E = h$ ,

$$y = \pm \sqrt{2h + x^2 - \frac{1}{2}x^4}.$$

**Question: What is the complete phase portrait of a double well? (continued)**

Therefore

| $h$            | level-set geometry                                     |
|----------------|--------------------------------------------------------|
| $h < -1/4$     | empty                                                  |
| $h = -1/4$     | the two center equilibria                              |
| $-1/4 < h < 0$ | two disjoint periodic ovals                            |
| $h = 0$        | the saddle together with two homoclinic-orbit closures |
| $h > 0$        | one periodic oval surrounding all three equilibria     |

The two nonconstant homoclinic trajectories on  $h = 0$  are not periodic because approach to the saddle takes infinite time.

**Visualization: Energy surface for the double well**

For

$$w = E(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

the critical points and heights are

| $(x, y)$  | $E(x, y)$ | surface type |
|-----------|-----------|--------------|
| $(-1, 0)$ | $-1/4$    | minimum      |
| $(1, 0)$  | $-1/4$    | minimum      |
| $(0, 0)$  | $0$       | saddle       |

Horizontal slices satisfy

|                |                                  |
|----------------|----------------------------------|
| $h < -1/4$     | empty                            |
| $h = -1/4$     | the two minima                   |
| $-1/4 < h < 0$ | two ovals                        |
| $h = 0$        | figure-eight saddle-energy level |
| $h > 0$        | one outer oval                   |

and project to the corresponding phase-plane energy levels.

**Definition: Heteroclinic trajectory**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $\mathbf{F} : U \rightarrow \mathbb{R}^2$ , and let  $p, q \in U$  be distinct equilibria of  $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$ . A solution  $\mathbf{z} : \mathbb{R} \rightarrow U$  is **heteroclinic from  $p$  to  $q$**  if

$$\lim_{t \rightarrow -\infty} \mathbf{z}(t) = p, \quad \lim_{t \rightarrow \infty} \mathbf{z}(t) = q.$$

Its image is a **heteroclinic trajectory** or **heteroclinic orbit**.

**Observation: Saddle connections are not periodic**

Every homoclinic or heteroclinic solution has an equilibrium limit as  $t \rightarrow \infty$ . A periodic solution that has a limit as  $t \rightarrow \infty$  must be constant: periodicity repeats every point on its trajectory arbitrarily far in the future, so every such point must equal the limit. A homoclinic solution is nonconstant by definition, and a heteroclinic solution has two distinct limiting equilibria. Therefore neither trajectory is periodic.

**Recall: Involution**

Let  $U$  be a set. A map  $R : U \rightarrow U$  is an **involution** if

$$R(R(\mathbf{z})) = \mathbf{z}.$$

**Recall: Reversing symmetry**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $\mathbf{F} : U \rightarrow \mathbb{R}^2$ , and let  $R : U \rightarrow U$  be a differentiable involution. The map  $R$  is a **reversing symmetry** for  $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$  if

$$\mathbf{F}(R(\mathbf{z})) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z}).$$

**Observation: A reverser reverses solution arcs**

If  $R$  is a reversing symmetry and  $\mathbf{z}(t)$  is a solution, then  $R(\mathbf{z}(-t))$  is also a solution.

**Observation: Reflection-reversibility test**

For reflection across the  $x$ -axis,

$$R(x, y) = (x, -y).$$

Writing  $\mathbf{F} = (f, g)$ , reversibility becomes

$$f(x, -y) = -f(x, y), \quad g(x, -y) = g(x, y).$$

**Proposition: Undamped one-dimensional mechanics is reversible**

Let  $I \subseteq \mathbb{R}$  be open, let  $m > 0$ , and let  $F : I \rightarrow \mathbb{R}$  be continuous. The system on  $I \times \mathbb{R}$

$$\dot{x} = y, \quad \dot{y} = \frac{1}{m}F(x)$$

is reversible under  $R(x, y) = (x, -y)$ .

**Proof.** Here

$$\mathbf{F}_{\text{sys}}(x, y) = \left(y, \frac{1}{m}F(x)\right), \quad DR = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore

$$\mathbf{F}_{\text{sys}}(R(x, y)) = \left(-y, \frac{1}{m}F(x)\right) = -DR\mathbf{F}_{\text{sys}}(x, y).$$

**Theorem: Nonlinear centers in reflection-reversible systems**

Let  $U \subseteq \mathbb{R}^2$  be a reflection-invariant open neighborhood of the origin, and let  $\mathbf{F} : U \rightarrow \mathbb{R}^2$  be continuously differentiable. Suppose the origin is an equilibrium, the system is reversible under  $R(x, y) = (x, -y)$ , and  $D\mathbf{F}(0, 0)$  has eigenvalues  $\pm i\omega$  with  $\omega > 0$ . Then every sufficiently small non-equilibrium trajectory is closed; the origin is a nonlinear center.

**Proof.** Reversibility forces the linearization to have the form

$$J = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}, \quad ab = -\omega^2 < 0.$$

After reversing time if necessary, take  $a > 0 > b$ . On a punctured neighborhood of the origin, define polar variables by

$$r^2 = x^2 + y^2, \quad x = r \cos \theta, \quad y = r \sin \theta,$$

choosing  $\theta$  continuously along each solution. Then

$$\dot{\theta} = \frac{xy - yx}{r^2} = \frac{bx^2 - ay^2 + o(r^2)}{r^2} \quad (r \rightarrow 0).$$

Since  $a > 0 > b$ , choose a disk of radius  $R > 0$  and constants  $c, C > 0$  such that, for  $0 < r \leq R$ ,

$$\dot{\theta} \leq -c < 0, \quad \|\mathbf{F}(\mathbf{z})\| \leq C\|\mathbf{z}\|.$$

The angular bound also shows that there is no other equilibrium in the disk. An angular change of  $2\pi$  takes at most  $2\pi/c$ , provided the trajectory stays in the disk. To justify that, use

$$\left| \frac{d}{dt} \log r \right| = \left| \frac{\mathbf{z} \cdot \mathbf{F}(\mathbf{z})}{r^2} \right| \leq C.$$

Thus, while the trajectory is in the disk,

$$r(t) \leq r(0)e^{Ct} \quad (t \geq 0).$$

If  $r(0) < Re^{-2\pi C/c}$ , this bound prevents exit before time  $2\pi/c$ ; compactness gives continuation up to that time. The trajectory therefore reaches two consecutive crossings of  $y = 0$ , separated by an angle of  $\pi$ . Reflecting the arc between the crossings and reversing its arrows produces the continuation from the second crossing back to the first. Uniqueness closes the orbit and prevents self-intersections. The loop surrounds the origin because its angle changes by  $2\pi$ . Uniqueness also places the original initial point on this same periodic orbit.

**Question: Does reversibility turn a linear center into a nonlinear center?**

Consider

$$\dot{x} = y - y^3, \quad \dot{y} = -x - y^2.$$

Find the equilibria and classify them.

**Solution.** The first component is odd in  $y$ , and the second is even in  $y$ , so  $R(x, y) = (x, -y)$  is a reverser. At the origin,

$$D\mathbf{F}(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

whose eigenvalues are  $\pm i$ . The reversible-center theorem makes the origin a nonlinear center. The equilibrium equations are

$$y(1 - y^2) = 0, \quad -x - y^2 = 0.$$

Thus the other equilibria are  $(-1, 1)$  and  $(-1, -1)$ . Since

$$D\mathbf{F}(x, y) = \begin{pmatrix} 0 & 1 - 3y^2 \\ -1 & -2y \end{pmatrix},$$

the determinant at either point is  $-2 < 0$ . Both are saddles.

**Question: Where is the homoclinic loop?**

Consider

$$\dot{x} = y, \quad \dot{y} = x - x^2.$$

Find an exact energy equation for the homoclinic loop in  $x \geq 0$ .

**Solution.** The origin is a saddle, and

$$E(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{3}x^3$$

is conserved because

$$\dot{E} = y(x - x^2) + (-x + x^2)y = 0.$$

The saddle energy is  $E = 0$ . Hence the upper branch satisfies

$$y = x\sqrt{1 - \frac{2x}{3}}, \quad 0 < x \leq \frac{3}{2},$$

and reaches the  $x$ -axis at  $(3/2, 0)$ . Reflection across the  $x$ -axis with time reversal gives the lower branch

$$y = -x\sqrt{1 - \frac{2x}{3}}, \quad 0 < x \leq \frac{3}{2}.$$

The two branches approach the origin as  $t \rightarrow \pm\infty$  and form a homoclinic loop. More precisely, the upper branch leaves the origin as  $t \rightarrow -\infty$  and reaches  $(3/2, 0)$  in finite time; the solution then follows the lower branch and approaches the origin as  $t \rightarrow +\infty$ .

**Question: Does reversibility imply conservation?**

Consider

$$\dot{x} = -2 \cos x - \cos y, \quad \dot{y} = -2 \cos y - \cos x$$

on  $\mathbb{R}^2$ . The vector field is  $2\pi$ -periodic in each coordinate. Show that the system is reversible but not conservative.

**Solution.** With  $R(x, y) = (-x, -y)$ ,  $DR = -I$ , and cosine even,

$$\mathbf{F}(R\mathbf{z}) = \mathbf{F}(\mathbf{z}) = -DR\mathbf{F}(\mathbf{z}),$$

so the system is reversible. In the representative cell  $[-\pi, \pi) \times [-\pi, \pi)$ , its equilibria are

$$(\sigma\pi/2, \rho\pi/2), \quad \sigma, \rho \in \{-1, 1\}.$$

The Jacobian is

$$D\mathbf{F}(x, y) = \begin{pmatrix} 2 \sin x & \sin y \\ \sin x & 2 \sin y \end{pmatrix}.$$

At  $(-\pi/2, -\pi/2)$ , the eigenvalues are  $-3, -1$ , so it is an asymptotically stable node and therefore has an open attracting basin. A conservative system cannot have an open attracting basin. Therefore this reversible system is not conservative.

**Visualization: Conservation and reversibility**

| structure      | identity                                                          | consequence                         |
|----------------|-------------------------------------------------------------------|-------------------------------------|
| first integral | $\nabla E \cdot \mathbf{F} = 0$                                   | trajectories remain on level sets   |
| reverser       | $\mathbf{F}(R\mathbf{z}) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z})$ | reflected arcs carry reversed time  |
| both           | two independent checks                                            | energy geometry plus orbit symmetry |

**Observation: Purely imaginary eigenvalues are inconclusive**

Purely imaginary eigenvalues of the Jacobian do not, by themselves, imply the existence of a first integral, a reversing symmetry, or a nonlinear center.

# The Nonlinear Pendulum: Geometry before Formulas

Thursday, November 12, 2026

## Idea: Dimensionless pendulum model

For a pendulum of length  $L > 0$  in gravitational acceleration  $g_0 > 0$ ,

$$\frac{d^2\theta}{dt^2} + \frac{g_0}{L} \sin \theta = 0.$$

Set

$$\omega_0 = \sqrt{\frac{g_0}{L}}, \quad \tau = \omega_0 t, \quad v = \frac{d\theta}{d\tau}.$$

With dots denoting differentiation with respect to  $\tau$ , the dimensionless system is

$$\dot{\theta} = v, \quad \dot{v} = -\sin \theta.$$

A dimensionless time interval  $T$  corresponds to physical time

$$T \sqrt{\frac{L}{g_0}}.$$

## Recall: Circle as an angular state space

The circle is the quotient

$$S^1 = \mathbb{R}/(2\pi\mathbb{Z}).$$

Thus  $\theta$  and  $\theta + 2\pi k$  represent the same angular position for every  $k \in \mathbb{Z}$ .

## Definition: Physical phase space of the pendulum

The **physical phase space** of the pendulum is

$$S^1 \times \mathbb{R},$$

where  $\theta \in S^1$  is angular position and  $v \in \mathbb{R}$  is angular velocity.

**Definition: Universal cover of the pendulum phase space**

Define

$$p : \mathbb{R}^2 \rightarrow S^1 \times \mathbb{R}, \quad p(\theta, v) = ([\theta], v),$$

where  $[\theta]$  is the equivalence class of  $\theta$  modulo  $2\pi$ . The plane  $\mathbb{R}^2$ , together with the projection  $p$ , is the **universal cover** used for the pendulum phase space. In this cover, the angle coordinate is allowed to take every real value.

**Visualization: Cylinder and universal cover**

The quotient map identifies vertical sides of every strip of width  $2\pi$ :

$$p(\theta, v) = p(\theta + 2\pi k, v), \quad k \in \mathbb{Z}.$$

$\mathbb{R}^2$  with  $\theta$ -period  $2\pi$

↓ identify  $\theta = -\pi$  with  $\theta = \pi$

$$S^1 \times \mathbb{R}$$

**Definition: Contractible closed curve**

A closed curve in  $S^1 \times \mathbb{R}$  is **contractible** if it can be continuously deformed to a point within  $S^1 \times \mathbb{R}$ . Otherwise it is **noncontractible**.

**Question: Which physical equilibria does the pendulum have?**

Find all equilibria of the following system and determine what linearization says about each one:

$$\dot{\theta} = v, \quad \dot{v} = -\sin \theta.$$

**Solution.** The equilibrium equations give

$$(\theta_*, v_*) = (k\pi, 0), \quad k \in \mathbb{Z}.$$

Modulo  $2\pi$ , the two equilibria are  $(0, 0)$  and  $(\pi, 0)$ . Since

$$DF(\theta, v) = \begin{pmatrix} 0 & 1 \\ -\cos \theta & 0 \end{pmatrix},$$

we have

$$DF(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \lambda = \pm i,$$

and

$$DF(\pi, 0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \pm 1.$$

Thus linearization is inconclusive at  $(0, 0)$ , while  $(\pi, 0)$  is a hyperbolic saddle. In displacement coordinates  $(\theta - \pi, v)$ , its stable and unstable eigenspaces are

$$E^s = \text{span}\{(1, -1)\}, \quad E^u = \text{span}\{(1, 1)\}.$$

**Theorem: Conservation of pendulum energy**

Every solution of

$$\dot{\theta} = v, \quad \dot{v} = -\sin \theta$$

conserves

$$E(\theta, v) = \frac{1}{2}v^2 - \cos \theta.$$

**Proof.** Along a solution,

$$\dot{E} = v\dot{v} + \sin \theta \dot{\theta} = v(-\sin \theta) + \sin \theta v = 0.$$

Therefore  $E(\theta(\tau), v(\tau))$  is constant. □

**Definition: Lift of a pendulum motion**

Let  $z : \mathbb{R} \rightarrow S^1 \times \mathbb{R}$  be a pendulum motion. A **lift** of  $z$  is a continuous curve

$$\tilde{z} = (\tilde{\theta}, v) : \mathbb{R} \rightarrow \mathbb{R}^2$$

such that

$$p(\tilde{z}(\tau)) = z(\tau) \quad (\tau \in \mathbb{R}).$$

**Definition: Libration**

A nonconstant pendulum motion is a **libration** if it has a lift  $\tilde{z} = (\tilde{\theta}, v)$  for which some  $T > 0$  satisfies

$$\tilde{\theta}(\tau + T) = \tilde{\theta}(\tau) \quad (\tau \in \mathbb{R}),$$

and  $\tilde{\theta}$  oscillates between two turning angles whose difference is less than  $2\pi$ .

**Definition: Rotation**

A pendulum motion is a **rotation** if it has a lift satisfying

$$\tilde{\theta}(\tau + T) = \tilde{\theta}(\tau) + 2\pi m$$

for every  $\tau \in \mathbb{R}$ , for some  $T > 0$  and  $m \in \{-1, 1\}$ .

**Recall: Separatrix**

A **separatrix** is an invariant trajectory that separates regions of the phase space having qualitatively different motions.

**Visualization: Pendulum energy levels in the universal cover**

For every  $k \in \mathbb{Z}$ , the energy levels repeat with period  $2\pi$  in  $\theta$ . On the energy level  $E = h$ ,

$$v = \pm \sqrt{2(h + \cos \theta)}.$$

| $h$          | level-set geometry                        | motion              |
|--------------|-------------------------------------------|---------------------|
| $h < -1$     | empty                                     | impossible          |
| $-1$         | $(2k\pi, 0)$                              | hanging equilibrium |
| $-1 < h < 1$ | closed oval about $(2k\pi, 0)$            | libration           |
| $1$          | saddles and their connecting separatrices | separatrix motion   |
| $h > 1$      | one graph above and one below the axis    | rotation            |

**Visualization: The separatrix**

At  $E = 1$ ,

$$v^2 = 2(1 + \cos \theta) = 4 \cos^2\left(\frac{\theta}{2}\right).$$

For  $-\pi < \theta < \pi$ , the upper trajectory branch is

$$v = 2 \cos\left(\frac{\theta}{2}\right).$$

It approaches  $(-\pi, 0)$  as  $\tau \rightarrow -\infty$  and  $(\pi, 0)$  as  $\tau \rightarrow +\infty$ . The lower branch is

$$v = -2 \cos\left(\frac{\theta}{2}\right), \quad -\pi < \theta < \pi,$$

with the opposite limits as  $\tau \rightarrow \pm\infty$ . The endpoints are equilibria: they belong to the closures of these trajectories but are never reached in finite time.

**Proposition: The hanging equilibrium is a nonlinear center**

The equilibrium  $(0, 0)$  of the undamped pendulum is a nonlinear center.

**Proof.** The origin is isolated, and the conserved energy satisfies

$$\nabla E(0, 0) = (0, 0), \quad D^2 E(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The Hessian is positive definite, so the definite-Hessian center criterion applies.  $\square$

**Visualization: Pendulum portrait on the cylinder**

Under the projection  $p(\theta, v) = ([\theta], v)$ , for  $k \in \mathbb{Z}$ ,

| universal cover $\mathbb{R}^2$ | cylinder $S^1 \times \mathbb{R}$ |
|--------------------------------|----------------------------------|
| libration oval                 | contractible closed orbit        |
| rotation graph                 | noncontractible closed orbit     |
| heteroclinic saddle connection | homoclinic orbit                 |
| $(2k\pi, 0)$                   | one hanging equilibrium          |
| $((2k+1)\pi, 0)$               | one inverted equilibrium         |

**Observation: Energy threshold**

For the undamped pendulum, the energy level  $E = 1$  separates libration from rotation.

**Question: How fast must the pendulum start to rotate?**

Suppose

$$\theta(0) = 0, \quad v(0) = v_0.$$

Classify the motion from  $v_0$ .

**Solution.** The conserved energy is

$$h = E(0, v_0) = \frac{1}{2}v_0^2 - 1.$$

Therefore

| $ v_0 $         | $h$          | motion            |
|-----------------|--------------|-------------------|
| 0               | -1           | equilibrium       |
| $0 <  v_0  < 2$ | $-1 < h < 1$ | libration         |
| $ v_0  = 2$     | $h = 1$      | separatrix motion |
| $ v_0  > 2$     | $h > 1$      | rotation          |

For  $v_0 > 0$  the angle initially increases; for  $v_0 < 0$  it initially decreases.

**Proposition: Energy dissipation in the damped pendulum**

Let  $b > 0$ . Every solution of

$$\ddot{\theta} + b\dot{\theta} + \sin \theta = 0$$

satisfies

$$\dot{E} = -bv^2 \leq 0, \quad E(\theta, v) = \frac{1}{2}v^2 - \cos \theta, \quad v = \dot{\theta}.$$

**Proof.** The first-order system is

$$\dot{\theta} = v, \quad \dot{v} = -bv - \sin \theta.$$

Hence

$$\dot{E} = v(\dot{v} + \sin \theta) = -bv^2.$$

□

**Theorem: Damping forbids nonconstant periodic motion**

If  $b > 0$ , every solution of

$$\ddot{\theta} + b\dot{\theta} + \sin \theta = 0$$

whose state  $([\theta], v)$  is periodic on  $S^1 \times \mathbb{R}$  is an equilibrium.

**Proof.** Set

$$v = \dot{\theta}, \quad E(\theta, v) = \frac{1}{2}v^2 - \cos \theta.$$

If the state has period  $T > 0$ , its energy also has period  $T$ , even if a lift of  $\theta$  changes by a multiple of  $2\pi$ . Hence

$$0 = E(\theta(T), v(T)) - E(\theta(0), v(0)) = -b \int_0^T v(\tau)^2 d\tau.$$

Thus  $v \equiv 0$ . The equation then gives  $\sin \theta \equiv 0$ , so the solution is an equilibrium.  $\square$

**Visualization: Damped equilibrium types**

For the damped system

$$\dot{\theta} = v, \quad \dot{v} = -bv - \sin \theta, \quad b > 0,$$

the characteristic polynomial at the hanging equilibrium is

$$\lambda^2 + b\lambda + 1.$$

| $b$         | local type                            |
|-------------|---------------------------------------|
| $0 < b < 2$ | asymptotically stable spiral          |
| $b = 2$     | asymptotically stable degenerate node |
| $b > 2$     | asymptotically stable node            |

At the inverted equilibrium, the Jacobian has determinant  $-1 < 0$ , so the equilibrium is a saddle for every  $b > 0$ .

**Observation: Direction of energy flow**

For the damped pendulum

$$\dot{\theta} = v, \quad \dot{v} = -bv - \sin \theta, \quad b > 0,$$

with energy

$$E(\theta, v) = \frac{1}{2}v^2 - \cos \theta,$$

every solution satisfies

$$\dot{E} = -bv^2.$$

Therefore, away from  $v = 0$ ,

$$\dot{E} = -bv^2 < 0.$$

Thus damped trajectories cross the undamped energy contours toward smaller values of  $E$ .

# Exam 2 Review: Choosing the Right Geometric Tool

Tuesday, November 17, 2026

## Visualization: State space determines the first calculation

| state space                    | first calculation                                             | global structure                                                             |
|--------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------|
| $S^1$                          | $f(\theta) = 0$ and $\text{sign } f$                          | $f \neq 0$ :<br>$T = \left  \int_0^{2\pi} \frac{d\theta}{f(\theta)} \right $ |
| $\mathbb{R}^2, \dot{z} = Az$   | $\tau = \text{tr } A, \Delta = \det A,$<br>$\tau^2 - 4\Delta$ | eigendirections and orientation                                              |
| $\mathbb{R}^2, \dot{z} = F(z)$ | $F(z) = 0, DF(z_*)$                                           | nullclines, invariant sets,<br>energy, symmetry                              |
| $S^1 \times \mathbb{R}$        | $F(z) = 0, DF(z_*)$                                           | periodic identification and<br>energy levels, when available                 |

## Observation: Scope of nonlinear linearization

At a hyperbolic equilibrium of a  $C^1$  nonlinear planar system, the nonlinear flow is locally topologically equivalent to its linearization. In particular, the linearization correctly determines whether the equilibrium is a sink, a source, or a saddle. At a nonhyperbolic equilibrium, the theorem is inconclusive.

## Question: A phase-locking threshold on the circle

For

$$\dot{\theta} = \mu - \sin \theta, \quad \theta \in S^1,$$

find and classify all fixed points for every  $\mu \in \mathbb{R}$ .

**Solution.** Let

$$f(\theta) = \mu - \sin \theta.$$

If  $|\mu| < 1$ , define

$$\alpha = \arcsin \mu, \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}.$$

The two fixed points modulo  $2\pi$  are

$$\theta_s = \alpha, \quad \theta_u = \pi - \alpha.$$

Since

$$f'(\theta) = -\cos \theta,$$

**Question: A phase-locking threshold on the circle (continued)**

we have

$$f'(\theta_s) < 0, \quad f'(\theta_u) > 0.$$

Thus  $\theta_s$  is asymptotically stable and  $\theta_u$  is unstable. If  $|\mu| > 1$ , there are no fixed points, and every solution rotates around the circle. At  $\mu = 1$  the fixed points coalesce at  $\theta = \pi/2$ ; at  $\mu = -1$  they coalesce at  $\theta = 3\pi/2$ . In both cases

$$f = 0, \quad f_\theta = 0, \quad f_{\theta\theta} \neq 0, \quad f_\mu = 1 \neq 0,$$

so each threshold is a saddle-node bifurcation on  $S^1$ . At  $\mu = 1$  the fixed point attracts from the left and repels from the right; at  $\mu = -1$  it repels from the left and attracts from the right. Therefore each threshold fixed point is half-stable and Lyapunov unstable.

**Question: Rotation period beyond phase locking**

For

$$\dot{\theta} = \mu - \sin \theta, \quad \theta \in S^1, \quad |\mu| > 1,$$

compute the positive rotation period.

**Solution.** If  $\mu > 1$ , then  $\mu - \sin \theta > 0$  and

$$\begin{aligned} T &= \int_0^{2\pi} \frac{d\theta}{\mu - \sin \theta} \\ &= \int_{-\pi/2}^{3\pi/2} \frac{d\phi}{\mu - \cos \phi} \quad (\phi = \theta - \pi/2) \\ &= \int_{-\pi}^{\pi} \frac{d\phi}{\mu - \cos \phi} \quad \text{by } 2\pi\text{-periodicity} \\ &= \int_{-\infty}^{\infty} \frac{2 ds}{(\mu - 1) + (\mu + 1)s^2} \quad \left(s = \tan \frac{\phi}{2}\right) \\ &= \frac{2\pi}{\sqrt{\mu^2 - 1}}. \end{aligned}$$

If  $\mu < -1$ , the orientation reverses and the positive period is again

$$T = \frac{2\pi}{\sqrt{\mu^2 - 1}}.$$

**Notation: Trace-determinant data**

For a real  $2 \times 2$  matrix  $A$ , write

$$\tau = \operatorname{tr} A, \quad \Delta = \det A, \quad D = \tau^2 - 4\Delta.$$

Here  $D$  is the discriminant of the characteristic polynomial

$$\lambda^2 - \tau\lambda + \Delta.$$

**Visualization: Trace–determinant classification**For  $\dot{z} = Az$ ,

| condition                     | type                         |
|-------------------------------|------------------------------|
| $\Delta < 0$                  | saddle                       |
| $\Delta > 0, D > 0, \tau < 0$ | asymptotically stable node   |
| $\Delta > 0, D > 0, \tau > 0$ | unstable node                |
| $\Delta > 0, D < 0, \tau < 0$ | asymptotically stable spiral |
| $\Delta > 0, D < 0, \tau > 0$ | unstable spiral              |
| $\Delta > 0, D < 0, \tau = 0$ | linear center                |

**Recall: Degenerate linear cases**

When  $D = 0$  and  $\Delta > 0$ , eigenvectors distinguish a star from a degenerate node. When  $\Delta = 0$ , the kernel of  $A$  is nontrivial, so the linear system has a non-isolated set of equilibria.

**Question: Recover a complete linear portrait**

Classify and orient

$$\dot{z} = Az, \quad A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}.$$

**Solution.** We compute

$$\tau = 2, \quad \Delta = 5, \quad D = 2^2 - 4(5) = -16.$$

The eigenvalues are

$$\lambda = 1 \pm 2i,$$

so the origin is an unstable spiral. At  $(1, 0)$ ,

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

which points upward; the rotation is counterclockwise. Since

$$A = I + 2 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the solution operator is

$$e^{At} = e^t \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix}.$$

Thus radii grow like  $e^t$  and the angular velocity is 2.

**Lemma: An invariant zero set is a uniqueness barrier**

Let  $G, H : \mathbb{R}^2 \rightarrow \mathbb{R}$  be locally Lipschitz, and consider the system

$$\dot{x} = xG(x, y), \quad \dot{y} = H(x, y).$$

Then the line  $x = 0$  is invariant. No trajectory can cross from  $x > 0$  to  $x < 0$ .

**Proof.** Let  $y(t)$  solve

$$\dot{y} = H(0, y), \quad y(t_0) = y_0.$$

Then  $(0, y(t))$  solves the full system through  $(0, y_0)$ . Local uniqueness forces every solution through that state to remain on the axis. A trajectory crossing from  $x > 0$  to  $x < 0$  would meet the axis and then leave it, which is impossible.  $\square$

**Question: When does an imaginary linear pair give a nonlinear center?**

Analyze

$$\dot{x} = y, \quad \dot{y} = -x - x^3.$$

Determine the global phase curves and classify the origin.

**Solution.** The origin is the only equilibrium. Its Jacobian is

$$DF(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

with eigenvalues  $\pm i$ , so linearization alone is inconclusive. Define

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

Then

$$\begin{aligned} \dot{E} &= y\dot{y} + (x + x^3)\dot{x} \\ &= y(-x - x^3) + (x + x^3)y \\ &= 0. \end{aligned}$$

For every  $h > 0$ , there is a unique number  $A_h > 0$  satisfying

$$\frac{1}{2}A_h^2 + \frac{1}{4}A_h^4 = h.$$

The level  $E = h$  consists of the two graph branches

$$y = \pm \sqrt{2h - x^2 - \frac{1}{2}x^4}, \quad |x| \leq A_h,$$

which meet only at  $(\pm A_h, 0)$ . Hence the level is a compact simple closed curve. Moreover,

$$\nabla E = (x + x^3, y)$$

vanishes only at the origin, which does not lie on a positive level. Therefore every level  $E = h$ ,  $h > 0$ , is a periodic orbit, and the origin is a nonlinear center.

**Question: Period of the nonlinear oscillator**

For

$$\dot{x} = y, \quad \dot{y} = -x - x^3,$$

set up the period of the orbit through  $(A, 0)$ , where  $A > 0$ .

**Solution.** For

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

the orbit through  $(A, 0)$  has energy

$$E_A = \frac{1}{2}A^2 + \frac{1}{4}A^4.$$

On the branch with  $y > 0$ ,

$$y = \sqrt{2E_A - x^2 - \frac{1}{2}x^4}, \quad dt = \frac{dx}{y}.$$

Reflection symmetry about both coordinate axes makes the four quarter-orbit travel times equal. Hence

$$T(A) = 4 \int_0^A \frac{dx}{\sqrt{2E_A - x^2 - \frac{1}{2}x^4}}.$$

With  $x = A \sin \phi$ ,

$$2E_A - x^2 - \frac{1}{2}x^4 = A^2 \cos^2 \phi \left( 1 + \frac{A^2}{2} + \frac{A^2}{2} \sin^2 \phi \right),$$

so

$$T(A) = 4 \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 + \frac{A^2}{2} + \frac{A^2}{2} \sin^2 \phi}}.$$

In particular,

$$\lim_{A \rightarrow 0} T(A) = 2\pi.$$

**Question: What changes when linear damping is added?**

For

$$\dot{x} = y, \quad \dot{y} = -x - x^3 - \varepsilon y, \quad \varepsilon > 0.$$

Compute  $\dot{E}$  and determine whether the damped system can have a nonconstant periodic orbit.

**Solution.** For

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

we obtain

$$\dot{E} = y(-x - x^3 - \varepsilon y) + (x + x^3)y = -\varepsilon y^2 \leq 0.$$

If a nonconstant solution were periodic with period  $T$ , then

$$0 = E(x(T), y(T)) - E(x(0), y(0)) = -\varepsilon \int_0^T y(t)^2 dt,$$

**Question: What changes when linear damping is added? (continued)**

so  $y \equiv 0$ . The equation

$$\dot{y} = -x - x^3 = -x(1 + x^2)$$

then forces  $x \equiv 0$ . Therefore the damped system has no nonconstant periodic orbit.

**Recall: Regular level set**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $E \in C^1(U)$ , and let  $c \in \mathbb{R}$ . The level set  $E = c$  is **regular** if

$$\nabla E(z) \neq 0 \quad \text{at every } z \in U \text{ satisfying } E(z) = c.$$

**Visualization: Hypotheses and their valid conclusions**

Let  $U \subseteq \mathbb{R}^2$  be open and let  $F : U \rightarrow \mathbb{R}^2$  be a vector field.

| calculation                                                                                                                                                                | valid conclusion                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| $z_*$ is an equilibrium,<br>$DF(z_*)$ has eigenvalues $\pm i$<br>$H \in C^1(U)$ , $H^{-1}(0)$ regular,<br>$\nabla H \cdot F = 0$ on $H^{-1}(0)$ ,<br>$F$ locally Lipschitz | linearization is inconclusive                                          |
| $E \in C^1(U)$ , $\dot{E} = 0$                                                                                                                                             | $H^{-1}(0)$ is invariant                                               |
| $E \in C^1(U)$ ,<br>$\dot{E} < 0$ away from equilibria                                                                                                                     | each trajectory stays in one connected<br>component of an energy level |
| $F$ locally Lipschitz                                                                                                                                                      | no nonconstant periodic orbit                                          |
|                                                                                                                                                                            | solution curves that meet agree after a time shift                     |

**Visualization: Invalid converses and overclaims**

Let  $F$  be the vector field, and let  $H$  and  $E$  be differentiable real-valued functions on its phase space.

$$\lambda = \pm i \not\Rightarrow \text{nonlinear center,}$$

$$H^{-1}(0) \text{ invariant} \not\Rightarrow H^{-1}(0) \text{ is the global basin boundary,}$$

$$\dot{E} = 0 \not\Rightarrow \text{every connected component of an energy level is a periodic orbit,}$$

$$\text{numerical portrait} \not\Rightarrow \text{proof.}$$

# Limit Cycles and Self-Sustained Oscillation

Tuesday, November 24, 2026

## Recall: Periodic solution and closed orbit

A nonconstant solution  $z : \mathbb{R} \rightarrow \mathbb{R}^2$  is **periodic** if there is a number  $T > 0$  such that

$$z(t + T) = z(t)$$

for every  $t \in \mathbb{R}$ . Its least positive period is its **period**. Its image

$$\Gamma = \{z(t) : 0 \leq t \leq T\}$$

is its **closed orbit**.

## Definition: Limit cycle

A closed orbit  $\Gamma$  is a **limit cycle** if it is isolated: there is a neighborhood of  $\Gamma$  containing no other closed orbit.

## Notation: Distance from an orbit

For  $z \in \mathbb{R}^2$  and a compact orbit  $\Gamma \subset \mathbb{R}^2$ , write

$$\text{dist}(z, \Gamma) = \inf_{p \in \Gamma} \|z - p\|.$$

## Definition: Orbital Lyapunov stability

A limit cycle  $\Gamma$  of  $\dot{z} = F(z)$  is **orbitally Lyapunov stable** if, for every  $\varepsilon > 0$ , there is a  $\delta > 0$  such that every solution  $z(t)$  with

$$\text{dist}(z(0), \Gamma) < \delta$$

is defined for every  $t \geq 0$  and satisfies

$$\text{dist}(z(t), \Gamma) < \varepsilon \quad \text{for all } t \geq 0.$$

**Definition: Attracting limit cycle**

A limit cycle  $\Gamma$  of  $\dot{z} = F(z)$  is **attracting** if there is a  $\delta > 0$  such that every solution  $z(t)$  with

$$\text{dist}(z(0), \Gamma) < \delta$$

is defined for every  $t \geq 0$  and satisfies

$$\text{dist}(z(t), \Gamma) \rightarrow 0 \quad (t \rightarrow \infty).$$

**Definition: Asymptotic orbital stability**

A limit cycle is **asymptotically orbitally stable** if it is both orbitally Lyapunov stable and attracting.

**Definition: Half-stable limit cycle**

A planar limit cycle  $\Gamma$  separates the plane into an inside and an outside. It is **half-stable** if it attracts every sufficiently close trajectory on one side, whereas every sufficiently close trajectory on the other side eventually leaves some fixed neighborhood of  $\Gamma$ .

**Notation: Stable limit cycle**

In these notes, **stable limit cycle** means asymptotically orbitally stable limit cycle.

**Visualization: Closed families versus isolated cycles**

| phase portrait               | isolated? | amplitude after perturbation |
|------------------------------|-----------|------------------------------|
| linear center                | no        | new amplitude persists       |
| undamped pendulum librations | no        | new energy persists          |
| attracting limit cycle       | yes       | returns to selected orbit    |

**Proposition: Radial construction principle**

Let  $I \subseteq (0, \infty)$  be an open interval, let  $R : I \rightarrow \mathbb{R}$  and  $\Omega : I \times S^1 \rightarrow \mathbb{R}$  be locally Lipschitz, and consider

$$\dot{r} = R(r), \quad \dot{\theta} = \Omega(r, \theta), \quad (r, \theta) \in I \times S^1.$$

Suppose  $a \in I$  is an isolated zero of  $R$  and

$$\Omega(a, \theta) \neq 0$$

for every  $\theta \in S^1$ . Then  $r = a$  is a limit cycle.

**Proof.** Since  $R(a) = 0$ , uniqueness makes  $r = a$  invariant. The continuous nonzero function  $\Omega(a, \theta)$  has one sign on the connected circle, so the solution traverses  $r = a$  periodically with period

$$T = \left| \int_0^{2\pi} \frac{d\theta}{\Omega(a, \theta)} \right|.$$

On any closed orbit,  $r(t)$  is periodic. The scalar autonomous equation  $\dot{r} = R(r)$  has no nonconstant periodic solution, so  $r(t)$  must be a constant zero of  $R$ . Isolation of the zero  $a$  therefore excludes every nearby closed orbit, including noncircular ones.  $\square$

**Corollary: Radial stability test**

Let  $I \subseteq (0, \infty)$  be open, let  $R : I \rightarrow \mathbb{R}$  and  $\Omega : I \times S^1 \rightarrow \mathbb{R}$  be locally Lipschitz, and consider

$$\dot{r} = R(r), \quad \dot{\theta} = \Omega(r, \theta).$$

Suppose  $a \in I$  is an isolated zero of  $R$  and  $\Omega(a, \theta) \neq 0$  for every  $\theta \in S^1$ . Then  $r = a$  is a limit cycle with the following classification, according to the signs of  $R(r)$  for  $r$  sufficiently close to  $a$ :

| $R(r)$ for $r < a$ | $R(r)$ for $r > a$ | type                               |
|--------------------|--------------------|------------------------------------|
| $R(r) > 0$         | $R(r) < 0$         | stable                             |
| $R(r) < 0$         | $R(r) > 0$         | not orbitally Lyapunov stable      |
| $R(r) > 0$         | $R(r) > 0$         | half-stable: attracts from inside  |
| $R(r) < 0$         | $R(r) < 0$         | half-stable: attracts from outside |

**Proof.** The distance to the circle  $r = a$  is  $|r - a|$ . Apply the scalar phase-line test to  $\dot{r} = R(r)$ : motion toward  $a$  decreases this distance to zero; motion away from  $a$  eventually leaves a fixed neighboring annulus. These are precisely the stated orbital classifications.  $\square$

**Question: The canonical attracting limit cycle**

Analyze

$$\dot{r} = r(1 - r^2), \quad \dot{\theta} = 1.$$

Interpret these as polar equations in the plane, so all coordinate pairs  $(0, \theta)$  represent the single Cartesian origin. Find and classify every nonnegative invariant radius.

**Solution.** The radial equilibria are  $r = 0$  and  $r = 1$ . The sign chart is

| $r$       | $(0, 1)$ | $1$ | $(1, \infty)$ |
|-----------|----------|-----|---------------|
| $\dot{r}$ | +        | 0   | -             |

so the origin repels nearby nonzero states and  $r = 1$  is an attracting limit cycle with period  $T = 2\pi$ .

**Question: Solve the canonical radial equation**

For  $t \geq 0$ , solve

$$\dot{r} = r(1 - r^2), \quad r(0) = r_0 > 0.$$

**Solution.** Set  $q = r^2$ . Then

$$\dot{q} = 2q(1 - q).$$

For  $u = 1/q$ ,

$$\dot{u} = -\frac{\dot{q}}{q^2} = 2(1 - u),$$

and hence

$$u(t) = 1 + (r_0^{-2} - 1)e^{-2t}.$$

Since  $r(t) > 0$ ,

$$r(t) = \frac{1}{\sqrt{1 + (r_0^{-2} - 1)e^{-2t}}} \rightarrow 1 \quad (t \rightarrow \infty).$$

**Question: What Cartesian system has the canonical cycle?**

Convert

$$\dot{r} = r(1 - r^2), \quad \dot{\theta} = 1$$

to Cartesian coordinates.

**Solution.** With  $x = r \cos \theta$  and  $y = r \sin \theta$ , we have

$$\dot{x} = \dot{r} \cos \theta - r \dot{\theta} \sin \theta, \quad \dot{y} = \dot{r} \sin \theta + r \dot{\theta} \cos \theta.$$

Since  $r^2 = x^2 + y^2$ ,

$$\begin{aligned} \dot{x} &= x(1 - x^2 - y^2) - y, \\ \dot{y} &= y(1 - x^2 - y^2) + x. \end{aligned}$$

Conversely,

$$r\dot{r} = x\dot{x} + y\dot{y}, \quad r^2\dot{\theta} = x\dot{y} - y\dot{x},$$

which recover the radial and angular equations for  $r > 0$ .

**Question: Two limit cycles**

Classify the positive invariant radii of

$$\dot{r} = r(r^2 - 1)(4 - r^2), \quad \dot{\theta} = 1.$$

**Solution.**

|           |        |   |        |   |        |
|-----------|--------|---|--------|---|--------|
| $r$       | (0, 1) | 1 | (1, 2) | 2 | (2, ∞) |
| $\dot{r}$ | −      | 0 | +      | 0 | −      |

so the cycle  $r = 1$  repels nearby radii on both sides and is not orbitally Lyapunov stable, whereas  $r = 2$  is a stable limit cycle. The origin attracts every state with  $0 < r < 1$ .

**Question: A half-stable limit cycle**

Classify the positive invariant radii of

$$\dot{r} = r(r^2 - 1)^2(4 - r^2), \quad \dot{\theta} = 1.$$

**Solution.**

|           |        |   |        |   |        |
|-----------|--------|---|--------|---|--------|
| $r$       | (0, 1) | 1 | (1, 2) | 2 | (2, ∞) |
| $\dot{r}$ | +      | 0 | +      | 0 | −      |

so  $r = 1$  attracts from the inside and repels from the outside: it is half-stable. The cycle  $r = 2$  remains stable.

**Theorem: Homogeneous linear systems have no limit cycles**

No planar homogeneous linear system

$$\dot{z} = Az$$

has a limit cycle.

**Proof.** Suppose  $z(t)$  is a nonconstant periodic solution. For every  $c > 0$ ,

$$\frac{d}{dt}(cz(t)) = A(cz(t)),$$

so  $cz(t)$  is another nonconstant periodic solution. Its maximum distance from the origin is  $c$  times that of the original orbit, so the two orbits are distinct when  $c \neq 1$ . Taking  $c$  arbitrarily close to 1 produces closed orbits arbitrarily close to the original one. Thus no closed orbit is isolated.  $\square$

**Question: What does the van der Pol energy balance prove?**

For

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0, \quad \mu > 0,$$

compute the derivative of the linear-oscillator energy and state its exact consequence.

**Solution.** Let

$$v = \dot{x}, \quad E(x, v) = \frac{1}{2}v^2 + \frac{1}{2}x^2.$$

Then

$$\begin{aligned} \dot{E} &= v(\dot{v} + x) \\ &= -\mu(x^2 - 1)v^2. \end{aligned}$$

Hence

$$\begin{array}{l|l} |x| < 1, v \neq 0 & \dot{E} > 0 \\ |x| > 1, v \neq 0 & \dot{E} < 0 \\ |x| = 1 \text{ or } v = 0 & \dot{E} = 0 \end{array}$$

Energy is injected exactly when  $|x| < 1$  and  $v \neq 0$ , dissipated exactly when  $|x| > 1$  and  $v \neq 0$ , and instantaneously unchanged when  $|x| = 1$  or  $v = 0$ . This pointwise sign change supplies an amplitude-selection mechanism, but the calculation alone does not prove existence, uniqueness, or stability of a periodic orbit.

**Idea: Self-sustained oscillation**

In a radial or energy-balance description of a self-sustained oscillator,

small amplitude  $\longrightarrow$  net outward motion,

large amplitude  $\longrightarrow$  net inward motion,

suggests that the dynamics may select an intermediate amplitude. This is a heuristic: it does not by itself prove that a periodic orbit exists or that such an orbit is isolated, unique, or stable. Each of those conclusions requires additional arguments.

# How to Rule Out Periodic Orbits

Tuesday, December 1, 2026

## Idea: Periodic return versus monotone change

If  $z(t)$  has period  $T > 0$  and  $V$  is a single-valued function of state, then

$$V(z(T)) - V(z(0)) = 0.$$

Therefore either inequality

$$V(z(T)) - V(z(0)) < 0 \quad \text{or} \quad V(z(T)) - V(z(0)) > 0$$

rules out a nonconstant periodic orbit.

## Definition: Gradient system

Let  $U \subseteq \mathbb{R}^2$  be open. A system is a **gradient system** on  $U$  if there is a single-valued function  $V \in C^1(U)$  such that

$$\dot{z} = -\nabla V(z).$$

The function  $V$  is called a **potential** for the system.

## Observation: Potential decreases along a gradient flow

Along every solution of  $\dot{z} = -\nabla V(z)$ ,

$$\dot{V} = \nabla V \cdot \dot{z} = -\|\nabla V\|^2 = -\|\dot{z}\|^2 \leq 0.$$

## Recall: Simply connected planar domain

An open, path-connected set  $U \subseteq \mathbb{R}^2$  is **simply connected** if every closed curve in  $U$  can be continuously contracted to a point while remaining in  $U$ .

**Recall: Planar gradient test**

Let  $U \subseteq \mathbb{R}^2$  be open and let  $F = (f, g) \in C^1(U)$ . The identity

$$f_y = g_x$$

is necessary for  $F = -\nabla V$ . If  $U$  is simply connected, the identity is also sufficient: there is a function  $V \in C^2(U)$  such that  $F = -\nabla V$ .

**Theorem: Gradient flows have no closed orbit**

A gradient system has no nonconstant periodic orbit.

**Proof.** Suppose  $z(t)$  is  $T$ -periodic. Then

$$\begin{aligned} 0 &= V(z(T)) - V(z(0)) \\ &= \int_0^T \dot{V}(z(t)) dt \\ &= - \int_0^T \|\dot{z}(t)\|^2 dt. \end{aligned}$$

The continuous nonnegative integrand must vanish identically. Thus  $\dot{z} \equiv 0$ , so the solution is an equilibrium, not a nonconstant periodic orbit.  $\square$

**Question: Construct a potential**

Show that

$$\dot{x} = \sin y, \quad \dot{y} = x \cos y$$

has no closed orbit.

**Solution.** We seek  $V$  with

$$V_x = -\sin y, \quad V_y = -x \cos y.$$

Integrating the first equation with respect to  $x$  gives

$$V(x, y) = -x \sin y + C(y).$$

Then

$$V_y = -x \cos y + C'(y).$$

The second equation requires  $C'(y) = 0$ , so we may take

$$V(x, y) = -x \sin y.$$

Hence  $F = -\nabla V$ , and the gradient-flow theorem rules out every nonconstant periodic orbit.

**Definition: Strict Lyapunov function**

Let  $z_*$  be an equilibrium of  $\dot{z} = F(z)$  on an open set  $U$ . A function  $V \in C^1(U)$  is a **strict Lyapunov function for  $z_*$  on  $U$**  if

$$V(z_*) = 0, \quad V(z) > 0 \quad (z \neq z_*),$$

and

$$\dot{V}(z) = \nabla V(z) \cdot F(z) < 0 \quad (z \neq z_*).$$

**Observation: Strict decrease forbids periodic return**

If  $V$  is a strict Lyapunov function on  $U$ , then no nonconstant periodic orbit is contained in  $U$ .

**Theorem: Strict Lyapunov stability criterion**

Suppose  $F$  is locally Lipschitz on  $U$  and  $V$  is a strict Lyapunov function for  $z_*$  on a neighborhood  $U$ . If  $c > 0$  is chosen so that

$$D_c = \{z \in U : V(z) \leq c\}$$

is compact and contains  $z_*$  in its interior, then  $z_*$  is asymptotically stable. More precisely, every solution with  $z(0) \in D_c$  is defined for all  $t \geq 0$ , remains in  $D_c$ , and converges to  $z_*$ . Moreover,  $z_*$  is Lyapunov stable, and  $D_c$  is contained in its basin of attraction.

**Corollary: Global strict Lyapunov criterion**

Let  $F$  be locally Lipschitz on  $\mathbb{R}^2$ , let  $z_*$  be an equilibrium, and let  $V$  be a strict Lyapunov function for  $z_*$  on  $\mathbb{R}^2$ . If

$$V(z) \rightarrow \infty \quad (\|z\| \rightarrow \infty),$$

then every solution remains in the compact sublevel set determined by its initial value, extends to every  $t \geq 0$ , and converges to  $z_*$ . Thus  $z_*$  is globally asymptotically stable.

**Question: Monotone energy without a gradient field**

Show that

$$\ddot{x} + (\dot{x})^3 + x = 0$$

has no nonconstant periodic solution.

**Solution.** Set  $y = \dot{x}$ . Then

$$\dot{x} = y, \quad \dot{y} = -x - y^3.$$

For

$$E(x, y) = \frac{1}{2}(x^2 + y^2),$$

we have

$$\dot{E} = x\dot{x} + y\dot{y} = xy + y(-x - y^3) = -y^4 \leq 0.$$

If a solution had period  $T > 0$ , then

$$0 = E(x(T), y(T)) - E(x(0), y(0)) = - \int_0^T y(t)^4 dt.$$

Hence  $y \equiv 0$ , and  $\dot{y} = -x$  then forces  $x \equiv 0$ . Thus the only periodic solution is the equilibrium.

**Question: Choose a Lyapunov function by canceling a mixed term**

Analyze

$$\dot{x} = -x + 4y, \quad \dot{y} = -x - y^3.$$

**Solution.** Let

$$V(x, y) = x^2 + ay^2, \quad a > 0.$$

Then

$$\begin{aligned} \dot{V} &= 2x(-x + 4y) + 2ay(-x - y^3) \\ &= -2x^2 + (8 - 2a)xy - 2ay^4. \end{aligned}$$

Choose  $a = 4$ . Then

$$V = x^2 + 4y^2, \quad \dot{V} = -2x^2 - 8y^4 < 0$$

away from  $(0, 0)$ . Moreover,

$$V(x, y) \rightarrow \infty \quad \left( \sqrt{x^2 + y^2} \rightarrow \infty \right).$$

Therefore every sublevel set  $\{V \leq c\}$ ,  $c \geq 0$ , is compact and positively invariant. Hence the origin is globally asymptotically stable and no nonconstant periodic orbit exists.

**Definition: Dulac multiplier**

Let  $U \subseteq \mathbb{R}^2$  be open and let  $F \in C^1(U, \mathbb{R}^2)$ . A function  $B \in C^1(U)$  is a **Dulac multiplier on  $U$**  if either

$$\nabla \cdot (BF) > 0 \quad \text{throughout } U$$

or

$$\nabla \cdot (BF) < 0 \quad \text{throughout } U.$$

**Theorem: Bendixson–Dulac criterion**

Let  $U \subseteq \mathbb{R}^2$  be open and simply connected. Let  $F \in C^1(U)$ , and suppose there is a function  $B \in C^1(U)$  such that

$$\nabla \cdot (BF) > 0$$

throughout  $U$ , or such that

$$\nabla \cdot (BF) < 0$$

throughout  $U$ . Then no nonconstant closed orbit lies entirely in  $U$ .

**Proof.** Suppose a closed orbit  $C$  lies in  $U$ , and let  $D$  be its interior. Simple connectivity gives  $D \subset U$ . If  $n$  is the outward unit normal, the planar divergence theorem gives

$$\iint_D \nabla \cdot (BF) \, dA = \oint_C BF \cdot n \, ds.$$

The right side is zero because  $F$  is tangent to its trajectory. The left side is nonzero because its integrand has a strict constant sign. This is a contradiction.  $\square$

**Question: A Dulac multiplier in the positive quadrant**

Consider

$$\dot{x} = x(2 - x - y), \quad \dot{y} = y(4x - x^2 - 3)$$

on

$$U = \{(x, y) : x > 0, y > 0\}.$$

Use  $B(x, y) = 1/(xy)$  to rule out closed orbits in  $U$ .

**Solution.** We have

$$BF = \left( \frac{2 - x - y}{y}, \frac{4x - x^2 - 3}{x} \right).$$

Therefore

$$\begin{aligned} \nabla \cdot (BF) &= \frac{\partial}{\partial x} \left( \frac{2 - x - y}{y} \right) + \frac{\partial}{\partial y} \left( \frac{4x - x^2 - 3}{x} \right) \\ &= -\frac{1}{y} < 0. \end{aligned}$$

The domain  $U$  is open and simply connected, and  $B, F \in C^1(U)$ . Bendixson–Dulac therefore excludes every closed orbit contained in  $U$ . The multiplier is singular on the coordinate axes, so this calculation does not apply to a region containing either axis.

**Question: An exponential Dulac multiplier**

Show that

$$\dot{x} = y, \quad \dot{y} = -x - y + x^2 + y^2$$

has no closed orbit in  $\mathbb{R}^2$ .

**Solution.** Let

$$B(x, y) = e^{-2x}.$$

Then

$$\begin{aligned} \nabla \cdot (BF) &= \frac{\partial}{\partial x} (e^{-2x}y) + \frac{\partial}{\partial y} (e^{-2x}(-x - y + x^2 + y^2)) \\ &= -2e^{-2x}y + e^{-2x}(-1 + 2y) \\ &= -e^{-2x} < 0. \end{aligned}$$

The plane is simply connected and  $B, F$  are continuously differentiable, so Bendixson–Dulac rules out every closed orbit.

**Corollary: One-sign Dulac divergence forbids two nested periodic orbits**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $B, F \in C^1(U)$ , and suppose  $\nabla \cdot (BF)$  has a strict constant sign on  $U$ . Then  $U$  cannot contain two distinct periodic orbits  $C_1$  and  $C_2$  for which the compact annular region between  $C_1$  and  $C_2$  is contained in  $U$ .

**Proof.** Let  $D \subset U$  be the compact annular region between two such orbits. The divergence theorem on  $D$  gives

$$\iint_D \nabla \cdot (BF) dA = \oint_{C_1} BF \cdot n ds + \oint_{C_2} BF \cdot n ds = 0,$$

because  $F$  is tangent to both boundary components. The area integral is nonzero, a contradiction.  $\square$

**Visualization: Three periodic-orbit nonexistence criteria**

| structure             | sufficient condition                                                  | contradiction                            |
|-----------------------|-----------------------------------------------------------------------|------------------------------------------|
| $\dot{z} = -\nabla V$ | $\dot{V} = -\ \dot{z}\ ^2$                                            | $V(z(T)) - V(z(0)) < 0$                  |
| general flow          | $\dot{V} < 0$                                                         | $V(z(T)) - V(z(0)) < 0$                  |
| planar flow           | $\nabla \cdot (BF)$ has a strict sign<br>on a simply connected domain | nonzero area flux<br>=zero boundary flux |

**Observation: Failure of a criterion**

Failure to find a Lyapunov function  $V$  or a Dulac multiplier  $B$  gives no conclusion about whether a periodic orbit exists.

# Proving That a Periodic Orbit Exists

Thursday, December 3, 2026

## Definition: Positive semi-orbit

Let  $z(t; z_0)$  be a solution defined for every  $t \geq 0$  and satisfying  $z(0; z_0) = z_0$ . Its **positive semi-orbit** is

$$\gamma^+(z_0) = \{z(t; z_0) : t \geq 0\}.$$

## Definition: Omega-limit set

Let  $z(t; z_0)$  be a solution defined for every  $t \geq 0$  and satisfying  $z(0; z_0) = z_0$ . Its **omega-limit set** is

$$\omega(z_0) = \left\{ p \in \mathbb{R}^2 : \begin{array}{l} \text{there is a sequence } t_n \rightarrow \infty \\ \text{such that } z(t_n; z_0) \rightarrow p \end{array} \right\}.$$

## Proposition: Basic omega-limit properties

Let  $U \subseteq \mathbb{R}^2$  be open, let  $F : U \rightarrow \mathbb{R}^2$  be locally Lipschitz, and let  $z(t; z_0)$  solve  $\dot{z} = F(z)$ . If  $\gamma^+(z_0)$  lies in a compact subset of  $U$ , then  $\omega(z_0)$  is nonempty, compact, and invariant.

## Theorem: Poincaré–Bendixson theorem

Let  $U \subseteq \mathbb{R}^2$  be open and let  $F \in C^1(U)$ . Suppose the positive semi-orbit of  $z_0$  is contained in a compact subset of  $U$ . If  $\omega(z_0)$  contains no equilibrium of

$$\dot{z} = F(z),$$

then  $\omega(z_0)$  is a periodic orbit.

## Recall: Positive invariance

A set  $D$  is **positively invariant** for  $\dot{z} = F(z)$  if every solution satisfies

$$z(0) \in D \implies z(t) \in D \quad \text{for every } t \geq 0 \text{ for which the solution is defined.}$$

**Definition: Trapping region**

For a flow on an open set  $U \subseteq \mathbb{R}^2$ , a **trapping region** is a nonempty, compact, connected, positively invariant set  $D \subset U$ .

**Observation: Inward-pointing boundary test**

Let  $D$  be a closed planar region bounded by finitely many smooth arcs meeting at corners, and let  $F$  be locally Lipschitz on a neighborhood of  $D$ . At a smooth point of  $\partial D$ , let  $n$  be the outward unit normal. The strict inequality

$$F \cdot n < 0$$

means that the field crosses into  $D$ . To allow tangency, require  $F \cdot n \leq 0$  at every smooth boundary point. At a corner, require this inequality for the outward normal of every incident smooth side. These conditions prevent a solution from leaving  $D$  through its boundary.

**Corollary: Trapping-region criterion**

Let  $F \in C^1(U)$  for an open set  $U \subseteq \mathbb{R}^2$ , and let  $D \subset U$  be a trapping region. If  $D$  contains no equilibrium of  $\dot{z} = F(z)$ , then  $D$  contains at least one periodic orbit.

**Proof.** Choose  $z_0 \in D$ . Positive invariance keeps the solution in the compact set  $D \subset U$ . The continuation theorem therefore makes the solution global forward in time, and

$$\omega(z_0) \subseteq D.$$

The set  $\omega(z_0)$  contains no equilibrium, so the Poincaré–Bendixson theorem makes it a periodic orbit in  $D$ .  $\square$

**Visualization: Boundary directions for an annular trap**

For

$$D = \{(r, \theta) \in (0, \infty) \times S^1 : r_- \leq r \leq r_+\}, \quad 0 < r_- < r_+,$$

the outward normals and inward conditions are

| boundary  | outward normal | inward condition |
|-----------|----------------|------------------|
| $r = r_+$ | $e_r$          | $\dot{r} < 0$    |
| $r = r_-$ | $-e_r$         | $\dot{r} > 0$    |

Here  $e_r$  denotes the radial unit vector pointing toward increasing  $r$ . On the inner boundary, “outward from the annulus” means “into the hole.”

**Question: Construct a trapping annulus**

Consider

$$\dot{r} = r(1 - r^2) + \mu r \cos \theta, \quad \dot{\theta} = 1, \quad \theta \in S^1, \quad 0 < \mu < 1.$$

Prove that the system has a periodic orbit.

**Solution.** Choose the radii

$$r_- = \sqrt{\frac{1-\mu}{2}}, \quad r_+ = \sqrt{2(1+\mu)}.$$

On  $r = r_-$ , the bound  $\cos \theta \geq -1$  gives

$$\begin{aligned} \dot{r} &= r_-(1 - r_-^2 + \mu \cos \theta) \\ &\geq r_-(1 - r_-^2 - \mu) \\ &= \frac{r_-}{2}(1 - \mu) > 0. \end{aligned}$$

On  $r = r_+$ , the bound  $\cos \theta \leq 1$  gives

$$\begin{aligned} \dot{r} &= r_+(1 - r_+^2 + \mu \cos \theta) \\ &\leq r_+(1 - r_+^2 + \mu) \\ &= -r_+(1 + \mu) < 0. \end{aligned}$$

Thus

$$D = \{(r, \theta) : r_- \leq r \leq r_+, \theta \in S^1\}$$

is a trapping annulus. The map

$$(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$$

identifies  $(0, \infty) \times S^1$  with the punctured plane, so  $D$  is a planar annulus and the planar trapping-region criterion applies. Since  $\dot{\theta} = 1$ , the system has no equilibrium in  $D$ . The trapping-region criterion gives at least one periodic orbit in  $D$ . It gives no uniqueness or stability conclusion.

**Question: Construct a polygonal trap for the glycolysis model**

For

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y, \quad a, b > 0,$$

construct a compact positively invariant polygon in the closed nonnegative quadrant  $\{(x, y) : x \geq 0, y \geq 0\}$ .

**Solution.** Choose

$$Y > \frac{b}{a}$$

and let  $D$  have vertices

$$(0, 0), \quad (b + Y, 0), \quad (b, Y), \quad (0, Y).$$

**Question: Construct a polygonal trap for the glycolysis model (continued)**

Its four sides satisfy:

| side            | calculation                                | direction                 |
|-----------------|--------------------------------------------|---------------------------|
| $x = 0$         | $\dot{x} = ay \geq 0$                      | right or tangent          |
| $y = 0$         | $\dot{y} = b > 0$                          | up                        |
| $y = Y$         | $\dot{y} = b - (a + x^2)Y \leq b - aY < 0$ | down                      |
| $x + y = b + Y$ | $\dot{x} + \dot{y} = b - x \leq 0$         | toward $x + y \leq b + Y$ |

On the diagonal,  $x \geq b$ . At its top endpoint  $x = b$ , the diagonal derivative is zero but the top-side inequality is strict. Thus the inward inequality holds for every side incident to each boundary point, so  $D$  is positively invariant.

**Recall: Hyperbolic source**

An equilibrium of a planar system is a **hyperbolic source** if both eigenvalues of its Jacobian have positive real parts.

**Question: When does the glycolysis equilibrium repel?**

For

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y, \quad a, b > 0,$$

find the equilibrium and determine when it is a hyperbolic source.

**Solution.** Let

$$F(x, y) = (-x + ay + x^2y, b - ay - x^2y).$$

Adding the equations gives

$$\dot{x} + \dot{y} = b - x.$$

Thus every equilibrium satisfies  $x_* = b$ , and substitution gives the unique equilibrium

$$z_* = (x_*, y_*) = \left(b, \frac{b}{a + b^2}\right).$$

The Jacobian is

$$DF(x, y) = \begin{pmatrix} -1 + 2xy & a + x^2 \\ -2xy & -(a + x^2) \end{pmatrix}.$$

At  $z_*$ , the determinant and trace of the Jacobian are

$$\Delta = \det DF(z_*) = a + b^2 > 0$$

and

$$\tau = \operatorname{tr} DF(z_*) = -\frac{b^4 + (2a - 1)b^2 + a + a^2}{a + b^2}.$$

Let  $q = b^2$ . The condition  $\tau > 0$  is

$$q^2 + (2a - 1)q + a + a^2 < 0.$$

The roots of the quadratic are

$$q_{\pm} = \frac{1 - 2a \pm \sqrt{1 - 8a}}{2}.$$

**Question: When does the glycolysis equilibrium repel? (continued)**

Therefore  $z_*$  is a hyperbolic source when

$$0 < a < \frac{1}{8}, \quad q_- < b^2 < q_+.$$

**Definition: Strict exit neighborhood**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $F : U \rightarrow \mathbb{R}^2$  be continuous, and let  $z_* \in U$  be an equilibrium of  $\dot{z} = F(z)$ . A compact neighborhood  $E \subset U$  of  $z_*$  with smooth boundary is a **strict exit neighborhood** if

$$F(z) \cdot n_E(z) > 0 \quad (z \in \partial E),$$

where  $n_E$  is the outward unit normal of  $E$ .

**Lemma: Lyapunov matrix for a repelling linear system**

Let  $A$  be a real  $2 \times 2$  matrix whose eigenvalues have positive real parts. Define

$$P = \int_0^\infty e^{-A^\top s} e^{-As} ds.$$

The integral converges to a symmetric positive definite matrix satisfying

$$A^\top P + PA = I.$$

Here  $I$  is the  $2 \times 2$  identity matrix.

**Proof.** The eigenvalues of  $-A$  have negative real parts, so  $e^{-As}$  decays exponentially and the integral converges. For every  $\xi \neq 0$ ,

$$\xi^\top P \xi = \int_0^\infty \|e^{-As} \xi\|^2 ds > 0.$$

Moreover,

$$\begin{aligned} A^\top P + PA &= - \int_0^\infty \frac{d}{ds} (e^{-A^\top s} e^{-As}) ds \\ &= I. \end{aligned}$$

□

**Proposition: Quadratic exit neighborhood of a planar hyperbolic source**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $F \in C^2(U)$ , and let  $z_* \in U$  be an equilibrium of  $\dot{z} = F(z)$ . Suppose every eigenvalue of

$$A = DF(z_*)$$

has positive real part. Define

$$P = \int_0^\infty e^{-A^\top s} e^{-As} ds$$

and

$$V(z) = (z - z_*)^\top P (z - z_*).$$

Then  $\dot{V} > 0$  sufficiently near  $z_*$ , except at  $z_*$  itself. Consequently, for every sufficiently small  $\rho > 0$ , the ellipsoid  $\{z : V(z) \leq \rho\}$  is a strict exit neighborhood.

**Proof.** Set

$$\xi = z - z_*.$$

Taylor's theorem gives

$$F(z_* + \xi) = A\xi + R(\xi), \quad R(\xi) = O(\|\xi\|^2) \quad (\xi \rightarrow 0).$$

Using  $A^\top P + PA = I$ , we obtain

$$\begin{aligned} \dot{V} &= 2\xi^\top PF(z_* + \xi) \\ &= \xi^\top (A^\top P + PA)\xi + 2\xi^\top PR(\xi) \\ &= \|\xi\|^2 + O(\|\xi\|^3) \quad (\xi \rightarrow 0). \end{aligned}$$

For sufficiently small nonzero  $\xi$ , the remainder has absolute value at most  $\frac{1}{2}\|\xi\|^2$ . Hence

$$\dot{V} \geq \frac{1}{2}\|\xi\|^2 > 0.$$

Because  $P$  is positive definite, sufficiently small ellipsoids  $V \leq \rho$  lie in this neighborhood and in  $U$ . On their boundaries,  $\nabla V \neq 0$  and the outward unit normal is

$$n = \frac{\nabla V}{\|\nabla V\|},$$

so

$$F \cdot n = \frac{\dot{V}}{\|\nabla V\|} > 0.$$

Hence  $V \leq \rho$  is a strict exit neighborhood. □

**Question: Force a periodic orbit in the glycolysis model**

Assume

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y,$$

where

$$0 < a < \frac{1}{8}, \quad b > 0, \quad q_- < b^2 < q_+, \quad q_{\pm} = \frac{1 - 2a \pm \sqrt{1 - 8a}}{2}.$$

Let  $Y > b/a$ , and let  $D$  be the positively invariant polygon with vertices

$$(0, 0), \quad (b + Y, 0), \quad (b, Y), \quad (0, Y).$$

Prove that the system has a periodic orbit.

**Solution.** Let

$$F(x, y) = (-x + ay + x^2y, b - ay - x^2y).$$

The unique equilibrium is

$$z_* = \left(b, \frac{b}{a + b^2}\right),$$

and it is a hyperbolic source.

Because

$$0 < \frac{b}{a + b^2} < \frac{b}{a} < Y,$$

the point  $z_*$  satisfies all four defining inequalities for  $D$  strictly:  $x_* > 0$ ,  $y_* > 0$ ,  $y_* < Y$ , and  $x_* + y_* < b + Y$ . Hence  $z_* \in \text{int}(D)$ .

By the hyperbolic-source proposition, choose a strict exit ellipsoid  $E$  about  $z_*$  lying inside  $\text{int}(D)$ . Remove its interior:

$$K = D \setminus \text{int}(E).$$

The outer boundary still points inward or tangent. On the inner boundary, the outward normals satisfy  $n_K = -n_E$ , so

$$F \cdot n_K = -F \cdot n_E < 0.$$

Thus  $K$  is positively invariant. Since  $D$  and  $E$  are convex and  $E$  lies strictly inside  $D$ ,  $K$  is a compact connected annulus. Its only possible equilibrium was removed with  $E$ . The trapping-region criterion therefore gives a periodic orbit in  $K$ .

**Question: Check the standard glycolysis parameters**

For

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y,$$

let

$$a = 0.08, \quad b = 0.6.$$

Classify the equilibrium and state the conclusion justified by an annular trapping region.

**Solution.** Let

$$F(x, y) = (-x + ay + x^2y, b - ay - x^2y).$$

The unique equilibrium is

$$z_* = \left(b, \frac{b}{a + b^2}\right) = \left(0.6, \frac{15}{11}\right).$$

**Question: Check the standard glycolysis parameters (continued)**

The determinant of its Jacobian is

$$\Delta = \det DF(z_*) = a + b^2 = 0.08 + 0.36 = 0.44.$$

Also,

$$\begin{aligned} b^4 + (2a - 1)b^2 + a + a^2 &= 0.1296 - 0.3024 + 0.08 + 0.0064 \\ &= -0.0864, \end{aligned}$$

so

$$\tau = \operatorname{tr} DF(z_*) = \frac{0.0864}{0.44} \approx 0.1964 > 0.$$

Finally,

$$\tau^2 - 4\Delta < 0.$$

The equilibrium is an unstable spiral. The annular trapping region proves that at least one periodic orbit exists. It does not prove that the orbit is unique, isolated, or stable.

**Observation: Planar restriction**

The Poincaré–Bendixson conclusion is special to planar flows; it does not extend to autonomous systems in dimensions three and higher.

**Visualization: What the standard planar criteria establish**

| critereon                                                               | conclusion                                  |
|-------------------------------------------------------------------------|---------------------------------------------|
| $\dot{V} < 0$ on every non-equilibrium state                            | no periodic orbit                           |
| $\nabla \cdot (BF)$ has one strict sign<br>on a simply connected domain | no periodic orbit in that domain            |
| compact trap with no equilibrium                                        | at least one periodic orbit                 |
| Poincaré–Bendixson alone                                                | no conclusion about uniqueness or stability |

# Liénard Systems and Relaxation Oscillations

Tuesday, December 8, 2026

## Definition: Liénard equation

Let  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  be continuous. A **Liénard equation** is a second-order equation

$$\ddot{x} + f(x)\dot{x} + g(x) = 0.$$

## Recall: Position–velocity form

For the Liénard equation

$$\ddot{x} + f(x)\dot{x} + g(x) = 0,$$

set  $v = \dot{x}$ . Its position–velocity system is

$$\dot{x} = v, \quad \dot{v} = -g(x) - f(x)v.$$

## Notation: Liénard energy function

Write

$$G(x) = \int_0^x g(u) du, \quad H(x, v) = \frac{1}{2}v^2 + G(x).$$

## Observation: Liénard energy balance

Along a solution of the position–velocity system,

$$\dot{H} = v\dot{v} + g(x)\dot{x} = -f(x)v^2.$$

Thus  $f(x) > 0$  removes energy and  $f(x) < 0$  supplies energy whenever  $v \neq 0$ .

**Theorem: Liénard's theorem**

Let  $f, g \in C^1(\mathbb{R})$ . Assume:

1.  $g$  is odd and  $g(x) > 0$  for every  $x > 0$ ;
2.  $f$  is even;
3. the odd function

$$F(x) = \int_0^x f(u) du$$

has exactly one positive zero  $a$ ;

4.  $F(x) < 0$  for  $0 < x < a$ ;
5. for  $x > a$ , the function  $F$  is positive and nondecreasing, and

$$F(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

Then

$$\ddot{x} + f(x)\dot{x} + g(x) = 0$$

has exactly one limit cycle surrounding the origin, and that limit cycle is stable.

**Question: Apply Liénard's theorem to the van der Pol oscillator**

Prove that, for every  $\mu > 0$ ,

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

has exactly one stable limit cycle.

**Solution.** Here

$$f(x) = \mu(x^2 - 1), \quad g(x) = x.$$

Both functions are continuously differentiable,  $g$  is odd and positive for  $x > 0$ , and  $f$  is even. Moreover,

$$\begin{aligned} F(x) &= \int_0^x \mu(u^2 - 1) du \\ &= \mu \left( \frac{x^3}{3} - x \right) \\ &= \frac{\mu x}{3} (x^2 - 3). \end{aligned}$$

The only positive zero is

$$a = \sqrt{3}.$$

For  $0 < x < \sqrt{3}$ ,  $F(x) < 0$ . For  $x > \sqrt{3}$ ,  $F(x) > 0$  and

$$F'(x) = \mu(x^2 - 1) > 0.$$

Also,

$$F(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

Every hypothesis of Liénard's theorem holds. Hence the van der Pol oscillator has exactly one stable limit cycle for every  $\mu > 0$ .

**Definition: Planar slow-fast system**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $\varepsilon_0 > 0$ , and let

$$Q \in C^2(U \times (-\varepsilon_0, \varepsilon_0), \mathbb{R}), \quad P \in C^1(U \times (-\varepsilon_0, \varepsilon_0), \mathbb{R}).$$

A **planar slow-fast system in fast time** is

$$x' = Q(x, y, \varepsilon), \quad y' = \varepsilon P(x, y, \varepsilon), \quad 0 < \varepsilon < \varepsilon_0,$$

where  $(x, y) \in U$  and the prime denotes differentiation with respect to the fast time  $s$ .

**Definition: Critical manifold**

For a planar slow-fast system in fast time

$$x' = Q(x, y, \varepsilon), \quad y' = \varepsilon P(x, y, \varepsilon), \quad (x, y) \in U,$$

the **critical manifold** of the singular problem  $\varepsilon = 0$  is

$$\mathcal{C}_0 = \{(x, y) \in U : Q(x, y, 0) = 0\}.$$

**Definition: Attracting branch**

For a scalar fast variable  $x$ , a connected  $C^1$  curve  $\mathcal{B} \subseteq \mathcal{C}_0$  is an **attracting branch** if

$$Q_x(x, y, 0) < 0$$

for every  $(x, y) \in \mathcal{B}$ .

**Definition: Repelling branch**

For a scalar fast variable  $x$ , a connected  $C^1$  curve  $\mathcal{B} \subseteq \mathcal{C}_0$  is a **repelling branch** if

$$Q_x(x, y, 0) > 0$$

for every  $(x, y) \in \mathcal{B}$ .

**Definition: Fold**

A point  $(x_*, y_*) \in \mathcal{C}_0$  is a **generic fold**, or **knee**, if

$$Q_x(x_*, y_*, 0) = 0, \quad Q_{xx}(x_*, y_*, 0) \neq 0, \quad Q_y(x_*, y_*, 0) \neq 0.$$

**Idea: Relaxation-oscillation geometry**

For a planar slow–fast family, a **relaxation oscillation** is a family of periodic motions that, as  $\varepsilon \downarrow 0$ , spends most of each period near attracting branches of the critical manifold and connects those slow portions by transitions on the fast time scale.

**Question: Derive the large-parameter slow–fast coordinates**

To study the limit  $\mu \rightarrow \infty$ , rewrite

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

so that its critical manifold is visible.

**Solution.** Define

$$F_0(x) = \frac{x^3}{3} - x, \quad y = \frac{\dot{x}}{\mu} + F_0(x).$$

Since

$$F'_0(x) = x^2 - 1,$$

the van der Pol equation gives

$$\dot{y} = \frac{\ddot{x} + \mu(x^2 - 1)\dot{x}}{\mu} = -\frac{x}{\mu}.$$

Rearranging the definition of  $y$  gives

$$\boxed{\dot{x} = \mu(y - F_0(x)), \quad \dot{y} = -\frac{x}{\mu}.}$$

With fast time  $s = \mu t$  and  $\varepsilon = \mu^{-2}$ , this becomes

$$x' = y - F_0(x), \quad y' = -\varepsilon x.$$

Here the prime denotes differentiation with respect to  $s$ . The critical manifold is

$$\mathcal{C}_0 = \{(x, y) : y = F_0(x)\}.$$

**Question: Classify the critical cubic**

For

$$\dot{x} = \mu(y - F_0(x)), \quad \dot{y} = -\frac{x}{\mu}, \quad \mu > 0, \quad F_0(x) = \frac{x^3}{3} - x,$$

classify the branches and folds of the critical manifold  $y = F_0(x)$ .

**Solution.** With  $s = \mu t$  and  $\varepsilon = \mu^{-2}$ , the fast-time system has

$$Q(x, y, 0) = y - F_0(x), \quad Q_x(x, y, 0) = 1 - x^2.$$

Hence

$$\begin{array}{l|l} |x| > 1 & \text{attracting branch} \\ |x| < 1 & \text{repelling branch} \\ x = \pm 1 & \text{folds} \end{array}$$

**Question: Classify the critical cubic (continued)**

At  $x = \pm 1$ ,

$$Q_x = 0, \quad Q_{xx} = -2x \neq 0, \quad Q_y = 1 \neq 0,$$

so both folds are generic. Their coordinates are

$$(1, -2/3) \quad \text{and} \quad (-1, 2/3).$$

**Definition: Singular relaxation cycle**

For this singular slow-fast problem, the **singular relaxation cycle** is the oriented piecewise closed curve consisting of segments on the attracting branches of the critical manifold, traversed in the direction determined by  $\dot{y} = -x/\mu$ , and horizontal fast jumps between those branches.

**Visualization: Geometry of the singular relaxation cycle**

The critical cubic is

$$y = F_0(x) = \frac{x^3}{3} - x.$$

Its folds are

$$(1, -2/3) \quad \text{and} \quad (-1, 2/3).$$

Horizontal fast jumps have the limiting endpoints

$$(1, -2/3) \longrightarrow (-2, -2/3),$$

$$(-1, 2/3) \longrightarrow (2, 2/3),$$

because

$$F_0(x) + \frac{2}{3} = \frac{1}{3}(x-1)^2(x+2)$$

and

$$F_0(x) - \frac{2}{3} = \frac{1}{3}(x+1)^2(x-2).$$

Since

$$\dot{y} = -\frac{x}{\mu},$$

the oriented singular cycle is

| segment           | $x$ -range          | direction |
|-------------------|---------------------|-----------|
| right slow branch | $2 \rightarrow 1$   | downward  |
| lower fast jump   | $1 \rightarrow -2$  | left      |
| left slow branch  | $-2 \rightarrow -1$ | upward    |
| upper fast jump   | $-1 \rightarrow 2$  | right     |

The slow segments take time of order  $\mu$ ; the fast jumps take time of order  $\mu^{-1}$  away from neighborhoods of the fold points.

**Question: Compute the leading relaxation period**

Estimate the period  $T(\mu)$  of the van der Pol limit cycle as  $\mu \rightarrow \infty$ .

**Solution.** Introduce the slow time

$$\tau = \frac{t}{\mu}.$$

In the singular reduced problem,

$$0 = y - F_0(x), \quad \frac{dy}{d\tau} = -x.$$

Thus, along either smooth outer branch,

$$y = F_0(x),$$

and therefore

$$F'_0(x) \frac{dx}{d\tau} = -x.$$

Since  $F'_0(x) = x^2 - 1$ ,

$$\frac{d\tau}{dx} = -\frac{x^2 - 1}{x}.$$

The right slow segment runs from  $x = 2$  to  $x = 1$ , so

$$\begin{aligned} T_{\text{right}}(\mu) &= \mu \int_1^2 \left( x - \frac{1}{x} \right) dx + o(\mu) \\ &= \mu \left[ \frac{x^2}{2} - \ln x \right]_1^2 + o(\mu) \\ &= \mu \left( \frac{3}{2} - \ln 2 \right) + o(\mu) \quad (\mu \rightarrow \infty). \end{aligned}$$

By symmetry, the left slow segment has the same leading duration. The fast jumps contribute  $o(\mu)$  to the period. Therefore

$$\boxed{T(\mu) = \mu(3 - 2 \ln 2) + o(\mu)} \quad (\mu \rightarrow \infty).$$

**Visualization: Conclusions and their hypotheses**

| conclusion                              | hypothesis               | scope                     |
|-----------------------------------------|--------------------------|---------------------------|
| one stable limit cycle                  | all Liénard hypotheses   | $\mu > 0$                 |
| slow outer-branch motion and fast jumps | $\mu \rightarrow \infty$ | leading singular geometry |
| $T(\mu) = \mu(3 - 2 \ln 2) + o(\mu)$    | $\mu \rightarrow \infty$ | leading period            |

**Observation: Failure of a Liénard hypothesis**

Failure of a Liénard hypothesis makes the theorem inconclusive; it does not prove that a limit cycle is absent.

**Question: Small parameter in the van der Pol oscillator**

For the van der Pol oscillator

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0,$$

suppose  $\mu > 0$  is small. Which conclusions remain valid, and which approximation is not justified?

**Answer.** Liénard's theorem still gives exactly one stable limit cycle. The large- $\mu$  relaxation approximation is not justified.

**Question: Equilibrium-free trapping annulus**

A compact trapping annulus contains no equilibrium. What follows?

**Answer.** The Poincaré–Bendixson theorem gives at least one periodic orbit, but not uniqueness, isolation, or stability.

**Question: Two positive zeros of the Liénard integral**

For a Liénard equation, suppose

$$F(x) = \int_0^x f(u) du$$

has two positive zeros. What does Liénard's theorem imply?

**Answer.** Liénard's theorem is inconclusive because its one-positive-zero hypothesis fails.

**Question: Is the critical cubic invariant at a finite parameter?**

For fixed  $\mu > 0$ , is

$$y = F_0(x), \quad F_0(x) = \frac{x^3}{3} - x,$$

an invariant curve of

$$\dot{x} = \mu(y - F_0(x)), \quad \dot{y} = -\frac{x}{\mu}?$$

**Solution.** Set  $h(x, y) = y - F_0(x)$ . Along a solution,

$$\begin{aligned} \dot{h} &= \dot{y} - F'_0(x)\dot{x} \\ &= -\frac{x}{\mu} - \mu(x^2 - 1)(y - F_0(x)). \end{aligned}$$

On the cubic  $h = 0$ , this reduces to  $\dot{h} = -x/\mu$ . Thus the vector field crosses the cubic at every point with  $x \neq 0$ . Consequently, for finite  $\mu$ , the cubic is the exact  $x$ -nullcline but is not an invariant curve. It is the critical manifold of the singular problem and approximates the slow portions of the orbit when  $\mu$  is sufficiently large, away from the folds.

# Cumulative Final Review: A Geometric Decision Framework

Thursday, December 10, 2026

| Visualization: State space, diagnostic data, and global method |                                                                                       |                                                                                        |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| system                                                         | diagnostic data                                                                       | global method                                                                          |
| $\dot{x} = f(x)$                                               | $f(x_*) = 0$ , sign $f$                                                               | phase line and uniqueness                                                              |
| $\dot{\theta} = f(\theta)$ on $S^1$                            | $f(\theta_*) = 0$ , sign $f$                                                          | $f \neq 0$ : $T = \left  \int_0^{2\pi} \frac{d\theta}{f(\theta)} \right $              |
| $\dot{z} = Az$<br>$\dot{z} = F(z)$ in $\mathbb{R}^2$           | $\text{tr } A$ , $\det A$ , $(\text{tr } A)^2 - 4 \det A$<br>$F(z_*) = 0$ , $DF(z_*)$ | eigendirections and exact linear flow<br>invariance, energy, symmetry, global theorems |

## Observation: Hypothesis for the circle-period formula

The circle-period formula applies when  $f(\theta) \neq 0$  for every  $\theta \in S^1$ ; continuity then forces  $f$  to have one sign on  $S^1$ .

## Observation: Scope of nonlinear linearization

For  $C^1$  nonlinear planar systems, a hyperbolic Jacobian gives only the local portrait near the equilibrium.

## Recall: Local conclusion

A **local conclusion** holds on a neighborhood of a specified state.

## Recall: Global conclusion

A **global conclusion** holds on an explicitly stated entire domain.

**Recall: Necessary condition**

A condition is **necessary** for a conclusion if the conclusion implies the condition.

**Recall: Sufficient condition**

A condition is **sufficient** for a conclusion if the condition implies the conclusion.

**Question: A scalar saddle-node**

Analyze

$$\dot{x} = \lambda + x^2$$

for every real  $\lambda$ .

**Solution.** If  $\lambda < 0$ , the equilibria are

$$x_- = -\sqrt{-\lambda}, \quad x_+ = \sqrt{-\lambda}.$$

Since  $f'(x) = 2x$ , the equilibrium  $x_-$  is asymptotically stable and  $x_+$  is unstable. The phase line is

$$\begin{array}{c|cccccc} x & (-\infty, x_-) & x_- & (x_-, x_+) & x_+ & (x_+, \infty) \\ \hline \dot{x} & + & 0 & - & 0 & + \end{array}$$

At  $\lambda = 0$ , the origin attracts from the left and repels from the right:

$$\begin{array}{c|ccc} x & (-\infty, 0) & 0 & (0, \infty) \\ \hline \dot{x} & + & 0 & + \end{array}$$

Thus the origin is half-stable and Lyapunov unstable. If  $\lambda > 0$ , there is no equilibrium and  $\dot{x} > 0$  everywhere.

**Question: Saddle-node bottleneck time**

Let  $A > 0$  and  $\lambda > 0$ . For

$$\dot{x} = \lambda + x^2,$$

compute the travel time from  $x = -A$  to  $x = A$  and its leading behavior as  $\lambda \downarrow 0$ .

**Solution.** The travel time is

$$\begin{aligned} T_A(\lambda) &= \int_{-A}^A \frac{dx}{\lambda + x^2} \\ &= \frac{2}{\sqrt{\lambda}} \arctan\left(\frac{A}{\sqrt{\lambda}}\right) \sim \frac{\pi}{\sqrt{\lambda}} \quad (\lambda \downarrow 0). \end{aligned}$$

**Question: The saddle-node on a circle**

Let  $a > 0$ . Analyze

$$\dot{\theta} = \omega - a \sin \theta, \quad \theta \in S^1,$$

for every  $\omega \in \mathbb{R}$ .

**Solution.** If  $|\omega| < a$ , define

$$\alpha = \arcsin(\omega/a), \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}.$$

The equilibria modulo  $2\pi$  are

$$\theta_s = \alpha, \quad \theta_u = \pi - \alpha.$$

Since

$$f'(\theta) = -a \cos \theta,$$

$\theta_s$  is asymptotically stable and  $\theta_u$  is unstable. At  $|\omega| = a$ , they coalesce in a saddle-node. At  $\omega = a$ , the fixed point  $\theta = \pi/2$  attracts from the left and repels from the right. At  $\omega = -a$ , the fixed point  $\theta = 3\pi/2$  attracts from the right and repels from the left. In either threshold case the fixed point is half-stable and Lyapunov unstable. If  $|\omega| > a$ , there is no equilibrium and the phase rotates with the sign of  $\omega$ .

**Question: Rotation period beyond the circle saddle-node**

Let  $a > 0$  and  $|\omega| > a$ . Compute the positive rotation period of

$$\dot{\theta} = \omega - a \sin \theta, \quad \theta \in S^1.$$

**Solution.** For  $\omega > a$ , set  $\phi = \theta - \pi/2$  and  $u = \tan(\phi/2)$ . Then

$$\begin{aligned} T &= \int_0^{2\pi} \frac{d\theta}{\omega - a \sin \theta} \\ &= \int_{-\pi/2}^{3\pi/2} \frac{d\phi}{\omega - a \cos \phi} \\ &= \int_{-\pi}^{\pi} \frac{d\phi}{\omega - a \cos \phi} \quad \text{by } 2\pi\text{-periodicity} \\ &= \int_{-\infty}^{\infty} \frac{2 du}{(\omega - a) + (\omega + a)u^2} \\ &= \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \end{aligned}$$

For  $\omega < -a$ , the orientation reverses and the positive period has the same formula.

**Recall: Trace–determinant notation**

For a real  $2 \times 2$  matrix  $A$ , write

$$\tau = \operatorname{tr} A, \quad \Delta = \det A, \quad D = \tau^2 - 4\Delta.$$

**Visualization: Trace–determinant classification**

For the linear system  $\dot{z} = Az$ ,

| condition                        | linear type                                 |
|----------------------------------|---------------------------------------------|
| $\Delta < 0$                     | saddle                                      |
| $\Delta > 0, D > 0$              | node; stability from sign $\tau$            |
| $\Delta > 0, D < 0, \tau \neq 0$ | spiral; stability from sign $\tau$          |
| $\Delta > 0, D < 0, \tau = 0$    | center                                      |
| $D = 0, \Delta > 0$              | star or degenerate node; check eigenvectors |
| $\Delta = 0$                     | non-isolated equilibrium set                |

Negative trace means asymptotically stable; positive trace means unstable whenever  $\Delta > 0$  and the equilibrium is hyperbolic.

**Question: The same Jacobian can hide different nonlinear behavior**

Compare

$$\dot{x} = y, \quad \dot{y} = -x$$

with

$$\dot{x} = y, \quad \dot{y} = -x - y(x^2 + y^2).$$

**Solution.** Both systems have

$$DF(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

and eigenvalues  $\pm i$ . For the first system,

$$E(x,y) = \frac{1}{2}(x^2 + y^2)$$

satisfies  $\dot{E} = 0$ , so its positive level sets are periodic orbits and the origin is a center.

For the second system,

$$\dot{E} = x\dot{x} + y\dot{y} = -y^2(x^2 + y^2) \leq 0.$$

Thus each solution stays in the closed disk determined by its initial energy and exists for all  $t \geq 0$ . Since its energy is nonincreasing and nonnegative, write

$$E(x(t), y(t)) \rightarrow L \geq 0.$$

Its omega-limit set is nonempty and invariant, and every point in it has energy  $L$ . If  $L > 0$ , energy must stay constant along the trajectories in this set, so  $\dot{E} = -2Ly^2 = 0$  forces  $y = 0$ . But on  $y = 0$  the equation  $\dot{y} = -x$  prevents any nonzero point from staying on that line. This contradicts invariance. Therefore  $L = 0$ , and

$$(x(t), y(t)) \rightarrow (0, 0) \quad (t \rightarrow \infty).$$

**Question: The same Jacobian can hide different nonlinear behavior (continued)**

Also,

$$\|(x(t), y(t))\| \leq \|(x(0), y(0))\|$$

proves Lyapunov stability. Thus the origin of the nonlinear system is globally asymptotically stable. Every solution converges to it, so there is no nonconstant periodic orbit.

**Question: A parameter creates a unique stable limit cycle**

Let  $\mu \in \mathbb{R}$ . Analyze

$$\dot{x} = \mu x - y - (x^2 + y^2)x,$$

$$\dot{y} = x + \mu y - (x^2 + y^2)y.$$

**Solution.** At the origin,

$$DF(0,0) = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}, \quad \lambda = \mu \pm i.$$

For  $r > 0$ ,

$$\begin{aligned} r\dot{r} &= x\dot{x} + y\dot{y} \\ &= \mu r^2 - r^4, \end{aligned}$$

so

$$\dot{r} = r(\mu - r^2).$$

Also,

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = 1.$$

Therefore

$$\begin{array}{l|l} \mu < 0 & \dot{r} < 0 \text{ for } r > 0, \quad r(t) \rightarrow 0 \\ \mu = 0 & \dot{r} = -r^3 < 0 \text{ for } r > 0, \quad r(t) \rightarrow 0 \\ \mu > 0 & \dot{r} > 0 \text{ for } 0 < r < \sqrt{\mu}, \quad \dot{r} < 0 \text{ for } r > \sqrt{\mu} \end{array}$$

Thus the origin is globally asymptotically stable for  $\mu \leq 0$  and unstable for  $\mu > 0$ . When  $\mu > 0$ , the invariant circle  $r = \sqrt{\mu}$  attracts every solution with  $r(0) > 0$  and has period  $2\pi$ . On every other positive-radius interval,  $r(t)$  is strictly monotone, so no other periodic orbit exists. Thus  $r = \sqrt{\mu}$  is the unique stable limit cycle.

**Visualization: Criteria for periodic-orbit questions**

| criterion                                                            | conclusion                       |
|----------------------------------------------------------------------|----------------------------------|
| $\nabla \cdot (BF) < 0$ away from equilibria                         | no periodic orbit                |
| $\nabla \cdot (BF)$ has one strict sign on a simply connected domain | no periodic orbit in that domain |
| compact trapping region with no equilibrium                          | at least one periodic orbit      |
| isolated closed orbit                                                | limit cycle                      |
| all Liénard hypotheses                                               | exactly one stable limit cycle   |

**Question: A scaled Liénard oscillator**

Let  $a > 0$  and  $\mu > 0$ . Prove that

$$\ddot{x} + \mu(x^2 - a^2)\dot{x} + x = 0$$

has exactly one stable limit cycle.

**Solution.** For Liénard's theorem,

$$f(x) = \mu(x^2 - a^2), \quad g(x) = x,$$

and

$$F(x) = \int_0^x f(u) du = \mu \left( \frac{x^3}{3} - a^2 x \right).$$

The function  $g$  is odd and positive for  $x > 0$ , and  $f$  is even. The only positive zero of  $F$  is

$$x = \sqrt{3}a.$$

Moreover,

$$F(x) < 0 \quad (0 < x < \sqrt{3}a),$$

and for  $x > \sqrt{3}a$ ,

$$F(x) > 0, \quad F'(x) = \mu(x^2 - a^2) > 0, \quad F(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

Thus there is exactly one stable limit cycle for every  $\mu > 0$ .

**Question: Period of the scaled relaxation oscillator**

Fix  $a > 0$ . Estimate the period of the stable limit cycle of

$$\ddot{x} + \mu(x^2 - a^2)\dot{x} + x = 0$$

as  $\mu \rightarrow \infty$ .

**Solution.** Set  $x = aX$  and divide the equation by  $a$ :

$$\ddot{X} + \mu a^2(X^2 - 1)\dot{X} + X = 0.$$

This is the van der Pol equation with parameter  $\nu = \mu a^2$ . The scaling changes amplitude but leaves time, and hence period, unchanged. Since  $a > 0$  is fixed,  $\mu \rightarrow \infty$  implies  $\nu \rightarrow \infty$ . The van der Pol period estimate therefore gives

$$T(\mu) = \nu(3 - 2 \ln 2) + o(\nu).$$

Substituting  $\nu = \mu a^2$  gives

$$\boxed{T(\mu) = \mu a^2(3 - 2 \ln 2) + o(\mu)} \quad (\mu \rightarrow \infty).$$

**Question: Audit a planar omega-limit conclusion**

For a  $C^1$  planar vector field, a positive semi-orbit lies in a compact subset of the vector field's domain, and its omega-limit set contains no equilibrium. What follows?

**Answer.** The Poincaré–Bendixson theorem makes the omega-limit set a periodic orbit.

**Question: Audit an annulus conclusion**

A compact annulus contains no equilibrium. Does a periodic orbit follow?

**Answer.** No. Positive invariance, or another form of forward confinement, must also be proved.

**Question: Audit a first-integral conclusion**

For a  $C^1$  planar system, a connected compact regular level component of a first integral contains no equilibrium. What follows?

**Answer.** The level component is a periodic orbit. Nearby energy levels may form a family, so the orbit need not be a limit cycle.

**Question: Audit a divergence conclusion**

The divergence of a  $C^1$  planar vector field is negative on an open, simply connected region containing a proposed closed orbit. What follows?

**Answer.** The Bendixson–Dulac criterion with  $B \equiv 1$  rules out the proposed closed orbit.

**Idea: Four layers of a complete qualitative analysis**

state space and parameters  $\Rightarrow$  equilibria and local data  
 $\Rightarrow$  global invariant structure  $\Rightarrow$  exact conclusion.